Formality as a Financing Decision

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Abstract

Can bank loans make informal firms formalize? We match the universe of bank loans, the social security records, and the economic censuses in Mexico to track formality, loans, and firm transitions over time. Three facts emerge. Three quarters of firms with a bank loan are informal; informal firms pay higher rates, post more collateral, and obtain smaller loans; and firms move between formality and informality in both directions. We develop a dynamic model in which firms that need a loan and are nearly indifferent between being formal and being informal change their status when the relative terms or availability of loans change. Using leave-one-out shift-share bank credit supply shocks, we find that a ten percent credit supply expansion raises the financial advantage of formal firms and, consequently, increases formalization by 0.5 percent and decreases formal-to-informal transitions by 0.4 percent. Our contribution is to show that bank loans affect informal firms and their formality decisions.

Keywords: informality, firm dynamics, financial frictions, credit supply, formalization.
JEL codes: O17, G21, L25

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1 Introduction

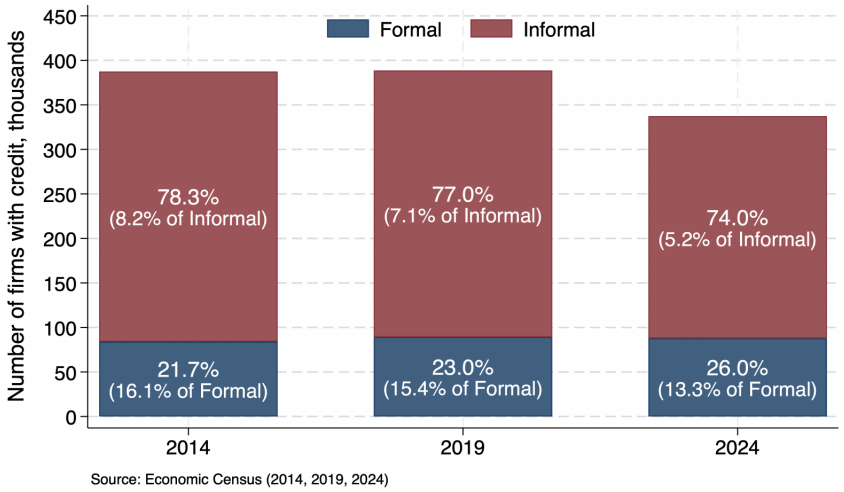
The effects of the financial sector on formal firms are well known in the literature. However, little is known about its effect on the informal sector. The reason is that there is little data on informality and the financial sector. This paper overcomes this data limitation, uncovers stylized facts linking informality and finance, and addresses the following research question: Can bank loans affect the formality decision of firms, especially of those already in the informal sector? We show that credit supply expansions reach informal firms but improvement in terms and volume is higher for formal ones. This differentiated effect increases the financial advantage of formal firms and changes the two-way formality transitions. A ten percent expansion raises formalization by 0.5 percent and lowers informalization by 0.4 percent.

This paper matches the universe of bank loans to social security administrative records at the firm level to uncover the formality of borrowers, the loan position of formal firms, and the difference in loan terms between formal and informal enterprises. It contains nearly 3.3 million firms, 1.3 million borrowing and 2.2 million formal, across 119 banks between 2005 and 2025 (Appendix A). However, informal firms without a bank loan escape this dataset. To overcome this problem, we merge these two datasets to the economic censuses, which is only possible at the market level, to observe all types of firms over time: formal and informal, borrowers and non-borrowers. The resulting dataset is unique because it contains formality, bank loans, and firm transitions on the same markets over twenty years. As a result, we can study a previously unexplored relationship: the effect of bank loans on informal firms.

With these data, we document three stylized facts. First, informal firms are inside the banking system, in contrast to what is commonly assumed. Figure 1 reveals that among all the firms holding a bank loan in Mexico, 74 percent are informal[1]. Therefore, informal firms are directly exposed to bank loan conditions. However, because the informal sector is so large, the share of informal firms with a bank loan represents only 5 percent, against 13 percent for formal firms.

[1]An establishment is informal when it reports zero social security contributions in the census while earning income, the census classification used throughout the paper (Section 3).

Figure 1: Borrowing shares by formality



Second, informal firms face worse loan terms than their formal counterparts. Table 1 demonstrates that, at the median, an informal firms pay an interest rate 0.7 percentage points higher, need 3 percentage points more collateral, and obtain 33.2 thousand dollars smaller loans than a formal firm. These differences hold for the 10th and 90th percentiles[2] and, in general, for the majority of the distribution.

Third, formality status is not permanent and it can change, potentially due to changes in the economic environment, like the financial conditions. Table 2 shows that between 11.6 and 21.8 percent of formal firms become informal in five years, while between 3.5 and 6.5 percent of informal firms formalize. In contrast to previous work that treats formality as an absorbing state from which firms cannot transition into informality (Meghir et al., 2015; Ulyssea, 2018), we see formality transitions as two-way[3].

To understand the plausible mechanisms that link formality decisions to the financial sector, we build a parsimonious dynamic model in which firms choose each period between formality, informality, and exit. Formality carries taxes and higher operating costs but better financial terms (lower marginal loan cost and more pledgeable collateral) while informality saves the taxes and has lower operating costs but faces detection costs that grow with size and worse financial terms. Therefore, the formality decision trades off the tax cost with the financial benefits of being formal. The model predicts that a relative improvement in the terms or availability of loans for formal firms with respect to the informal ones, should increase formalization and decrease formal-to-informal transitions, for firms that need additional loans.

[2]With the exception of collateral at the 90th percentile

[3]The formal-to-informal and informal-to-formal transitions are not an artifact of the definition of informality. Appendix B documents that these transitions exist under alternative definitions of informality: accounting for the intensive margin (Table 12), using the social security number (Table 13), or using the tax number (Table 14).

The model predictions are tested in the empirical section, where the challenge to claim causality is endogeneity in the form of reverse causality (formality affecting financial conditions and not the inverse) and unobserved common characteristics (only productive firms formalize and hold a bank loan). To overcome this challenge we rely on a Leave-One-Out Shift-Share Instrumental Variable (LOO-SSIV) with exogenous shift.

This method disentangles the changes in credit within firms that borrow from several banks into supply (bank fixed-effects) and demand (firm fixed-effects) (Khwaja and Mian, 2008; Amiti and Weinstein, 2018), leaving the market of interest outside of the estimation sample. Afterwards, it computes the predetermined bank shares in the market of interest. The final instrumental variable is the aggregation of the shift (bank fixed-effects), weighted by the share (bank market-shares), as in the quasi-experimental Shift-Share framework of Borusyak et al. (2021).[4] The instrument is strong, with a first stage F statistic above 265, and a one standard deviation shock roughly doubles five year market credit.

The model predictions are corroborated. Improving the formal advantage in loan terms and availability increases formalization and reduces formal-to-informal transitions. A credit supply expansion yields 25 percent less collateral, 19 percent more credit, and half a percentage point cheaper loans for formal firms four years after the shock; whereas for informal firms the volume response is smaller (16 percent), rates do not change (-0.1 percentage points, not significant), and collateral falls by less (13 percent). The credit supply shock amplifies the formal financial advantage and, in consequence, as predicted by the model, causally raises formalization by 0.5 percent and lowers formal-to-informal transitions by 0.4 percent, after a ten percent increase in credit. That is about 850 additional transitions from informal to formal and 600 fewer in the opposite direction across the markets in our sample. The results imply that policies that affect the relative loan advantage of formal firms, from credit guarantees to bank competition, can shift the share of formal firms. These main results are robust to balance, pretrend and placebo tests, the removal of the largest lender and of the largest markets, and a shock-level design with re-identified shocks (Section 7.3).

Additionally, for a ten percent increase in credit supply the stock of formal firms rises by 0.5 percent, the formal entry rises by 0.7 percent, and exit rises by 0.5 percent, mostly among informal firms. Regarding the number of firms holding a loan, it increases 0.4 percent among formal and 0.3 percent among informal firms. Thus, the shock changes the terms and size of the loan, not who can borrow. On the real side, and less precisely estimated, the formal wage bill rises by 0.2 percent and the formal employment by 0.2 percent, whereas the mean wage falls marginally. The value added falls by 0.5 percent in

[4]Inference uses the exposure robust standard errors of Adão et al. (2019). As a robustness, we report the wild cluster bootstrap probabilities (Cameron et al., 2008). The design rests on roughly seventeen effective bank shocks, no single bank carries more than 17 percent of the exposure weight, and the results survive rebuilding the instrument without the largest lender, at a cost in precision (Tables 5 and 33).

the informal sector and does not move in the formal one.

The paper's contribution lies in showing that bank loans also affect informal firms, and not only the formal ones, and that this effect has broader implications on formality shares, the labor market, and average productivity. Moreover, we contribute to three literatures, developed in Section 2. To the credit supply literature it adds the margin that previous work could not incorporate: informality. To the informality literature it adds a financial lever that moves the formality margin. To the literature on firm dynamics with financial frictions it adds quasi-experimental evidence on how financing terms shape the formality choice in a firm dynamics setting. The main limitation of this work is the absence of general equilibrium effects and competition in the loan, labor, and goods markets. More research is needed to address these limitations and to better understand the effects of the financial sector on informality.

The paper proceeds as follows. Section 2 reviews the literature. Section 3 describes the data and Section 4 the descriptive stylized facts. Section 5 builds the dynamic model, Section 6 develops the identification strategy, and Section 7 presents the results. Section 8 concludes.

2 Literature review

This paper relates to three strands of literature. The first is the effects of credit supply shocks on real outcomes. On the methods, to disentangle demand from supply in credit, Khwaja and Mian (2008) rely on multi-bank firms and Degryse et al. (2019) on credit changes at the market-sector level to include also single bank firms. Amiti and Weinstein (2018) extend the decomposition to granular bank shocks. Regarding the effects on advanced economies, Chodorow-Reich (2014) shows the reduction in employment of US firms connected to distressed banks, whereas Greenstone et al. (2020) find a small effect on labor after a county credit shock. Huber (2018) finds reductions in employment and output in German counties. In developing economies, Banerjee and Duflo (2014) states that small firms are credit constrained. In Mexico, Morais et al. (2019) use the supervisory data on the universe of bank loans to identify credit supply shocks from foreign monetary policy and find strong real effects on firms, Gutierrez et al. (2023) estimate that one standard deviation credit supply shock increases formal employment by 1.4 percentage points and Bruhn and Love (2014) argue that opening a bank branch raises informal business ownership, employment and income.[5]

However, this literature is constrained by the fact that administrative records do not

[5]In Brazil, formal employment increased after a reform that expanded credit (Fonseca and Van Doornik, 2022). Melcangi (2024) shows that even unconstrained firms increase their cash reserves and reduce employment in a credit crisis.

capture informal firms and household surveys do not follow the firm, its loans, or its formality status over time, so this literature is incapable to assessing whether credit affects the informal sector or the formality decision. Our contribution is to overcome this constraint by matching the universe of business loans to social security records and economic census. We show that loans also affect informal firms, not only formal ones, and the formality decision itself. In particular, we find that credit supply expansions increase formalization and decrease movement back to informality, increasing the formality share.[6]

The second strand of literature is on informality. La Porta and Shleifer (2014) reveal that informal firms are small, unproductive, and rarely formalize. Ulyssea (2020) presents an extensive survey on the causes and consequences of informality. In Mexico, Levy (2018) and Busso et al. (2012) position informality as the main source of misallocation and low productivity, while Alvarez and Ruane (2024) find that the informality growth had little effect on aggregate productivity. At the modeling level, Ulyssea (2018) separates the intensive and the extensive margins of informality, Meghir et al. (2015) and Bobba et al. (2021) integrate the formality choice into a job search model, and Haanwinckel and Soares (2021) rationalize informality through compensating differentials. Regarding empirical estimations on the causes of informality, Imbert and Ulyssea (2026) find that a labor supply shock, rural migration, affects the informality share.

However, the main margins studied are the fiscal and administrative costs of formality. In Brazil, Rocha et al. (2018) find that lowering taxes formalizes existing informal firms, with a low elasticity, and Almeida and Carneiro (2012) that stricter enforcement raises formal employment. In Mexico, simplifying registration has modest effects on formalization, mainly on wage earners opening a business rather than on informal firms (Bruhn, 2011; Kaplan et al., 2011). Moreover, policies that changed the relative cost of formality", like non-contributory health insurance or the 2019 border zone tax cut, reduced formalization of small firms or left formal employment unchanged (Bosch and Campos-Vazquez, 2014; Campos-Vazquez et al., 2020).[7] In contrast to this literature, we explore the financial margin at the market level in a causal way. Unlike the modest effects of previous policies, we find that formality status is sensitive to loan terms. Our model allows, guided by the data evidence, that firms switch in both directions the formality decision, whereas previous models do not allow a formal firm to become informal.[8]

The third strand of literature relates financial frictions to firm dynamics. Hopenhayn

[6]The instrument computation follows the quasi-experimental Shift-Share framework of Borusyak et al. (2021), with exposure robust inference from Adão et al. (2019) and diagnostics in the spirit of Goldsmith-Pinkham et al. (2020).

[7]Field experiments that reimburse the registration cost find similarly small effects (De Mel et al., 2013).

[8]Adom (2025) calibrates a dynamic model for Nigeria in which financial constraints delay the formalization of informal firms, whereas Dix-Carneiro et al. (2026) study the effects of trade on informality in Brazil.

(1992) provides the foundation of firm dynamics models, Evans and Jovanovic (1989) shows that wealth constrains entry into entrepreneurship, and Cooley and Quadrini (2001) and Gomes (2001) integrate the financial dimension into these dynamic models. Albuquerque and Hopenhayn (2004) and Clementi and Hopenhayn (2006) add borrowing limits based on limited enforcement. Buera et al. (2011), Arellano et al. (2012), Midrigan and Xu (2014) and Moll (2014) apply these models to development economics. Regarding the literature that links the formality choice to finance, Straub (2005) adds the credit market channel, where firms formalize to get access to credit. Antunes and Cavalcanti (2007) and Quintin (2008) connect the size of the informal economy to enforcement of credit contracts. D'Erasmo and Boedo (2012), D'erasmo (2016), Franjo et al. (2022) and Erosa et al. (2023) integrate the formality choice in an economy with heterogeneous firms, where they calibrate the financial disadvantage of informality and trace it to output, productivity, tax revenue. For Brazil, D'erasmo (2016) studies formalization after the credit expansion of 2000.

The empirical literature linking informality and finance is thinner. Beck et al. (2014) show that firm evasion is lower in places with more credit information and branch reach. Gandelman and Rasteletti (2016) finds that industries more dependent on finance have fewer informal jobs where credit is more present. Fajnzylber et al. (2011) find that the gains from formalizing in Brazil do not run through access to credit. However, none of this evidence observes the loan terms for informal firms or follows the formality status of firms over time and the calibrated models typically exclude informal firms from credit (Franjo et al., 2022). Our contribution is demonstrating that informal firms access credit and that the formality decision depends on the formal-informal gap of loan terms: interest rates, collateral and loan size. Moreover, instead of calibrating the financial gap, we measure it in the loan registry and change it with quasi-experimental variation, confirming the models' predictions.

3 Data

This work uses three microdata sources and the Bank of Mexico definition of local labor markets in Mexico. The first data source is the credit registry and includes the universe of bank loans to formal and informal firms, where amounts, costs, and collateral requirements are observed. The second source are the social security records that measure formal employment and wages. The third dataset is the economic census, which is a quinquennial enumeration of formal and informal establishments with fixed facilities, on which we can follow establishments over time.

The credit registry is generated by the banking commission (CNBV) from the regulatory report R04C that every bank has to file by law.[9] This data includes the universe of

[9]There is an aggregate, public version of this data available on the commission's website. We used the

commercial bank loans to businesses, from 2005 to date, at a monthly frequency. The characteristics of each loan are included, such as the amount, the interest rate, the collateral requirement, and the anonymized identity of the firm. It includes 119 banks and 1.3 million borrowing firms, between 2005 and 2025.

The social security records (IMSS) cover the universe of registered employers and their workers, formal by definition. Wages are daily base salaries, which include cash payments, bonuses, commissions, and payments in kind.[10]

The economic census (INEGI) surveys establishments with fixed physical facilities every five years, and the analysis uses the 2009, 2014, 2019, and 2024 waves.[11] An establishment in the economic census is informal if it reports positive income and workers but zero social security contributions, formal otherwise. The economic census allows us to identify entry, exit and formality transitions of establishments, as well as their value added.

The 777 Local Labor Markets (llm) in the country are constructed by the Bank of Mexico using commuting flows[12]. Credit is observed in 705 of them, and the estimation sample of the main specification in Section 6 keeps the 556 markets in which the instrument covers at least half of baseline credit, which translates into 1,390 market-year observations across the census windows.

To combine the datasets, first, we matched the credit registry and the social security records at the firm level. This match makes possible to identify the formality of the borrower, which is informal if it does not appears in the social security records, formal otherwise. The matched data contains 3.3 million firms. Notice that the credit-social security merged data includes formal firms with and without loans, and informal firms with loans; however, it does not capture informal firms that do not have a bank loan. To overcome this problem we merge the previous matched datasets to the economic census at the market level. By doing this we are able to identify all the formal and informal establishments, with or without loans, at a market level and follows them over time.[13]

The final dataset excludes agriculture, mining, construction, transport, financial services, and government. Borrowers whose main address is outside of the country are ex-

individual non-public version, accessed through the Bank of Mexico secured environment, EconLab, where all the information is anonymized.

[10]Wages are bottom coded at the minimum wage and top coded at twenty five times the Mexico City reference minimum. The unit of observation is a job rather than a person, so a worker holding two registered jobs contributes twice to our employment counts.

[11]The census is representative of localities with more than 2,500 inhabitants and non-agricultural sectors. Activity without a fixed physical facilities, like street vending, are excluded.

[12]The methodology and information is published in the Bank of Mexico's market analysis.

[13]Figure 8 in Appendix B compares the informality share of firms with bank loans from two different sources: the credit-social security matched registries and the economic census. The average level is similar, 76 percent against 81 percent in the census. However, at the state level the differences are higher, mainly due to the differences in the definition of establishment and firm.

cluded. The monthly credit registry and social security records are made annual by keeping only the January snapshot. We keep only observations with valid locality and activity codes. All the merges are made using anonymized versions of the tax number (RFC), social security number (IMSS), and borrower ID. All the data construction and definitions are collected in Appendix A.

The combination of these three datasets allows us to observe credit, formality, and firm dynamics in the same markets over twenty years, something unique in a dataset. The effects of bank loans on informal firms is relatively unexplored, mainly due to the unavailability of the data. In contrast, the uniqueness of this data allows us to uncover the fact that bank loans reach the informal sector, and the conditions under which this happens, which is a central argument of this paper.

4 Descriptive statistics

This section provides general descriptive statistics of the matched dataset. However three stylized facts stand out for their relevance and implications. The most important and poorly documented descriptive fact from the data is that informal firms are not outside the banking system, corroborated by two different sources. Table 1 reveals that in 2024, between 67 (credit-social security registry) and 74 (census) percent of firms with bank loans were informal. This means that changes in credit conditions affect informal firms directly.

Table 1: Descriptive statistics on firms, credit, and formality

	Formal	Informal	All
Panel A. Firms and bank credit: Economic Census 2024			
Firms	658,065	4,810,115	5,468,180
Share of firms (percent)	12.0	88.0	100.0
Firms with bank credit	87,788	249,588	337,376
Credit incidence (percent of firms)	13.3	5.2	6.2
Share of borrowers (percent)	26.0	74.0	100.0
Panel B. Credit terms by borrower status: credit registry, pooled 2005-2025			
Borrower years	1,089,934	4,132,779	5,222,713
Borrowers, 2024	83,334	171,635	254,969
Unique borrowers, 2005–2025	210,895	1,114,533	1,269,287
Banks per borrower, mean	1.35	1.14	1.19
Borrowers with two or more banks (percent)	23.9	10.2	13.1
Credit records per borrower, mean	4.32	2.32	2.74
Authorized credit per borrower (thousand dollars)			
10th percentile	5.9	2.2	2.6
Median	47.9	14.7	17.2
90th percentile	631.8	184.2	263.2
Interest rate (percent)			
10th percentile	9.3	10.0	9.9
Median	15.5	16.2	16.0
90th percentile	24.4	29.0	28.0
Mean	16.1	18.5	17.9
Relationships without collateral (percent, 2010-2024)	80.2	88.1	86.2
Collateral to authorized credit, collateralized relationships			
10th percentile	0.05	0.06	0.05
Median	0.45	0.48	0.47
90th percentile	1.45	1.31	1.36

Notes: Panel A source is the 2024 Economic Census, where informal is defined as an establishment with positive income and workers but zero social security contributions. Panel B is the January snapshot and the source is the administrative social security records matched with the universe of loans at the firm level. Formal in panel B is defined as firm that appears in the social security records, and therefore paid them. Percentiles are computed within each column. Authorized credit is the sum of authorized lines across the borrower's banks, in nominal pesos converted at 19 pesos per dollar. Interest rates are annualized gross rates at the bank relationship level, winsorized at the 1st and 99th percentiles within year. Collateral information exists from 2010. The ratio divides collateral value by the authorized line among relationships with positive collateral.

The second stylized fact, shown in Table 1, is that the loan contracting conditions are harsher for informal firms. At median values, an informal firm pays 0.7 percentage points more interest rate, requires 3 percentage points more collateral, and obtains a loan that is 33.2 thousand dollars smaller than a formal firm. However, the ratio of non-collateralized credit is 7.9 percentage points higher for informal firms. This translates into a financial advantage for formal firms.

Table 2: Establishment transitions across censuses

		Destination		
		Formal	Informal	Exit
2014 Origin	Formal	122,784	23,526	56,994
	(% of Formal)	60.4	11.6	28.0
	Informal	263,565	2,232,216	1,531,660
	(% of Informal)	6.5	55.4	38.0
2019 Origin	Formal	324,052	138,041	170,946
	(% of Formal)	51.2	21.8	27.0
	Informal	146,071	2,547,652	1,473,395
	(% of Informal)	3.5	61.1	35.4

Notes: An establishment is formal if has positive income and workers but zero social security contributions. Exit means the establishment is not observed in the following census. Source: Economic Censuses 2014, 2019 and 2024.

The third stylized fact, displayed in Table 2, is that the formality status of a firm is dynamic and can change in both directions. In a five year window, between 3.5 and 6.5 percent of the informal firms formalize and between 11.6 and 21.8 percent of the formal firms become informal. In contrast to previous literature, where the formal status is an absorbing state,[14] we allow formal transitions to happen in both directions. This implies that formality is not a one-time decision, is continuously re-optimized and is sensitive to changes in the economic environment, including potentially the bank credit conditions.

These three stylized facts guide the modeling decisions of Section 5, the logic behind the empirical strategy in Section 6, and the argumentative scaffold of the paper. The central idea is simple and argues that certain firms (those almost indifferent between being formal or being informal and that need a bank loan) decide to be formal or not based on the different loan conditions they would face as formal or informal. Therefore, for these firms formality is a financing decision.

In general, Tables 2 and 1 present additional descriptive facts and statistics. For instance, the high prevalence of informality, where 88% of the 5.5 million establishments are

[14]See for example Meghir et al. (2015); Imbert and Ulyssea (2026); Ulyssea (2018).

informal; or the higher loan incidence among formal firms, 13.3 percent in the formal sector against 5.2 percent in the informal one. Moreover, Table 3 contains descriptive statistics at the market level, the unit of inference. The median market contains 2,319 firms, 50 borrowers, and 2.5 million dollars of authorized loans. Finally, Appendix D shows, first, that informal firms hold less capital per worker, invest less, and carry less bank debt per worker at every size and age. Second, that 85 percent of borrowing firms in 2024 hold loans from a single bank. This last fact is consequential for the empirical strategy in Section 6 and motivates the robustness check in Table H.7.

Table 3: Estimation sample: local markets across five year windows

	Mean	SD	p10	Median	p90	N
Panel A. Pooled levels at the start of the window, per market						
All firms	8,874.6	29,241.7	415	2,319	20,316	1,394
Formal	1,094.3	3,872.7	13	138.5	2,695	1,394
Informal	7,780.3	25,628.2	393	2,159.5	17,436	1,394
Firms with bank credit (census)	572.8	1,793.1	3	119	1,331	1,394
Switchers, informal to formal	253.4	879.8	3	38.5	589	996
Switchers, formal to informal	229.0	814.4	3	37	537	996
Borrowers (registry)	556.8	3,714.0	4	50.5	919	1,394
Total authorized credit†	431.5	5,578.9	0.1	2.5	195.6	1,394
Collateral ratio	2.21	55.35	0.00	0.04	0.31	1,394
Interest rate (percent)	14.86	4.56	8.88	14.91	20.30	1,394
IMSS workers	26,655	142,866	37	1,316	47,088	1,394
Panel B. Instrument						
Shift share instrument (standardized)	0.248	1.197	−0.893	−0.261	2.063	1,394
Coverage	0.921	0.110	0.762	0.967	1.000	1,394

Notes: All statistics pool the 1,394 market window observations (556 markets, windows 2009–2014, 2014–2019, 2019–2024), unweighted. N smaller where undefined. Census is the source for firms by status and firms with bank credit. The registry os the source for borrowers, credit, collateral ratio and interest rate (credit weighted market averages). † Million dollars at 19 pesos per dollar. Coverage: share of baseline credit held by banks with an estimated shock. Five year changes of every outcome, and the full versions by source is in Tables 16 and 17.

5 A parsimonious model of formality and finance

The objective of this section is to build a parsimonious model that rationalizes the link between bank credit conditions and firms' formality decisions. The model is a stationary firm dynamics model in the tradition of Hopenhayn (1992), with two operating regimes as in Ulyssea (2018) and costly external finance as in Gomes (2001). The empirical facts of Section 4 guide the modeling decisions. For instance, we allow firms to switch back and

forth their formality status, as observed in Table 2, and informal firms to access bank loans (Table 1, panels A and B) under sector-specific loan terms (Table 1, panel B). The model identifies which firms respond when financial conditions change and signs their response. Moreover, it serves as the stepping stone for the empirical strategy of Section 6.

5.1 Environment

For simplicity, time is discrete and lasts two periods, $t \in \{1, 2\}$. The economy is populated by two types of risk-neutral entrepreneurs: incumbents and potential entrants. An incumbent entrepreneur owns a firm that belongs to a (formal or informal) sector $s \in \{i, f\}$ and holds $k_1 > 0$ installed capital at period 1, while a potential entrant entrepreneur is endowed with financial wealth $k_0 \geq 0$ and may or may not start a firm in sector $s \in \{i, f\}$ in period 2. Every entrepreneur discounts period 2 at factor $\beta \in (0, 1)$ and consumes her dividends and the depreciated capital, which depreciates at rate $\delta \in (0, 1)$. We assume that output is the *numéraire* good, the wage w is exogenous to the firm, and both output and wages before taxes are the same for formal and informal firms. These assumptions allow us to focus on the formality transitions caused by financial conditions and are first-order inconsequential for the conclusions.

An active firm in sector s with capital k and labor ℓ produces

$$y \;=\; \theta\, k^{\alpha} \ell^{\nu}, \qquad \alpha, \nu > 0, \qquad \alpha + \nu < 1, \tag{1}$$

where log productivity combines a permanent component v, drawn at birth and known to the firm, and transitory shocks that follow an AR(1) process (Appendix I). As shown below, the expectation of the profitability index $x \equiv \theta^{1/(1-\nu)}$ is the only object of this process that enters the firm's decisions.

A formal firm pays a proportional tax $\tau_y \in [0, 1)$ on revenue and a payroll tax $\tau_\ell \geq 0$ on its wage bill. An informal firm evades both taxes. Therefore, the only advantages of formality are financial, together with a survival advantage introduced below, and the only advantages of informality are fiscal and its lower fixed costs.[15] Static profits are

$$\pi_s(\theta, k) = \max_{\ell \geq 0} \left(1 - \tau_y^s\right) \theta k^{\alpha} \ell^{\nu} - \left(1 + \tau_\ell^s\right) w\, \ell, \tag{2}$$

with $\tau_y^f = \tau_y$, $\tau_\ell^f = \tau_\ell$ and $\tau_y^i = \tau_\ell^i = 0$. The model excludes on purpose the possibility of the intensive margin of informality, i.e. firms paying payroll taxes only on part of their

[15] For tractability, we exclude the size-dependent audit cost of informal firms used by Ulyssea (2018). Including it would make formality always optimal for large firms, regardless of financial conditions. For smaller firms, for which this cost is negligible, the exclusion is inconsequential, the financial advantage of formality and the conclusions of the model still hold.

total work force. This decision is data driven. As Table 12 shows, only 0.7 and 5.3 percent of firms used the intensive margin in 2014 and 2019, respectively.

Wages and taxes are paid within each period out of contemporaneous revenue and wealth. Investment, in contrast, must be paid one period before it becomes productive. Therefore, a firm raises external funds, denoted by $a \geq 0$, to cover any difference between the cost of desired capital and its internal funds.[16] External finance entails a sector-specific cost, so a firm raising an amount a incurs a financing cost of the form

$$\Phi_s(a) \;=\; \phi_s\, a, \qquad \phi_i \;>\; \phi_f \;\geq\; 0, \tag{3}$$

which is paid within period and on top of the repayment of the principal. The parameter ϕ_s is therefore interpreted as the spread in each sector between the marginal cost of external funds and the risk-free rate. The spread in the informal sector is higher because informal firms cannot pledge audited accounts, registered assets, or formal payroll as information collateral, in line with the data (Table 1, panel B).

Besides the sector difference in price, there is a difference in the amount of external funds available to a firm, which is limited by a collateral constraint on the capital being installed,

$$a \;\leq\; b_s\, k, \qquad 0 \leq b_i < b_f < 1. \tag{4}$$

This constraint means that lenders only finance a fraction b_s of the installed capital. For formal firms this fraction is higher because being formal raises pledgeability, which is consistent with Table 1.

Hence, the pair (ϕ_s, b_s) summarizes in reduced form the financial conditions faced by sector s. The annual results of Section 7.1 measure the effects of credit supply shocks on these two parameters, loan rates and collateral by sector. Given their importance, all the comparative statics below are taken with respect to these parameters.

A firm faces entry, $c_s^e \geq 0$, operating, $c_s \geq 0$ with $c_f > c_i \geq 0$, and switching costs, $c_{if}, c_{fi} \geq 0$. Additionally, in period two and before producing, a sector-specific exogenous death probability materializes, $\lambda_s \in [0, 1)$ with $\lambda_i \geq \lambda_f$.[17]

The timing is presented in detail in Appendix I. At time 1, and before observing any shock (ε_2 for productivity and λ_s for death), a firm or potential firm chooses her next period sector, $d \in \{\text{out, informal, formal}\}$. In the transition stage a firm chooses the next period optimal level of capital k_2 and the level of external funding $a \leq b_{s'} k_2$ at cost $\Phi_{s'}(a)$.

[16] External finance includes a wide range of funding sources. This paper focuses on bank loans and abstracts from the optimal choice of financing instruments. More broadly, the financing-cost function can be interpreted as a reduced-form representation of other sources of costly external finance, provided that their relevant features are captured by the same specification.

[17] The inequality $\lambda_i \geq \lambda_f$ is consistent with the data observed in Table 2.

Then, in period 2 and after observing the realization of shocks, active firms produce, sell the depreciated capital and pay taxes, wages, and any cost.

The internal funds available to a firm are the operating profit of the previous period plus the resale value of depreciated capital, net of the fixed costs and any other switching or entry fee:

$$n \;=\; \begin{cases} \pi_s(\theta_1, k_1) - c_s + (1-\delta)k_1 - c_{ss'}\mathbf{1}_{\{s' \neq s\}}, & \text{incumbent of sector } s, \\ k_0 - c^e_{s'}, & \text{entrant into } s', \end{cases} \tag{5}$$

with $c_{ss} \equiv 0$. Then, an exiting incumbent simply consumes $n^{\text{exit}} = \pi_s(\theta_1, k_1) - c_s + (1-\delta)k_1$ and a non-entrant consumes k_0.

5.2 Capital and investment decisions

After solving the static labor problem of the firm, we can re-express her profits as

$$\pi_s(\theta, k) = \Omega_s \, x \, k^{\gamma}, \tag{6}$$

where $\gamma \equiv \alpha/(1-\nu) \in (0,1)$, $x \equiv \theta^{1/(1-\nu)}$ is the profitability index, and $\Omega_s \equiv (1-\nu)\,(1-\tau^s_y)^{\frac{1}{1-\nu}} \left[\frac{\nu}{(1+\tau^s_\ell)w}\right]^{\frac{\nu}{1-\nu}}$ collects the technology, wage, and taxes paid in sector s. This expression is useful because profits are linear in x, so the single sufficient statistic in every decision below is the expectation of the profitability index, $\hat{x} \equiv E[x_2 \mid \mathcal{I}_1]$. All the statements are formally presented in Appendix I and all proofs are collected in Appendix J.

Consider now the problem of capital and external funds at the transition stage of a firm that decided to operate next period in sector s, with internal funds $n \geq 0$ and expected profitability index $\hat{x}$:

$$G_s(\hat{x}, n) \;\equiv\; \max_{k \geq 0} \; -k - \Phi_s\big(a\big) + \beta\Big[(1-\lambda_s)\,\Omega_s \hat{x}\, k^{\gamma} + (1-\delta)k\Big], \quad a = \max\{0, k-n\} \leq b_s k. \tag{7}$$

Therefore, the value of operating in sector s is $U_s(\hat{x}, n) = n + G_s(\hat{x}, n) - \beta(1-\lambda_s)c_s$. In other words, the entrepreneur consumes n minus the resources put into the firm and expects profit net of the fixed cost if it survives plus the depreciated capital resale.

Define the cost of capital as $R^0 \equiv 1 - \beta(1-\delta)$ when it is acquired with internal funds and as $R_s \equiv R^0 + \phi_s$ when capital is paid for with external funds borrowed under sector s conditions. Then, the corresponding optimal capital targets under these two prices are

$$\bar{k}_s(\hat{x}) = \left[\frac{\gamma\beta(1-\lambda_s)\Omega_s\hat{x}}{R^0}\right]^{\frac{1}{1-\gamma}} \quad > \quad \tilde{k}_s(\hat{x}) = \left[\frac{\gamma\beta(1-\lambda_s)\Omega_s\hat{x}}{R_s}\right]^{\frac{1}{1-\gamma}}. \tag{8}$$

Table 4 summarizes the solution to the capital-financing problem and distinguishes four types of firms. Cash-flow sensitivity refers to the marginal response of optimal capital to internal funds, $\partial k_s^*/\partial n$, holding $\hat{x}$ and s fixed.

Table 4: Four regions of optimal capital and external funds

Region	Funds (n) vs. k_s^*	Capital k_s^*	Borrowing a_s^*	Cash sensitivity
1: Unconstrained	$n \geq \bar{k}_s$	$\bar{k}_s(\hat{x})$	0	0
2: Constrained	$\tilde{k}_s \leq n < \bar{k}_s$	n	0	1
3: Borrower	$(1-b_s)\tilde{k}_s \leq n < \tilde{k}_s$	$\tilde{k}_s(\hat{x})$	$\tilde{k}_s - n$	0
4: Collateral-constrained	$n < (1-b_s)\tilde{k}_s$	$\dfrac{n}{1-b_s}$	$b_s k_s^*$	$\dfrac{1}{1-b_s}$

The first type is an unconstrained firm with sufficient internal funds to finance its internally financed capital target ($n \geq \bar{k}_s$). This firm does not borrow, and its cash-flow sensitivity is zero. The second type is a constrained non-borrower. Its internal funds are insufficient to attain the internally financed capital target, but borrowing is not optimal. The firm therefore invests all its internal funds, so its cash-flow sensitivity is one because each additional unit of internal funds increases capital by one unit. The third type is a borrower that attains the capital target associated with the higher cost of external finance. It borrows the difference between its chosen capital expenditure and its internal funds. Its cash-flow sensitivity is zero because additional internal funds reduce borrowing without changing its capital choice. The fourth type is a collateral-constrained firm. This firm borrows up to its collateral limit and invests less than its externally financed capital target. Its cash-flow sensitivity is $1/(1-b_s) > 1$. Each additional unit of internal funds supports $1/(1-b_s)$ additional units of capital, including $b_s/(1-b_s)$ units financed through additional borrowing.

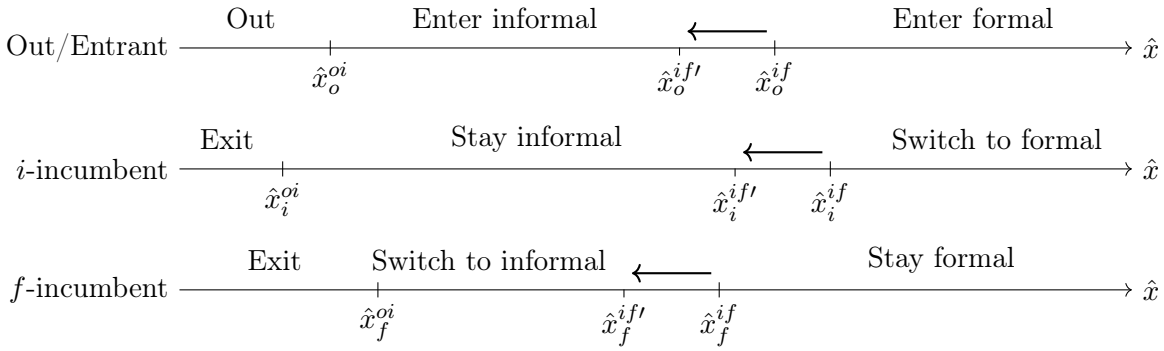
5.3 Sector decision

Define the origin $o \in \{\text{out, informal-incumbent, formal-incumbent}\}$, the destination $d \in \{\text{out, informal, formal}\}$, and the internal funds available before paying the origin-specific fee of each destination as n_o. Thus, the destination s leaves internal funds $n_{o,s} = n_o - \kappa_{os}$ inside the firm, where $\kappa_{os} \in \{c_s^e, c_{if}, c_{fi}, 0\}$ per (5). Then, the firm's formality decision compares

$$\Delta_o(\hat{x}, n_o) \equiv U_f(\hat{x}, n_o - \kappa_{of}) - U_i(\hat{x}, n_o - \kappa_{oi}). \tag{9}$$

There is a unique threshold above which a firm in origin o with internal funds n_o decides to be formal, defined by $\hat{x}_o^{if}$, such that $\Delta_o(\hat{x}_o^{if}, n_o) = 0$ (Appendix I). More generally, for each origin o and pair of destinations d, d' there exists a unique productivity threshold $\hat{x}_o^{dd'}$ that makes one of the destinations optimal.[18] The thresholds are ordered on the expected productivity line such that $\hat{x}_i^{oi} < \hat{x}_o^{oi} < \hat{x}_f^{oi} < \hat{x}_f^{if} < \hat{x}_o^{if} < \hat{x}_i^{if}$. Figure 2 summarizes these thresholds and previews the comparative statics below.

Figure 2: Sector thresholds by origin state and the effect of better formal credit terms



Notes: Each line orders the destinations of an origin state by expected profitability. Solid ticks are the thresholds at baseline. As an example, dashed ticks and arrows show where the formality thresholds move after a relative improvement in formal loan terms with respect to informal ones ($\Delta\phi_f < \Delta\phi_i \leq 0$ or $\Delta b_f > \Delta b_i \geq 0$) for firms that borrow. The movements of the thresholds between exit and informality are not drawn, for clarity, and no threshold moves for firms that do not borrow.

So far we have established two separate implications of the model. The first is an anatomy of financially constrained firms based on the firm's expected productivity and her internal funds. The second is an endogenous sector sorting based on the expected productivity of the firm. However, the relevant predictions of the model are in the interaction of these two implications. To see this, notice that for borrowers and collateral-constrained firms the sector productivity thresholds depend on the financial conditions (parameters ϕ_s and b_s), that is, $\hat{x}_o^{dd'}(\phi_s, b_s, \phi_{s'}, b_{s'})$ for all $o, d, d', s, s' \in \{\text{out, informal, formal}\}$, whereas for unconstrained firms they do not. Proposition I.10 in Appendix I.8 reports the sign of each threshold's response to the financial conditions for borrowing firms.[19]

These signs yield testable predictions that are tested against the empirical results in Section 7. A relative improvement of the loan terms for formal firms with respect to those

[18] Appendix I states the conditions under which the order, uniqueness and existence of these thresholds hold. In particular, Assumption 1 requires that the financing advantage of formality outweighs its fiscal cost net of the survival advantage.

[19] We do not draw implications or predictions for the exit margins. This is on purpose, the model does not incorporate competition among firms, which would make the modeling of the exit margin more realistic, and it was not built for that. It was built to make predictions about formalization and entry, not about exit.

faced by informal firms lowers every formality threshold, the arrows of Figure 2.[20] This means that relatively cheaper or more abundant credit in the formal sector ($\Delta\phi_f < \Delta\phi_i \leq 0$ or $\Delta b_f > \Delta b_i \geq 0$) increases formal entry, raises informal-to-formal switching, and reduces formal-to-informal switching. Anticipating the results, Section 7.1 shows that this is the case. A bank credit supply shock improves the loan terms of formal borrowers by more and, as the model predicts, formalization rises and de-formalization falls (Tables 6 and 7).

Notice that the size of the transitions after a change in loan terms depends on the density of firms near the thresholds, so the market level elasticities of Section 7 are large where many firms sit close to the formality margin.[21]

6 Empirical strategy

6.1 Shift-Share identification with exogenous shocks

We are interested in estimating the impact of credit supply shocks on firm-level transitions (formality status, entry, and exit) and other outcomes. The problem is that OLS estimates are contaminated by reverse causality and omitted-variable bias: markets with more informal-to-formal transitions may demand more credit, and an unobserved technology shock may simultaneously raise firm entry and credit volumes.

A Shift-Share design with an exogenous shifter overcomes this endogeneity. Consider the ideal experiment where additional funding is randomly allocated across banks, raising the credit supply of the selected banks. Because the selection is random, it is uncorrelated with firms' demand for credit and, hence, with firm transitions and other outcomes. In this setting, identification requires, as an exclusion restriction, that the lottery affect outcomes only through credit in equilibrium.

For these reasons, we implement a Leave-One-Out Shift-Share Instrumental Variable (LOO-SSIV) approach, following Amiti and Weinstein (2018) and Greenstone et al. (2020). First, we decompose yearly bank-firm credit growth into demand and supply components. We exclude from the sample the market m that we are interested in (the Leave-One-Out step) to eliminate the mechanical bias that would arise from estimating $\hat{\gamma}_{bt}^{-m}$ on data that include market m's own observations. Specifically, we estimate

$$\Delta \ln L_{bft} = \alpha_{ft}^{-m} + \gamma_{bt}^{-m} + \epsilon_{bft}^{-m}, \tag{10}$$

where Δ denotes the one-year log change, α_{ft}^{-m} is a firm-year fixed effect that absorbs

[20] When both sectors improve, the formality thresholds fall whenever the formal improvement is larger once weighted by the firm's borrowing in each sector, $a_f^*|\Delta\phi_f| > a_i^*|\Delta\phi_i|$ by the first row of Proposition I.10, with the analogous condition for collateral. When only formal terms improve, they fall unconditionally.

[21] Appendix I derives the transition semi-elasticities and Appendix K works out what the Shift-Share coefficient identifies through the lens of the model.

credit demand, and γ_{bt}^{-m} is a bank-year fixed effect that recovers the bank-specific supply shock. The regression is weighted by the total credit authorized from the bank to the firm over the five preceding years and the estimated bank-year effects are normalized to have zero mean within each year. Finally, we add the yearly shocks over the five-year window, $\hat{\gamma}_{bt,5}^{-m} = \sum_{k=0}^{4} \hat{\gamma}_{b,t-k}^{-m}$, to match the five year differences of the outcome variables.[22]

Notice that the estimation relies on the yearly credit variation across banks within a firm, so that only firms borrowing from two or more banks in a given year contribute to the estimation[23]. In what follows, we maintain the standard assumption that the supply shock identified on multi-bank firms also applies to single-bank firms. Relaxing this assumption yields similar results, as shown in the robustness exercise in Section 7.3, where we replace firm-year fixed effects with market-sector-year fixed effects as in Degryse et al. (2019). This change allows single-bank firms to also contribute to identification at the cost of potentially absorbing demand shifters into the estimated supply shocks. To validate that the bank effects capture supply rather than demand, Table 20 in Appendix E shows that bank shocks predict within-firm credit changes in the excluded market (Khwaja and Mian, 2008). A one log point higher bank shock raises the firm's credit from that bank by 0.13 log points relative to its credit from its other banks. At the firm level, the bank shocks affect both single-bank and multi-bank firms (0.011 and 0.092 log points per log point of the share-weighted bank shock), supporting the extension of the supply interpretation to single-bank firms.

Second, we construct bank shares at baseline in each market,

$$s_{bm(t-5)} = \frac{L_{bm(t-5)}}{\sum_{b'} L_{b'm(t-5)}}, \tag{11}$$

and define the Leave-One-Out Shift-Share instrument as

$$\hat{Z}_{mt}^{LOO} = \sum_{b \in \mathcal{B}_{mt}} s_{bm(t-5)} \, \hat{\gamma}_{bt,5}^{-m}, \tag{12}$$

where $\mathcal{B}_{mt}$ denotes the set of banks with at least one estimated shock in the window. Following the exogenous-shifter framework of Borusyak et al. (2021), identification requires i) many uncorrelated shocks with dispersed exposure and ii) that the shifters $\hat{\gamma}_{bt,5}^{-m}$ be as-good-as-randomly assigned across banks conditional on the included controls.

The as-good-as-random requirement would fail if banks receiving larger supply shocks systematically lend in markets with atypical transition dynamics. For example, if an oil boom simultaneously raises firm entry in oil-intensive markets and lending by oil-specialized

[22]The zero credit-weighted mean of the yearly shocks imply that adding over a year without a valid estimated shock is equivalent to assign to that year the average shock.

[23]From the pool of borrower firm-years, 13% hold loans from more than one bank and 87% borrow from a single bank. In terms of loan volume, 66% comes from multi-bank firms, while 34% comes from single-bank firms.

banks in those same markets. In this case, the estimated coefficient would partly capture sectoral demand rather than credit supply. We address these concerns by including controls that may correlate with both the shifter and the outcome (most directly, the market's sectoral composition of employment and of the lending banks' portfolios) and by conducting a battery of robustness checks (Section 7.3).

The many-uncorrelated-shocks condition requires exposure to be dispersed across many effectively independent bank shocks. In our context, the Mexican banking system is highly concentrated. Table 5 shows that the effective sample size, $\text{ESS} = 1/\sum_{b,t} w_{bt}^2$, in the pooled period is 16.6 and in the 2014-2019 window a single bank accounts for half of the exposure weight.[24] We argue, nevertheless, that this concentration is not a threat to our identification since the estimation does not rest on any single bank, no bank-window carries more than 17% of the pooled exposure weight. Moreover, Section 7.3 shows the point estimates are robust to dropping the largest bank altogether and to the unweighted specification, at the cost of precision.

Table 5: Descriptive statistics of the standardized bank shocks

	Exposure weighted			Unweighted					
Window	Mean	Std. dev.	IQR	Mean	Std. dev.	IQR	Nb. banks	ESS	W
2009-2014	0.20	0.59	1.12	−0.23	0.56	1.07	26	8.38	0.18
2014-2019	0.14	1.53	3.05	1.07	0.88	1.00	42	3.40	0.51
2019-2024	−0.35	0.39	0.53	−0.64	0.49	0.38	61	7.94	0.29
Pooled	0.00	1.00	1.17	0.00	1.00	1.19	67	16.56	0.17

Notes: The table describes the accumulated five-year bank supply shocks $\hat{\gamma}_{bt,5}$, estimated on the full sample rather than the leave-one-out counterparts used by the instrument, whose only difference is the exclusion of one market at a time. All statistics are computed over the bank-window cells with positive exposure in the estimation sample of the main specification (1,394 market-years, three five-year windows ending in 2014, 2019, and 2024).

Within each block (exposure-weighted and unweighted), shocks are standardized to zero mean and unit variance over those cells using the corresponding weights, so the pooled means and standard deviations equal 0 and 1 by construction. IQR is the difference between the 75th and 25th percentiles.

The weight w_{bt} is bank b's share of the baseline credit of the estimation sample markets in window t, normalized to sum to one within each window for the window rows and across all bank-window cells for the pooled row. $\text{ESS} = 1/\sum w_{bt}^2$ is the effective sample size of the shocks (Borusyak et al., 2021). W denotes the largest normalized weight.

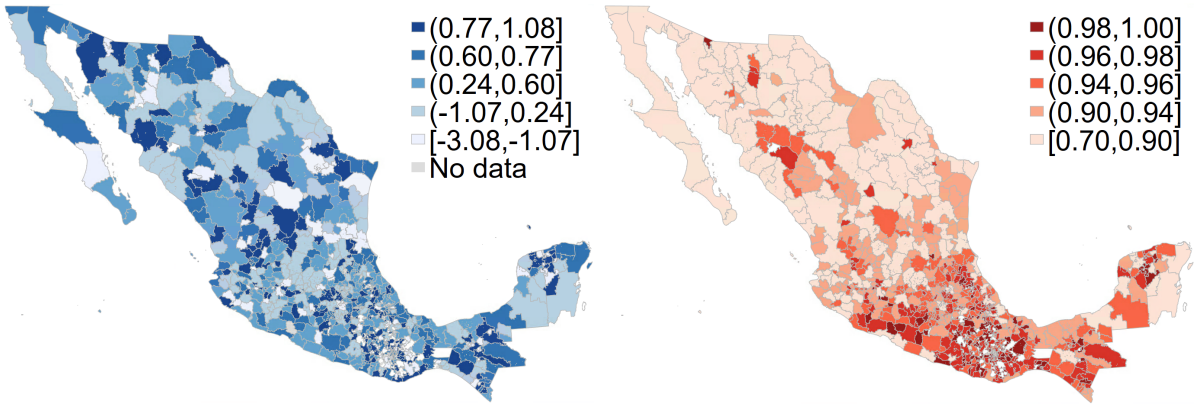
Nb. banks counts banks with an estimated shock and positive exposure in each window. The pooled row reports distinct banks (67 banks, across 129 bank-window cells).

Additionally, Section 7.3 shows the estimates are robust to dropping the largest bank.

[24] This is the descriptive statistics evidence suggested by Borusyak et al. (2021).

The left panel of Figure 3 displays the instrument on a map, where neighboring markets can be subject to very different magnitudes of the shock, which is where the identifying lies. Moreover, there is no visible geographical relation between the instrument and the baseline informality rate (right panel),[25] for which the balance evidence in Section 7.3 shows more formal evidence.

Figure 3: The identifying variation across local markets



The left panel maps the Shift-Share instrument across local labor markets. The right panel maps the baseline informality share. Gray areas have no credit data.

Once the instrument is built, the main estimating equation that relates local credit growth to outcome variables at the market level is:

$$\Delta_5 y_{mt} = \beta \, \Delta_5 \ln(C_{mt}) + X'_{m(t-5)}\Gamma + \lambda_t + \varepsilon_{mt}, \tag{13}$$

where $\Delta_5 y_{mt}$ is the five-year change in the outcome (such as the log number of firms by formality status, entry, wages, or value added); $\Delta_5 \ln(C_{mt})$ is the five-year log-change in credit in market m, instrumented by $\hat{Z}^{LOO}_{mt}$, which is standardized to unit variance; and λ_t captures the year fixed effects. The vector $X_{m(t-5)}$ contains the controls (used in every market level regression), which are: the sectoral composition of employment and of the lending banks' portfolios; the credit weighted average interest rate; the log numbers of banks and of loans; the credit share of the market's largest bank; and the sum of exposure shares, which is the share of credit granted by banks with a valid estimated supply shock, interacted with the year fixed effects. This last control accounts for the fact that exposure shares sum to less than one when some banks have no valid estimated supply shock, as prescribed by Borusyak et al. (2021) in the presence of incomplete shares. All controls vary at the market level and are fixed at their values at the start of each five year window. Regressions are weighted by the market's baseline credit $C_{m(t-5)}$.

Notice that the specification is estimated in five-year differences with updating exposure shares fixed at the start of each window, rather than in levels with market fixed effects. We

[25] Figure 10 in Appendix F plots the map of realized credit growth.

choose this specification because, when shares update across windows, unit fixed effects do not isolate shock variation over time, whereas the first difference specification does, and the updating shares keep the first stage strong (Borusyak et al., 2021).

Therefore, the corresponding first stage is

$$\Delta_5 \ln(C_{mt}) = \pi \hat{Z}_{mt}^{LOO} + X'_{m(t-5)}\Pi + \lambda_t + u_{mt}. \tag{14}$$

Some outcomes are available yearly and therefore we analyze their one-year changes, instead of five-year changes. For instance credit, interest rate, and collateral requirements. For this cases, the previous empirical strategy holds replacing every five-year index, change, and accumulation for their one-year counterpart.

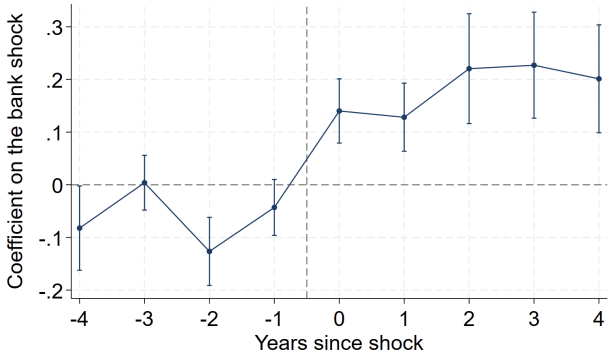
The coefficient of interest is β in Equation 13, which is the causal elasticity of the outcome with respect to credit at the market level, identified by exogenous bank level supply shocks. The interpretation is that one percent increase in credit supply implies an outcome change of $\hat{\beta}$-percent if the outcome is in logs, or of $\hat{\beta}/100$ percentage point if the outcome is in shares. Adão et al. (2019) argue that conventional standard errors over-reject in this setting because market residuals are correlated. In other words, markets with similar bank shares are exposed to the same shocks regardless of geographic distance. Following these authors suggestion, we report their exposure-robust standard errors, alongside standard errors clustered by market.

7 Main results

7.1 Annual effects on credit and loan terms

This section reports the main results of the IV estimates on credit and loan terms at an annual frequency. All the coefficients, standard errors, and specifications are Table 21 on Appendix G. Figure 4 reports the effect of the standardized IV on the log credit h years before and after the shock. The contemporaneous coefficient is 0.140 (s.e. 0.031). Therefore, a one standard deviation increase in the supply shock raises the market credit by 14 log points, about 15 percent, in the year of the shock. This means that for a market with the median baseline credit of the sample, 2.5 million dollars, the supply of credit increases roughly 375 thousand dollars. The effect is persistent over time, reaching a coefficient of 0.220 (0.053) two years after the shock and remaining at 0.201 (0.052) four years after.

Figure 4: Credit response to a bank credit supply shock

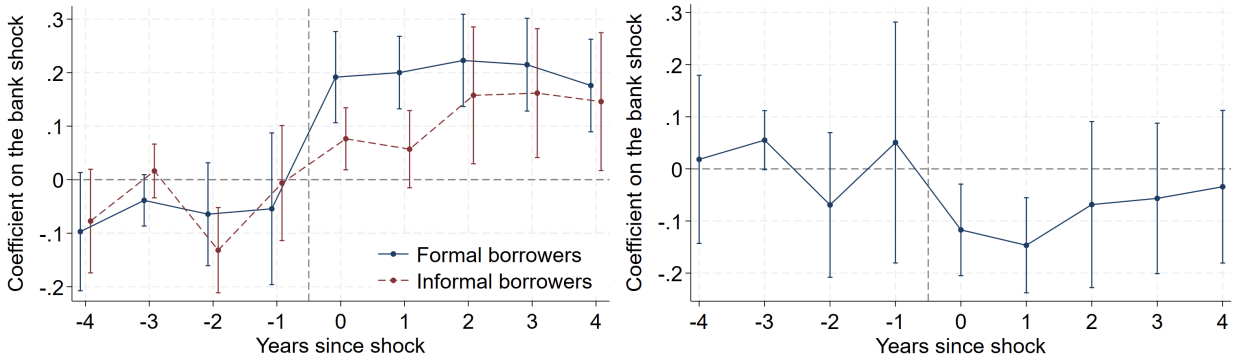


Notes: Reduced form coefficients on the standardized bank shock from separate annual market level regressions of log credit at each horizon h, with the annual analogs of the main controls, weighted by lagged market credit, on market years with instrument coverage of at least 0.5. 95 percent confidence intervals, s.e. clustered by market.

By formality, Figure 5 exposes that the credit supply shock affects the formal sector more strongly. The contemporaneous effect on the formal sector is 0.192 log points (s.e. 0.043), about 21 percent, which is two and a half times the effect on the informal sector, 0.076 (0.030). This difference, informal minus formal, is statistically significant the first two years of the shock, which falls contemporaneously to −0.117 (s.e. 0.045) and −0.147 (0.047) one year after, both significant at the 5 percent level. However, the gap narrows and is indistinguishable from zero from the second year on. The results support the heterogeneous effect by formality, changing the relative financial advantage of formality.

differentiated effect evidence supports the idea that a bank supply shock changes the relative financial benefits of formality, plausibly causing a re-optimization of the formality decision.

Figure 5: Credit response by formality of the borrower

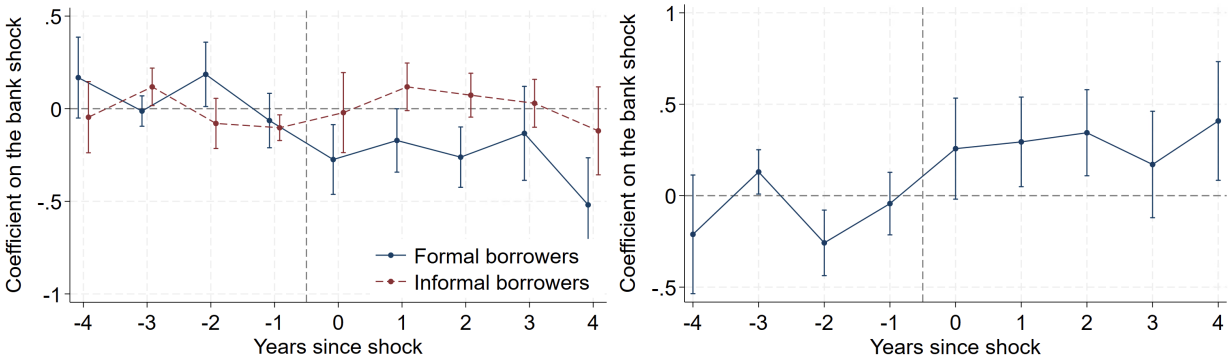


Notes: Same specification as Figure 4. Left: log credit to formal and to informal borrowers. Right: difference between the two, informal minus formal, from one regression per horizon on the difference in log credit. 95 percent confidence intervals.

It has been established that a credit supply shock has a differentiated effect on the

volume of credit to formal and informal firms. Now, we investigate if the supply shock has a differentiated effect on the loan terms, like interest rate and collateral requirements. Figure 6 displays the response of the lending rate, which falls only for formal firms. For a formal borrower, the rate falls contemporaneously by 0.27 percentage points (s.e. 0.10) and by 0.52 (0.13) four years later. In contrast, for an informal borrower the rate is unresponsive to the shock, −0.02 (0.11) the same year and −0.12 (0.12) four years later. In consequence, the difference, informal minus formal, plotted on the right panel, widens by 0.26 (0.14) percentage points the same year, 0.29 (0.13) one year after and 0.41 (0.17) four years later, significant at the 5 percent level at one, two and four years. This is evidence that the supply shock lowers the price of credit only for formal borrowers, increasing the financial cost advantage of being formal after the shock.

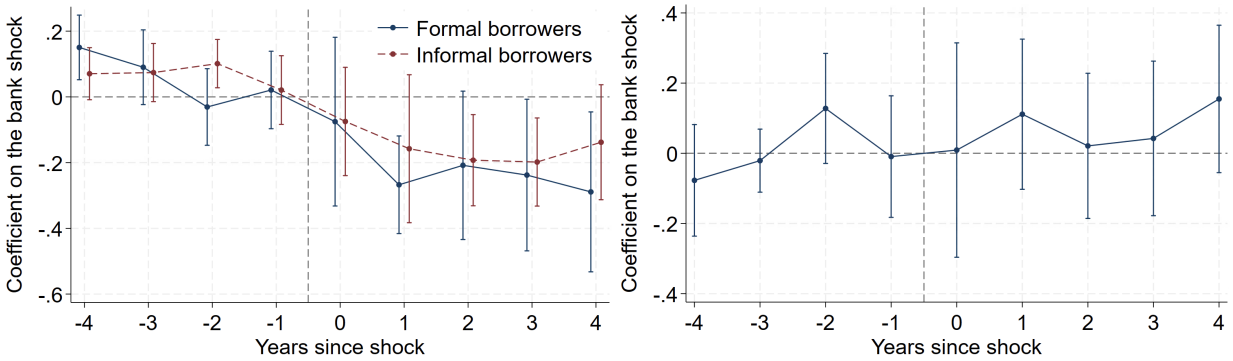
Figure 6: Interest rate response by borrower formality



Notes: Same specification as Figure 4, for the credit weighted lending rate in percentage points. Left: loans to formal and to informal borrowers in the market. Right: difference between the two, informal minus formal, from one regression per horizon. 95 percent confidence intervals.

Figure 7 repeats the exercise for collateral requirements and finds that collateral decreases equally for formal and informal firms after the shock. For a formal borrower, the collateral per peso of credit falls roughly a quarter, by 0.27 log points (s.e. 0.08) one year after the shock and by 0.29 (0.12) lower four years later. For informal borrowers the fall is marginally smaller, 0.19 (0.07) at two years and 0.14 (0.09) at four. However, this difference is not statistically different from zero, as displayed in the right panel. Hence, the shock relaxes collateral for both groups, with no detectable difference by formality.

Figure 7: Collateral response by borrower formality



Notes: Same specification as Figure 4, for log collateral posted per peso of credit, observed from 2010. Left: loans to formal and to informal borrowers in the market. Right: difference between the two, informal minus formal, from one regression per horizon. 95 percent confidence intervals.

Table 23 in Appendix G prints additional results. In particular, the fact that the credit supply is concentrated in the intensive margin (credit per loan), with no effects on the number of loans, borrowers, or loans per borrower. Additionally, there is little or no effect on the share of loans without collateral, formal employment, or wages.

Taken together, these results indicate that formal borrowers get more credit, cheaper and under less collateral requirements after a positive credit supply shock. In contrast, informal borrowers absorb less and later this credit supply shock and they do it at unchanged rates and with a smaller collateral reduction. The evidence of a differentiated effect supports the idea that a bank supply shock improves the relative financial benefit of being formal, plausibly causing a re-optimization of the formality decision, and increasing the number of informal firms formalizing, while reducing the number of formal to informal transitions. To test this conjecture, supported by the model in Section 5, the next subsection changes the time span of the analysis to the five year changes, the time-window available to follow firms over time, and explores the effect of credit supply shocks on firm transitions.

7.2 Five year effects on formality and firm outcomes

This section presents the main results of the IV estimates on formality and firm outcomes at the five-year horizon, the time window available for firm transitions. Our primary source of inference is the Adão et al. (2019) standard errors that account for the cross-market correlation induced by common shocks to bank credit supply. Additionally, we report the wild cluster bootstrap p-values in brackets (Cameron et al., 2008) as a nonparametric robustness check.

The first-stage displayed in Table 6 shows that the Shift-Share instrument, orthogonal to the local credit demand under the assumptions of the identification strategy, strongly

predicts credit in equilibrium. The coefficient on the standardized instrument is 1.11 (s.e. 0.07), which means that one standard deviation increase in the supply shock raises five-year credit growth by 0.70 log points, roughly a doubling of market credit over the window. Furthermore, the Kleibergen-Paap F exceeds 287 in every column, far above conventional weak-instrument thresholds.

Table 6: Shift-Share IV estimates of the effect of bank shocks on firm transitions

	Dependent variable: Five-year log change in the number of firm transitions			
	Informal→Formal	Formal→Informal	Formal→Formal	Informal→Informal
$\widehat{\Delta_5 \ln(C)}$	0.052***	−0.039***	0.040***	−0.006
	(0.017)	(0.014)	(0.009)	(0.004)
	[0.094]	[0.297]	[0.028]	[0.714]
FS Coeff.	1.129	1.128	1.128	1.128
	(0.067)	(0.066)	(0.066)	(0.066)
FS F	286.88	289.91	289.97	290.19
Mean	314	290	489	4,601
Obs.	957	958	970	996
Markets	522	527	535	548

Notes: Columns count surviving firms by formality status at the beginning and end of each five-year window (origin→destination), in logs. See Table 7 for specification and inference details. Wild cluster bootstrap p-values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The model in Section 5 predicts that an improvement in the financial conditions of formal firms relative to informal ones, as the one observed in Section 7.1, should increase the number of informal firms formalizing and decrease the formal to informal transitions. Table 6 puts to the test the predictions by estimating the response of the four origin-destination possibilities of surviving firms. The results confirm the model predictions: more firms formalize (0.052 percent for a one percent increase in credit supply) and fewer leave formality (0.039 percent). Additionally, more firms remain formal (0.040 percent), while there is no response on informal stayers (−0.006 percent, not significant). At the average market, a ten percent credit supply expansion translates roughly into 1.6 formalizations, one prevented switch into informality, and two additional formal stayers.

Regarding the stock of firms and the exit and entry margins, Table 7 implies that the credit supply changes the formality composition of firms, while leaving constant the total number of firms. In particular, a one percent increase in the five-year credit supply growth raises the stock of formal firms by 0.05 (s.e. 0.01) percent, while the informal stock (−0.002) and the total count (0.004) remain constant. At the mean market, where nearly nine out of ten firms are informal, a ten percent credit expansion adds about six formal firms, with

no effect for the informal firms. Aggregated on the 556 estimated markets, this translates to roughly 3,500 additional formal firms.

Table 7: Shift-Share IV estimates of the effect of bank shocks on the number of firms

	Dependent variable: Five-year log change in the number of firms								
	Existing			**Entering**			**Exiting**		
	Total	Formal	Informal	Total	Formal	Informal	Total	Formal	Informal
$\widehat{\Delta_5 \ln(C)}$	0.004	0.050***	−0.002	0.015	0.067***	0.012	0.048***	0.038***	0.055***
	(0.005)	(0.013)	(0.006)	(0.009)	(0.018)	(0.010)	(0.012)	(0.014)	(0.012)
	[0.791]	[0.018]	[0.915]	[0.548]	[0.047]	[0.639]	[0.006]	[0.235]	[0.002]
FS Coeff.	1.113	1.113	1.113	1.128	1.128	1.128	1.128	1.128	1.128
	(0.068)	(0.068)	(0.068)	(0.066)	(0.066)	(0.066)	(0.066)	(0.066)	(0.066)
FS F	265.81	265.76	265.81	290.19	289.92	290.19	290.19	290.00	290.19
Mean	10,069	1,250	8,823	4,366	458	3,917	3,170	349	2,832
Obs.	1,394	1,389	1,394	996	979	996	996	965	996
Markets	556	552	556	548	538	548	548	531	548

Notes: Two-stage least squares estimates at the market (LLM) × Economic Census five-year window level. Dependent variables are five-year log changes. Markets with a zero count in a window are excluded. The endogenous regressor is the five-year log change in market credit, instrumented by the leave-one-market-out shift-share credit-supply shock, standardized to unit variance. The Mean row reports the dependent variable's level in the estimation sample. All columns include year fixed effects and the baseline (start-of-window) controls: market sector shares and credit-weighted sector shares of the lending banks' portfolios; the credit-weighted average interest rate; log numbers of banks and of loans; the credit share of the market's largest bank; and the coverage share (baseline credit granted by banks with a valid estimated supply shock), entered as a main effect and interacted with the year fixed effects. The sample is restricted to market-years with coverage of at least 0.5 (Section 7.3 varies this threshold). AKM standard errors following Adão et al. (2019) in parentheses. Stars use the corresponding p-values. Wild cluster bootstrap p-values (Cameron et al., 2008), clustered at the market level, in brackets. The first-stage coefficient is reported with its market-clustered standard error; FS F is the Kleibergen-Paap rk Wald statistic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The mechanism operates mainly through the strong formal entry response (0.067). Additionally, the informal entry does not move (0.012, n.s.) and the exit rises everywhere, with a stronger effect in the informal sector (0.048 total, 0.038 formal, 0.055 informal). At the average market, a ten percent credit expansion is associated with about 15 (roughly 8,300 across the 548 markets in the sample) additional exits. We interpret the exit margin as a competition component in which the formal entrants contribute to the displacement of incumbent firms, predominantly informal, in a Schumpeterian way (Schumpeter, 1942). The model presented in Section 5 abstracts from this competition mechanism. Nevertheless, an extension of the model with competition as in Melitz (2003) is an interesting avenue

for future research.

To grasp the magnitude of the formal-entry response, let us put it in perspective. A supply-driven ten percent credit expansion incorporates about 3.1 firms to the existing 458 formal entrants in the average market. This raises the five-year formal startup rate ($458/1{,}250 = 36.6\%$) by roughly 0.25 percentage points, equivalent to about a third of the five-year 0.78 points U.S. startup-rate decline documented by Pugsley and Şahin (2019) (roughly 5 percentage points over 32 years). In other words, a one standard deviation increase in the credit supply shock, which roughly doubles market credit, raises the formal startup rate by 1.7 percentage points, more than two U.S. quinquennia of decline reversed.

Table 8 shows that the number of firms with bank credit rises for formal and informal firms similarly (0.042 and 0.034 per one percent of credit, statistically indistinguishable). The evidence on annual loan terms of Section 7.1 together with the one on Table 8 entail that a credit supply reaches a similar share of firms in both sectors. However, formal borrowers get more credit, cheaper and under less collateral requirements, while informal borrowers absorb less and later this credit supply shock and they do it at unchanged rates and with a smaller collateral reduction.

Table 8: Shift-Share IV estimates of the effect of shocks on the number of firms with credit

Dep. var.: Five-year log change, No. firms w/credit			
	Total	Formal	Informal
$\widehat{\Delta_5 \ln(C)}$	0.035***	0.042**	0.034**
	(0.011)	(0.018)	(0.014)
	[0.486]	[0.262]	[0.748]
FS Coefficient	1.113	1.113	1.113
	(0.068)	(0.068)	(0.068)
FS F	265.81	265.81	265.81
Mean	776	185	591
Obs.	1,394	1,394	1,394
Markets	556	556	556

Notes: See Table 7 for specification and inference details.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Finally, we study the effects of the credit supply shock on the labor market and on value added. Table 9 shows that credit expansions raise formal employment and the formal wage bill, while the average wage falls slightly and value added falls in the informal sector. The workers registered in the social security records, formal by definition, increase by 0.20 percent for a ten percent credit expansion, about 62 workers at the average market. The formal wage bill rises accordingly, by 0.24 percent, and the mean daily wage falls marginally,

by 0.12 percent, about 24 centavos on a mean of 204 MXN per day, probably due to marginal hires earning slightly below-average wages. Average and total value added rise by 0.29 and 0.32 percent, significant at the ten percent level. By formality, informal-sector value added falls by 0.49 percent (significant at the five percent level) while formal-sector value added does not move (0.28 percent, not significant).

Table 9: Shift-Share IV estimates of the effect of bank shocks on workers, wagebill, and value added

	Dependent variable: five-year log change in						
	Workers	Wage bill	Mean wage	Avg. VA	Total VA	VA formal	VA informal
$\widehat{\Delta_5 \ln(C)}$	0.020***	0.024**	−0.012***	0.029*	0.032*	0.028	−0.049**
	(0.005)	(0.009)	(0.004)	(0.015)	(0.017)	(0.031)	(0.021)
	[0.158]	[0.322]	[0.144]	[0.361]	[0.354]	[0.671]	[0.612]
FS Coeff.	1.113	1.113	1.113	1.113	1.113	1.113	1.113
	(0.068)	(0.068)	(0.068)	(0.068)	(0.068)	(0.068)	(0.068)
FS F	265.77	265.77	265.77	265.81	265.81	265.81	265.81
Mean†	30,967	12.7m	204	1.06m	22.5bn	16.8bn	5.7bn
Obs.	1,386	1,386	1,386	1,394	1,394	1,394	1,394
Markets	552	552	552	556	556	556	556

Notes: See Table 7 for specification and inference details. Wild cluster bootstrap p-values in brackets. The three labor-market columns are from IMSS records and therefore cover formal (registered) employment: "workers" is the market total of IMSS-registered workers, "wage bill" the market total of daily contribution-base wages, and "mean wage" the market average daily wage. The four value-added columns are from the Economic Census and cover all firms: "avg. VA" is average per-firm value added, "total VA" aggregate value added, and the last two columns split total value added by the census formality classifier, firms with positive income and workers but zero social security contributions. †Means in levels, in MXN (m = million, bn = billion): the mean daily wage is 204 MXN (≈ 10.7 USD); the mean daily wage bill 12.7 million MXN (≈ 0.67 million USD); average per-firm value added 1.06 million MXN (≈ 56,000 USD) and total market value added 22.5 billion MXN (≈ 1.2 billion USD), from census values recorded in thousands of current pesos. All conversions at 19 MXN/USD. Census worker counts, available only in the 2019 and 2024 waves, are reported in Appendix C, Table 15; the instrument is weak on that single-window subsample. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Three caveats on these results. First, none of the seven coefficients survives the wild cluster bootstrap (p-values between 0.14 and 0.67), so we read these responses as a complement to the firm-margin results rather than as stand-alone findings. Second, the value-added responses are sensitive to market trends: with the outcome's own lagged five-year change as a control, the aggregate and formal coefficients reverse sign while the informal decline strengthens (Appendix Table 25). The result we rely on is therefore the reallocation

of production away from the informal sector. Third, the design cannot speak to informal employment: census worker counts exist only for 2019 and 2024, a single window on which the instrument is weak (first-stage F of 1.9). Appendix Table 15 reports those estimates for completeness.

Two central lessons are drawn from the results presented in this section. First, when banks supply additional credit, they serve formal and informal firms but on better terms for the formal sector. Second, the formality decision of some formal and informal firms can be affected by relative changes in loan terms. Combined, these two lessons imply that the differentiated effect of the bank shocks translates, as predicted by the model, into changes in formality decisions. Firms close to the formality margin and in need of credit realize that the loan conditions improved for formal firms after the shock, inducing informal firms to formalize and discouraging formal firms to become informal.

7.3 Robustness checks

This section examines the robustness of the five-year results on three fronts: the identifying assumptions, the construction of the instrument, and the estimation sample. The main conclusion on formality survives these tests. A full account is given in Appendix H.

On the identifying assumption, we provide evidence that the instrument is as good as randomly assigned conditional on the controls. Following Borusyak et al. (2021) and Goldsmith-Pinkham et al. (2020), in Appendix Table 24 we regress the controls five and ten years before the baseline on the instrument. None of the coefficients is distinguishable from zero at the ten percent level. Moreover, Appendix Table 25 adds the dependent variable's own lagged five-year change as a control, where only aggregate and formal value-added responses reverse sign. Finally, Appendix Table 28 replaces each outcome with its five-year change before the shock window. The stock of firms shows a small positive pre-period correlation, common to both sectors, whereas the estimated effects load entirely on formal firms. Wages and value added show negative placebo coefficients, which we read as mean reversion in lending. We treat those responses with caution.

On the instrument construction, three variants of the instrument leave the results unchanged: the conventional version without the leave-one-out correction (Appendix Table 29), renormalizing the instrument by coverage (Appendix Table 32), and moving the coverage threshold from zero to 0.8 (Appendix Table 31). Additionally, Appendix Table 33 drops the largest lender from the shocks and exposure shares. The point estimates are unchanged, at the cost of a lower first stage (F of 35), confirming that the results do not rely on a single bank.

On the estimation sample, excluding the three largest metropolitan areas (Mexico City, Monterrey and Guadalajara), leaves the formality composition results intact (Appendix Table 27). Finally, Appendix Table 30 makes the analysis at the shock (bank) level, instead

of at the share (market) level, where the credit supply shocks are re-identified with market-sector-year fixed effects, instead of firm-year fixed effects, as in Degryse et al. (2019). This specification allows single-bank firms to also contribute to the estimation. Although nothing mechanical links this design to the market-level one, it reproduces the composition of the response.

As a whole, the main results are sound and are not the product of any single methodological choice. It survives the balance, pretrend and placebo tests; every reconstruction of the instrument; the removal of the largest lender and of the largest markets; and a shock-level design with re-identified shocks. Weights and population only matter for precision and the labor and value-added responses are the margins that depend on the specification, which is why the paper does not lean on them.

8 Conclusion

This paper matches the universe of bank loans, the administrative social security records, and the economic censuses to build a unique dataset that contains formality, bank loans, and firm transitions on the same markets over twenty years. As a result, we can study an unexplored relationship: the effect of bank loans on informal firms. With these data, we document three stylized facts. First, informal firms are inside the banking system, in contrast to what is commonly assumed. Roughly 70 percent of firms with credit were informal in 2024. Second, informal firms face worse loan terms than their formal counterparts. At the median, informal firms pay a 0.7 percentage points more interest rate, need 3 percentage points more collateral, and obtain 33.2 thousand dollars smaller loans than a formal firm. Third, firms move regularly between formality and informality, in both directions. In 2019, 3.5 percent of informal firms formalized and 22 percent of formal firms became informal five years later. These facts motivate the main argument: some firms, those close to indifference between being formal or informal and in need of a loan, reverse their formality status after a relative change in the terms or availability of loans.

We build a dynamic model in which firms choose whether to be formal. Their formality decision trades off the tax cost against the financial benefits of being formal. The model predicts that an improvement in the terms or availability of loans for formal firms, relative to informal ones, increases formalization and decrease formal-to-informal transitions, for firms that need additional loans. The empirical analysis confirms these predictions using Leave-One-Out Shift-Share bank credit supply shocks. Improving the formal advantage in the terms and availability of loans increases formalization and reduces formal-to-informal transitions. A credit supply expansion yields 25 percent less collateral, 19 percent more credit, and half a percentage point cheaper loans for formal firms four years after the shock. Whereas for informal firms the volume response is smaller (16 percent), rates do

not change (−0.1 percentage points, not significant), and collateral falls by less (13 percent). The credit supply shock amplifies the formal financial advantage and, in consequence, as predicted by the model, causally raises formalization by 0.5 percent and lowers formal-to-informal transitions by 0.4 percent, after a ten percent increase in credit. That is about 850 additional transitions from informal to formal and 600 fewer in the opposite direction across the markets in our sample.

The results imply that policies that affect the relative loan advantage of formal firms, from credit guarantees to bank competition, can shift the share of formal firms. The main limitation of this work is the absence of general equilibrium effects and competition in the loan, labor, and goods markets. More research is needed to address these limitations and to better understand the effects of the financial sector on informality.

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A Data sources, definitions and sample construction

The credit registry and the IMSS employer records are processed inside Banco de México's EconLab, and the economic census inside INEGI's microdata laboratory. The data at the individual level can not be exported, so we are able to merge the two sources only at the market level. Table 10 lists the definitions.

Table 10: Variable definitions by source

Object	Source and definition
Formality, registry base	A borrower is formal in a given year when its RFC appears in the IMSS employer records of that year in the same market and sector; otherwise it is informal. Firms in IMSS without a registry record are formal firms without bank credit.
Formality, census	EC, positive social security contributions (`J300A` > 0); zero income firms with zero contributions are coded formal.
Employment	EC, total persons engaged including owners and unpaid workers (`H001A`). IMSS, workers registered in January; in the matched base, the larger of the IMSS count and the registry's own count.
Wages	IMSS, January wages, bottom and top coded under IMSS rules (minimum wage and 25 times the reference wage).
Value added, wage bill and capital	EC, value added (`A131A`), remunerations (`J000A`) and end of year fixed assets at book value (`Q000A`).
Credit incidence, census	EC, the firm reports bank credit (`O511A`).
Credit, collateral and rate	R04C, January snapshot of every loan: authorized line, posted collateral and annual rate in percent, aggregated to borrower, bank, market, sector and year. The collateral field starts in 2010 and is empty in 2025.
Loan rates by status	R04C, unweighted over borrower years, by the IMSS registration of the borrower.
Firm dynamics	EC waves five years apart; exit is absence at the next wave, switching is measured among survivors, and entrants are firms aged two years or less.
Markets and sectors	Localities are mapped to municipalities and then to the public local labor market classification (777 markets). Sectors follow SCIAN; agriculture, mining, utilities, construction, transport, financial services and government are excluded.
Units	Pesos as reported in each source; dollar figures in the notes use 19 pesos per dollar.

The credit and social security registries enters as the January file of each year from 2005 to 2025. Loans with missing locality or activity code take the codes from other loans of the same borrower in the same file. Records that still lack a locality, an activity, an authorized line or a total exposure are dropped, as are records with a country code other than Mexico. Activities are mapped to SCIAN, and agriculture, mining, utilities, construction, transport, financial services and government are excluded. Localities are mapped to municipalities and municipalities to markets. The loan records are collapsed to one cell per borrower, bank, market, sector and year, with amounts summed, rates averaged and loans counted. In the social security records we drop workers with a missing or zero wage. A borrower with no IMSS employer record in a year is informal that year; IMSS firms without a registry record are formal firms without bank loans.

Table 11 follows the registry from the raw January files to the estimation sample. The upper block counts institutions, borrowers and borrower years over 2005–2025 after cleaning; the lower block counts the market windows that enter the five year regressions.

Table 11: From the raw credit registry to the estimation sample

Panel A. Credit registry					
Banks †					123
Banks after cleaning‡					119
Borrowing firms, registry†					1,141,464
Borrower firms, matched base§					1,269,292
Borrower-year, matched base, 2005-2025					5,222,771
Local markets					777
Local markets with registry credit					705
Panel B. Local markets by census year					
	2009	2014	2019	2024	All
Markets with an instrument	501	606	643	606	2,356
With coverage ≥ 0.5	295	408	498	533	1,734
Estimation sample¶	–	398	477	519	1,394

† January files 2005-2025, after dropping records with no locality, activity code, authorized credit line, or total exposure. ‡ We excluded agriculture, mining, utilities, construction, transport, financial services, and government sectors, as well as cases where activity or locality was not available or firms outside of the country. § Firms are anonymized and matched by the EconLab using IMSS number, tax identifier (RFC), and credit registry identifier. A firm present in several markets or sectors counts once per cell. ¶ Markets with coverage ≥ 0.5 at 2014, 2019 or 2024 whose five year change in establishments, change in credit, and controls are all defined; a five year change needs the market in the panel five years earlier, so no window ends at 2009. Coverage is the share of the market's baseline credit held by banks with a shock. The 2,356 market years belong to 683 markets and the estimation sample to 556. ¶ Markets with coverage ≥ 0.5 in which the five year changes in outcomes, credit controls are all observed. 2009 can not compute a five years change and therefore is empty. Coverage is the share of a market's baseline credit held by banks with a shock. The 2,356 market years belong to 683 markets, the estimation sample to 556.

B Firm transitions under alternative definitions of informality

Table 12: Establishment transition across censuses with the intensive margin

		Destination			
		Formal	Informal	Mix	Exit
2014 Origin	Formal	94,053	20,182	10,990	48,971
		(54.0%)	(11.6%)	(6.3%)	(28.1%)
	Informal	111,773	2,232,216	151,792	1,531,660
		(2.8%)	(55.4%)	(3.8%)	(38.0%)
	Mix	10,522	3,344	7,219	8,023
		(36.1%)	(11.5%)	(24.8%)	(27.6%)
2019 Origin	Formal	197,196	54,785	20,666	108,369
		(51.8%)	(14.4%)	(5.4%)	(28.4%)
	Informal	84,059	2,547,652	62,012	1,473,395
		(2.0%)	(61.1%)	(1.5%)	(35.4%)
	Mix	48,735	83,256	57,455	62,577
		(19.3%)	(33.0%)	(22.8%)	(24.8%)

Note: Percentages are calculated within each origin category. Formal: 100% of workers have social security (SS); Informal: 0%; Mix: strictly between 0% and 100%.

Table 13: Establishment transition across censuses with social security number definition

		Destination		
		Formal	Informal	Exit
2019 Origin	Formal	84,146	73,597	93,991
		(33.4%)	(29.2%)	(37.3%)
	Informal	142,684	2,261,091	1,331,140
		(3.8%)	(60.5%)	(35.6%)

Note: Percentages are calculated within each origin category. Informality is defined as an establishment without social security (IMSS) number. The table excludes 813,508 observations (17%) with missing responses to the question about the social security (IMSS) registration number.

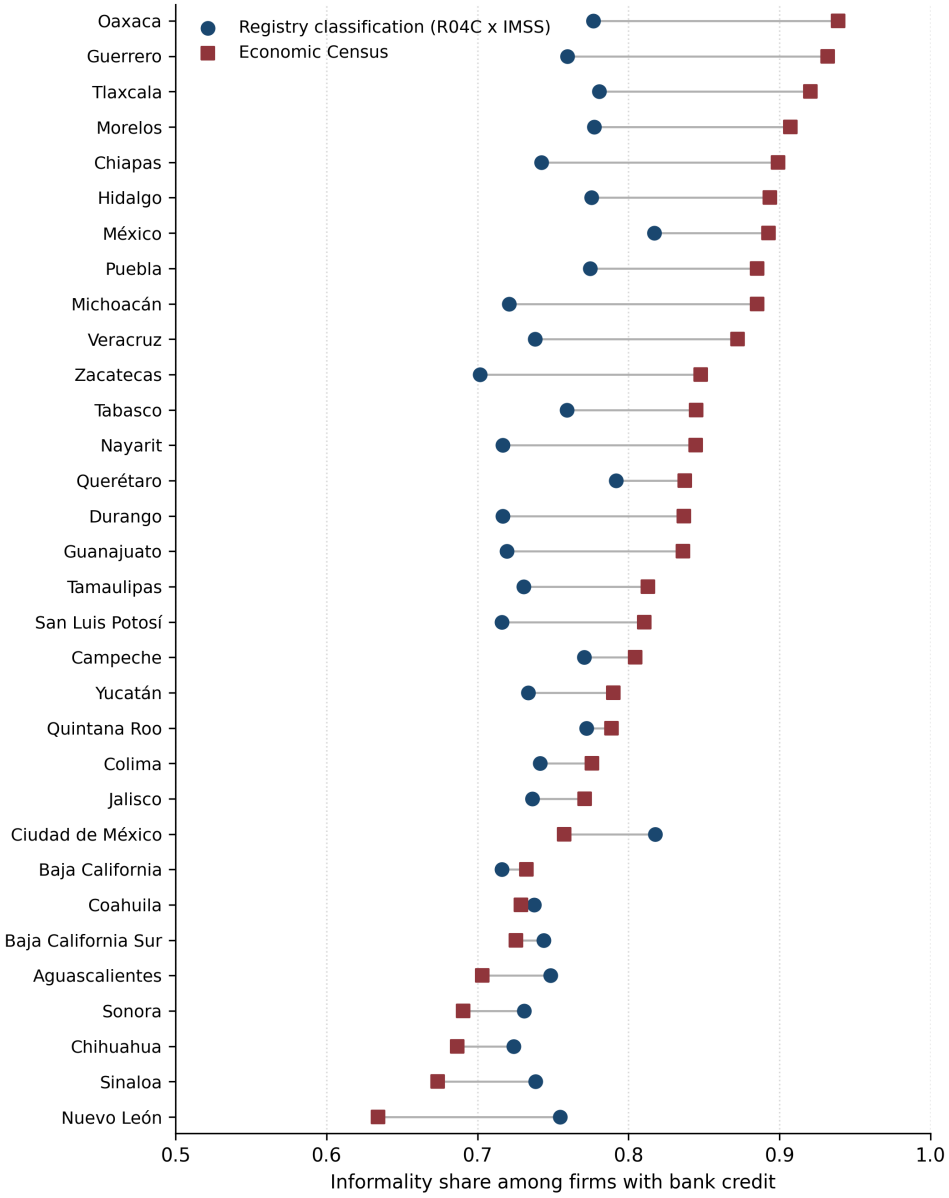
Table 14: Establishment transition across censuses with tax administrative number definition

		Destination		
		Formal	Informal	Exit
2019 Origin	Formal	749,911	284,018	475,078
		(49.7%)	(18.8%)	(31.5%)
	Informal	424,885	1,439,976	1,162,103
		(14.0%)	(47.6%)	(38.4%)

Note: Percentages are calculated within each origin category. Informality is defined as an establishment without tax (RFC) number. The table excludes 264,186 observations (5%) with missing responses to the question about the tax (RFC) number.

The census series in Figure 8 uses the census IMSS registration question, the closest counterpart to the administrative match; the establishment counts and transition tables elsewhere in the paper instead classify an establishment as informal when it reports zero social security payments while employing workers and earning income, and the two measures should not be read as the same object.

Figure 8: Registry classification of borrower informality against the census



Share of firms with bank credit classified as informal, by state, sorted by the census value. Registry: firms borrowing in the 2018 R04C whose RFC has no IMSS employer registration (474,237 firms). Census: establishments not registered with IMSS (question h424a) among those reporting bank credit, 2019 census. The classification, used only for loan pricing, gets the level of borrower informality approximately right (76 percent overall against a census state mean of 81) but compresses the cross-state gradient: it overstates informality where the census puts it lowest (Nuevo León, Sinaloa, Chihuahua, Sonora) and understates it where the census puts it highest (Oaxaca, Guerrero, Tlaxcala, Morelos).

C Census workers count

Table 15: Census worker counts

	Total	Formal	Informal
$\widehat{\Delta_5 \ln(C)}$	0.037	−0.071	0.074
	(0.082)	(0.193)	(0.204)
FS Coeff.	−2.184	−2.261	−2.184
	(1.589)	(1.454)	(1.589)
FS F	1.89	1.89	1.89
Obs.	519	496	519
Markets	519	496	519

Notes: Census worker counts exist only in the 2019 and 2024 waves, leaving a single five-year window (a cross-section of markets, so clusters equal observations). The first stage on this subsample is weak (F = 1.9), so these estimates are reported for completeness only and are not informative. AKM standard errors in parentheses.

D Additional descriptive statistics

Table 16: Estimation sample, census outcomes: establishments by formality and credit

	Mean	SD	p10	Median	p90	N
Panel A. Pooled levels at the start of the window, per market						
All firms	8,874.6	29,241.7	415	2,319	20,316	1,394
Formal	1,094.3	3,872.7	13	138.5	2,695	1,394
Informal	7,780.3	25,628.2	393	2,159.5	17,436	1,394
Firms with bank credit	572.8	1,793.1	3	119	1,331	1,394
Formal	136.6	547.7	0	12	280	1,394
Informal	436.2	1,277.0	1	95	1,035	1,394
Entrants	4,139.1	12,961.3	164	984	9,540	996
Formal	449.9	1,672.9	5	50	973	996
Informal	3,689.1	11,400.7	156	923.5	8,454	996
Exits	3,092.8	10,563.8	104	693	6,861	996
Formal	337.7	1,364.8	3	37	774	996
Informal	2,755.1	9,258.3	99	654.5	5,934	996
Switchers, informal to formal	253.4	879.8	3	38.5	589	996
Switchers, formal to informal	229.0	814.4	3	37	537	996
Stayers, formal	390.0	1,427.8	2	42	938	996
Stayers, informal	3,820.8	13,773.9	190	1,097	8,542	996
Panel B. Five year changes, log points						
All firms	0.144	0.134	0.029	0.146	0.270	1,394
Formal	0.127	0.335	−0.229	0.125	0.490	1,389
Informal	0.144	0.138	0.025	0.147	0.275	1,394
Firms with bank credit	0.907	1.718	−0.386	0.087	3.846	1,394
Formal	0.541	1.029	−0.439	0.211	2.095	1,394
Informal	1.089	2.080	−0.424	0.089	4.874	1,394
Entrants	0.076	0.255	−0.187	0.076	0.356	996
Formal	−0.029	0.520	−0.606	−0.016	0.511	979
Informal	0.082	0.259	−0.181	0.081	0.363	996
Exits	0.036	0.256	−0.211	0.044	0.300	996
Formal	−0.051	0.543	−0.693	−0.045	0.655	965
Informal	0.040	0.264	−0.211	0.046	0.306	996
Switchers, informal to formal	0.050	0.589	−0.693	0.061	0.690	957
Switchers, formal to informal	0.228	0.584	−0.434	0.222	0.916	958
Stayers, formal	0.250	0.407	−0.127	0.221	0.693	970
Stayers, informal	0.203	0.158	0.075	0.196	0.346	996

Notes: The establishments from the Economic Censuses are aggregated to local labor markets. All statistics pool the 1,394 market window observations (556 markets, windows 2009-2014, 2014-2019, 2019-2024), unweighted. N smaller where undefined. Formal: positive number of workers and income with social security contributions. Entrants, exits, switchers and stayers are measured between consecutive censuses as in Table 2.

Table 17: Estimation sample, registry and social security records: outcomes and instrument

	Mean	SD	p10	Median	p90	N
Panel A. Pooled levels at the start of the window, per market						
Borrowers	556.8	3,714.0	4	50.5	919	1,394
Formal	107.9	623.2	0	7	177	1,394
Informal	448.9	3,151.9	3	39	747	1,394
Number of loans	1,281.0	9,216.0	4	70	1,517	1,394
Total amount of authorized credit†	431.5	5,578.9	0.1	2.5	195.6	1,394
Average loan size‡	86.1	340.6	10.8	38.3	169.4	1,394
Loans per borrower	2.27	26.64	1.00	1.38	2.22	1,394
Collateral ratio	2.21	55.35	0.00	0.04	0.31	1,394
Interest rate (percent)	14.86	4.56	8.88	14.91	20.30	1,394
Banks in the market	10.1	9.0	2	7	21	1,394
IMSS workers	26,655	142,866	37	1,316	47,088	1,394
IMSS wage bill, daily‡	415.2	2,706.8	0.3	12.8	546.2	1,394
Panel B. Five year changes, log points						
Borrowers	0.298	0.888	−0.676	0.295	1.424	1,394
Formal	0.558	0.830	−0.340	0.693	1.542	1,130
Informal	0.211	0.928	−0.807	0.182	1.409	1,379
Number of loans	0.468	1.103	−0.646	0.224	2.015	1,394
Total amount of authorized credit	1.032	1.478	−0.509	0.901	2.774	1,394
Average loan size	0.565	1.045	−0.459	0.520	1.663	1,394
Loans per borrower	0.169	0.536	−0.337	0.111	0.761	1,394
Collateral ratio	0.702	2.455	−1.600	0.597	3.128	820
Interest rate§	0.150	5.072	−5.980	0.476	5.971	1,394
IMSS workers	0.216	0.448	−0.095	0.149	0.555	1,386
IMSS wage bill	0.582	0.542	0.180	0.490	1.016	1,386
Panel C. Instrument						
Shift share instrument (standardized)	0.248	1.197	−0.893	−0.261	2.063	1,394
Coverage	0.921	0.110	0.762	0.967	1.000	1,394

Notes: Same sample and layout as Table 16. Credit registry, as delivered: borrowers are firms with a positive authorized line at a regulated bank in January, formal if IMSS employers that year. Loans are outstanding credit. Collateral ratio and interest rate are credit weighted market averages. Banks are the ones lending in that market. IMSS: workers and daily wage bill of IMSS employers in the market. † million, ‡ thousand dollars at 19 pesos per dollar. § percentage points. Instrument standardized over all market windows with coverage. Coverage is the share of baseline credit held by banks with an estimated shock.

Table 18: Credit registry by year (January snapshot) and borrower status

	Formal borrowers					Informal borrowers				
Year	N	Credit†	Rate	No coll.	2+ banks	N	Credit†	Rate	No coll.	2+ banks
2005	2,516	31.6	16.00	–	5.0	21,428	21.1	16.62	–	2.0
2006	4,000	26.3	17.00	–	6.3	33,379	15.8	18.18	–	2.5
2007	5,717	21.1	16.30	–	6.8	50,164	13.7	18.00	–	3.0
2008	7,975	21.1	16.91	–	8.1	73,658	13.7	19.31	–	3.4
2009	9,568	18.8	18.10	–	8.7	78,573	13.2	19.50	–	3.8
2010	25,513	26.3	13.93	74.6	28.9	226,512	10.5	14.95	89.7	12.8
2011	28,888	26.3	14.07	75.4	29.3	228,728	10.9	15.38	88.9	13.0
2012	32,797	26.3	14.46	74.7	29.7	239,564	11.1	15.80	87.9	13.6
2013	38,581	26.3	14.79	77.4	28.3	259,762	11.6	16.24	88.9	14.0
2014	42,355	29.5	13.76	77.6	28.0	252,567	11.6	15.17	87.9	14.3
2015	46,402	32.6	13.24	80.6	27.8	252,332	12.8	14.59	88.9	14.3
2016	53,007	34.6	13.06	82.3	26.9	249,152	13.2	14.54	88.5	14.3
2017	80,833	48.5	16.00	79.6	21.3	279,487	15.2	17.55	83.7	7.5
2018	94,356	43.7	16.58	85.2	23.8	310,486	15.3	17.63	89.9	8.1
2019	99,452	46.7	17.21	81.8	23.7	299,635	15.7	18.34	88.2	8.2
2020	96,694	52.4	17.00	78.0	22.3	282,031	14.8	18.14	86.7	7.2
2021	90,647	52.6	14.96	76.8	21.2	246,308	15.8	15.85	86.0	6.8
2022	86,710	54.3	14.71	76.3	22.7	256,537	14.5	15.29	89.2	7.0
2023	77,402	67.1	15.69	92.0	23.3	154,974	22.1	15.69	92.6	9.9
2024	83,334	74.7	16.45	79.4	24.1	171,635	22.7	16.65	85.6	10.3
2025	83,187	84.2	16.20	–	25.8	165,867	26.6	15.95	–	12.4

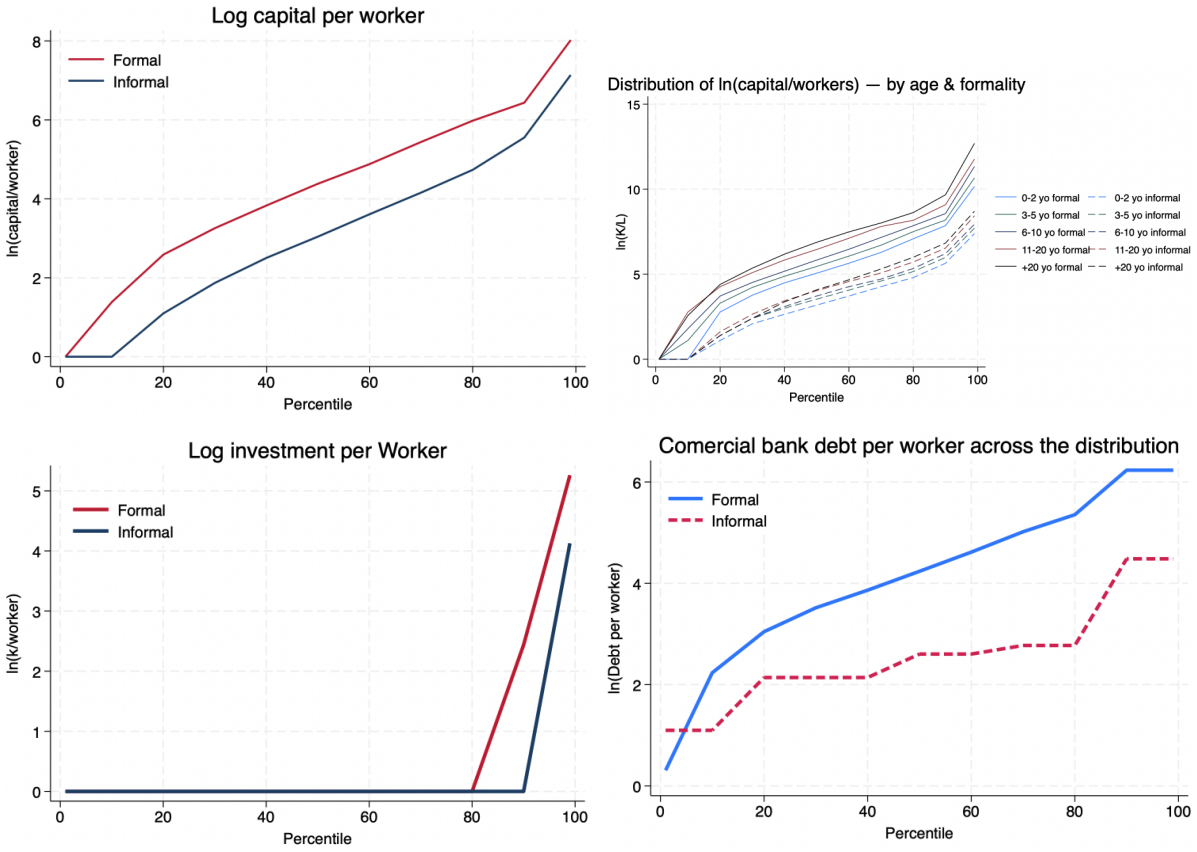
Notes: Credit registry matched to IMSS employer records. A borrower is formal in a given year if it appears as an IMSS employer that year. Borrowers: firms with a positive authorized line. Credit: median borrower authorized credit, †in nominal thousand dollars, at 19 pesos per dollar. Rate: median annualized gross rate at the bank relationship level, percent, winsorized at the 1st and 99th percentiles within year. No coll.: share of relationships without collateral, percent, reported for the years in which the registry carries the collateral field. 2+ banks: share of borrowers with two or more banks, percent.

Table 19: Estimation sample and all market windows with an instrument

	All market windows			Estimation sample, by window					
				2009-2014		2014-2019		2019-2024	
	Mean	Median	N	Mean	Median	Mean	Median	Mean	Median
Panel A. Pooled levels at the start of the window, per market									
All firms	7,244.4	1,699	1,748	8,980.7	2,675	8,564.4	2,113	9,078.4	2,271
Formal	882.9	88	1,748	1,096.9	159	1,072.9	139	1,112.1	120
Informal	6,361.6	1,605.5	1,748	7,883.8	2,453.5	7,491.4	1,989	7,966.3	2,128
With a loan (census)	465.5	73	1,748	98.4	10.5	787.6	208	739.1	199
Informal→Formal	212.4	25	1,204	-	-	236.1	44	269.2	34
Formal→Informal	191.9	23	1,204	-	-	207.7	30	248.5	42
Borrowers (registry)	470.5	33	1,662	218.7	28	610.9	58	766.4	66
Total credit†	345.3	1.3	1,748	28.9	0.7	663.2	2.2	526.3	6.0
Collateral ratio	1.87	0.03	1,662	-	-	0.11	0.09	5.84	0.10
Interest rate (%)	15.01	14.91	1,662	15.89	15.87	12.36	12.57	16.36	16.25
IMSS workers	22,360	837	1,674	25,687	1,642	25,289	1,195	28,654	1,168
Panel B. Instrument									
Shift share IV (std.)	0.000	−0.449	2,356	−0.225	−0.211	1.788	1.927	−0.804	−0.809
Coverage	0.702	0.938	2,356	0.937	0.991	0.925	0.975	0.905	0.945

Notes: First three columns include every market window for which the instrument is defined, unweighted, N per row (Panel B includes three windows, panel A includes two windows since there is no 2004 information). Last six columns are the estimation sample of the Table‘3 by window, with 398, 477 and 519 markets per window, respectively. Switchers (F→I and I→F) are undefined for the 2009 baseline. Variables and sources as in that table. † Million dollars at 19 pesos per dollar.

Figure 9: Capital, investment, and debt by formality



Census capital per worker by size and by age, census investment per worker, and survey bank debt per worker, formal against informal firms.

E Within-firm validation of the bank credit supply shocks

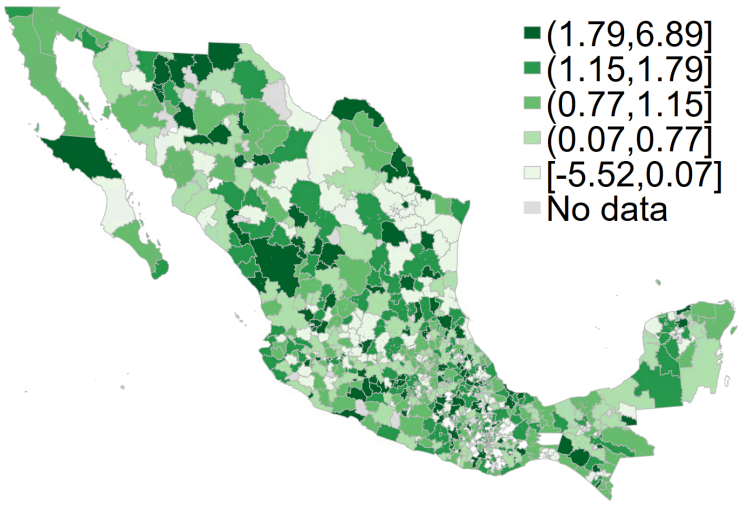
Table 20: Firm-level responses to bank credit supply shocks

	Coefficient (s.e.)	Mean	N
Within-firm credit growth on bank shock†	0.131*** (0.021)	0.038	1,020,447
weighted by lagged credit	0.454*** (0.052)	−0.425	1,020,447
Firm credit growth on its banks' shocks	0.016*** (0.004)	0.035	3,704,022
two or more banks	0.092*** (0.013)	0.067	430,148
single bank	0.011*** (0.004)	0.030	3,272,491

Khwaja and Mian (2008) estimates on the annual firm-bank panel. Rows 1 and 2: log credit growth of a firm-bank pair on the bank's credit supply shock (log points), with firm-by-year fixed effects, among firms with two or more banks. The second row weights pairs by lagged credit. Rows 3 to 5: a firm's log credit growth on the standardized lagged-share-weighted average of its banks' shocks, with market-by-year fixed effects, among firm-years whose banks with a shock hold at least half of its lagged credit. Standard errors clustered by market. †Clustered by bank, the standard error is 0.045. Means of the dependent variable in each sample, weighted where the regression is. N: firm-bank-years in rows 1 and 2, firm-years in rows 3 to 5. Rates winsorized at 1 and 99 percent by year (Appendix A). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F Map of credit growth

Figure 10: Five year credit growth across markets



Realized five year log change in bank credit by local labor market, the outcome side of the first stage.

G Annual shift-Share IV estimates

Table 22: Annual event study coefficients by horizon

		Credit		Interest rate (pp)		Collateral ratio	
h	Credit	Formal	Informal	Formal	Informal	Formal	Informal
−4	−0.082**	−0.097*	−0.077	0.168	−0.046	0.151***	0.071*
	(0.041)	(0.056)	(0.049)	(0.111)	(0.098)	(0.050)	(0.040)
−3	0.004	−0.039	0.016	−0.013	0.118**	0.090	0.074
	(0.027)	(0.024)	(0.026)	(0.042)	(0.052)	(0.058)	(0.045)
−2	−0.127***	−0.064	−0.132***	0.185**	−0.080	−0.031	0.101***
	(0.033)	(0.049)	(0.041)	(0.089)	(0.069)	(0.059)	(0.037)
−1	−0.043	−0.054	−0.006	−0.064	−0.103***	0.021	0.021
	(0.027)	(0.072)	(0.055)	(0.075)	(0.035)	(0.060)	(0.053)
0	0.140***	0.192***	0.076***	−0.274***	−0.021	−0.075	−0.075
	(0.031)	(0.043)	(0.030)	(0.096)	(0.110)	(0.131)	(0.084)
1	0.128***	0.200***	0.057	−0.172**	0.118*	−0.267***	−0.158
	(0.033)	(0.035)	(0.037)	(0.087)	(0.066)	(0.076)	(0.115)
2	0.220***	0.223***	0.158**	−0.262***	0.073	−0.208*	−0.193***
	(0.053)	(0.044)	(0.065)	(0.083)	(0.060)	(0.115)	(0.071)
3	0.227***	0.215***	0.162***	−0.133	0.029	−0.238**	−0.198***
	(0.051)	(0.044)	(0.061)	(0.129)	(0.066)	(0.118)	(0.068)
4	0.201***	0.176***	0.146**	−0.519***	−0.120	−0.289**	−0.138
	(0.052)	(0.044)	(0.066)	(0.130)	(0.121)	(0.124)	(0.089)

Coefficients plotted in Figures 4, 5, 6 and 7: reduced-form estimates on the annual market panel of the outcome h years after the shock on the IV, one regression per horizon and column, with the annual analogs of the main specification's controls, weighted by the market's lagged credit, on market-years with instrument coverage of at least 0.5. Standard errors clustered by market in parentheses. Stars from the corresponding two-sided tests. Credit columns in logs, interest rates in percentage points, the collateral ratio in logs. Market-years per regression range from 7,697 to 10,002 in the credit column, 5,859 to 7,840 in the formal credit and rate columns, 7,622 to 9,915 in the informal ones, 2,805 to 4,314 and 3,780 to 5,689 in the formal and informal collateral columns. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 23: Annual event study coefficients by horizon: additional outcomes

	Years since shock, h								
	−4	−3	−2	−1	0	1	2	3	4
Number of loans (log)									
Formal	0.36	1.45	−3.41	−1.65*	−0.80	−0.73	−0.07	0.01	0.52
	(0.82)	(1.18)	(2.50)	(0.95)	(0.83)	(0.96)	(0.99)	(1.00)	(1.04)
Informal	0.52	1.91	−4.18	−2.07	−3.22*	−3.28	−0.62	−0.12	0.06
	(1.00)	(1.54)	(2.73)	(1.38)	(1.95)	(2.07)	(3.26)	(2.81)	(3.27)
Number of borrowers (log)									
Formal	0.11	1.25	−3.26	−1.66*	−0.69	−0.93	−0.49	−0.31	0.17
	(0.87)	(1.10)	(2.50)	(0.98)	(0.74)	(0.81)	(0.82)	(0.82)	(0.88)
Informal	0.41	1.86	−4.20	−2.00	−3.23*	−3.22	−0.88	−0.50	−0.33
	(1.02)	(1.52)	(2.70)	(1.44)	(1.87)	(1.97)	(3.10)	(2.61)	(3.07)
Loans per borrower (log)									
Formal	−0.25	0.20	−0.15	0.02	−0.10	0.20	0.42*	0.31	0.35
	(0.18)	(0.17)	(0.10)	(0.14)	(0.18)	(0.22)	(0.25)	(0.26)	(0.30)
Informal	0.12	0.05	0.02	−0.07	0.02	−0.05	0.26	0.38	0.39
	(0.09)	(0.10)	(0.08)	(0.09)	(0.13)	(0.17)	(0.20)	(0.24)	(0.25)
Credit per loan (log)									
Formal	−10.06*	−5.31**	−3.03	−3.79	19.96***	20.74***	22.35***	21.48***	17.07***
	(5.58)	(2.55)	(4.84)	(7.70)	(3.74)	(2.85)	(3.88)	(3.80)	(3.70)
Informal	−8.25*	−0.28	−8.99***	1.44	10.86***	8.98***	16.38***	16.29***	14.53***
	(4.47)	(2.42)	(2.98)	(5.03)	(2.95)	(3.40)	(4.75)	(4.84)	(5.37)
Share of loans without collateral									
Formal	−0.06	0.13	−0.20	0.07	−0.79***	−0.50	−0.83***	−0.70**	−0.76**
	(0.10)	(0.19)	(0.13)	(0.14)	(0.30)	(0.30)	(0.30)	(0.33)	(0.34)
Informal	−0.14**	0.14	−0.05	0.07	−0.58**	−0.18	−0.38***	−0.26	−0.36*
	(0.07)	(0.12)	(0.06)	(0.07)	(0.23)	(0.17)	(0.14)	(0.17)	(0.20)
Formal sector, IMSS (log)									
Employment	0.19	0.33*	0.25	−0.07	0.10	0.09	0.53*	0.59*	0.22
	(0.16)	(0.19)	(0.16)	(0.17)	(0.18)	(0.21)	(0.32)	(0.32)	(0.43)
Mean wage	0.22**	0.24**	0.06	0.22***	0.01	−0.02	0.16	0.16	0.04
	(0.09)	(0.12)	(0.11)	(0.07)	(0.08)	(0.16)	(0.17)	(0.20)	(0.22)

Same specification as Table 22, one regression per horizon and row. Coefficients and standard errors are multiplied by 100, so log outcomes read as percent and the share of loans without collateral in percentage points. Loans, borrowers, loans per borrower and credit per loan are market totals or ratios for loans to formal and to informal borrowers. The share without collateral is the share of loans with no posted collateral. Employment and mean wage are the market's IMSS registered workers and their mean wage. Standard errors clustered by market in parentheses. Market years per regression range from 5,859 to 7,840 for the formal borrower series, 7,622 to 9,915 for the informal ones and 7,738 to 9,969 for employment and wages. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H Robustness checks

This appendix examines the robustness of the five-year results. We proceed in three steps. First, we test the identifying assumption directly. We check that the exposure shares are balanced on predetermined market characteristics, we control for lagged outcome changes, and we run placebo regressions on pre-period outcomes. Second, we vary the construction of the instrument. We drop the Leave-One-Out correction, renormalize by coverage, move the coverage threshold from zero to 0.8, and rebuild the instrument without the largest lender. Third, we vary the population and the estimator. We exclude the three metropolitan areas, drop the credit weights, compare the IV estimates with their OLS and reduced-form counterparts, and re-estimate everything at the bank level with re-identified shocks. The corresponding tables follow. One result survives every exercise: credit supply shifts the composition of firms toward formality without changing their number.

A central identifying assumption of our design is that the bank credit supply shocks aggregated by the Shift-Share instrument are as-good-as-randomly assigned conditional on the controls in our main specification. Following the balance tests suggested by Borusyak et al. (2021) and Goldsmith-Pinkham et al. (2020), we regress each of the seven predetermined market controls of the main specification on the Leave-One-Out Shift-Share instrument, conditional on year fixed effects and the coverage control and weighted by baseline market credit. Table 24 reports the results with each control measured five and ten years before the baseline period, that is, ten and fifteen years before the outcome wave. None of the fourteen coefficients is statistically distinguishable from zero at the ten percent level (the smallest p-value is 0.12) and the point estimates are economically small. As a complementary dynamic check, Table 25 re-estimates the main specifications adding the dependent variable's own lagged five-year change as a control. The core results are essentially unchanged, with the formal-firm response at 0.048 against 0.050 in the baseline; the exceptions are the aggregate and formal value-added responses, which reverse sign under this control and on which we therefore do not lean, while the value-added result we emphasize, the reallocation of production away from the informal sector, strengthens under it.

As a model-free check on our identification, Table 26 reports, for every outcome, the naive OLS regression on credit growth and the reduced-form regression on the standardized Leave-One-Out instrument, with the same controls, weights, and sample restrictions as the main IV specification and standard errors clustered at the market level. The reduced form confirms that the identifying correlation is present in the raw data, with significance concentrated in the same margins the paper emphasizes (formal firms 0.056, s.e. 0.016; formal entry 0.075; informal exit 0.062), and the OLS-IV comparison shows why instrumenting matters: OLS is attenuated relative to IV on the composition margins (the formal stock at 0.025 against 0.050, formalization at 0.031 against 0.052), the pattern expected when demand shocks dominate observational credit variation.

The credit-weighted regression places disproportionate weight on the three largest metropolitan areas (Mexico City, Monterrey, and Guadalajara). To address the concern that the main estimates reflect dynamics specific to these urban banking centers, Table 27 re-estimates every specification excluding them, nine of the 1,394 market-year observations. The composition core is essentially unchanged, with the formal-firm response at 0.049 against 0.050 and formal entry identical at 0.067, and several margins strengthen outside the metros (the employment response rises to 0.028 and the informal value-added decline deepens to -0.115). Two results soften and are disclosed as such: the informal-to-formal switching flow loses significance, with the composition shift operating instead through formal retention (0.055), fewer switches into informality (-0.049), and a declining count of informal stayers (-0.025), and the mean-wage decline disappears, indicating that the wage-composition effect of marginal hires is a metropolitan phenomenon. Dropping the largest markets therefore leaves the formalization core intact.

As a direct test of the identifying assumption that our Shift-Share instrument is uncorrelated with pre-existing dynamics in local outcomes, Table 28 reports placebo IV estimates in which the dependent variable is the five-year log change of each outcome before the shock window, $\Delta_5 Y_{t-5} = \ln Y_{t-5} - \ln Y_{t-10}$, with the endogenous regressor, instrument, controls, weights, and sample condition unchanged. The answer is margin-specific and disclosed in full. The stock of existing firms displays a positive pre-period correlation that is common to the two sectors rather than differential (formal 0.016, informal 0.019), whereas the main specification's estimated effects load entirely on formal firms. Moreover, the pre-period correlation is not what drives the estimates: adding the lagged outcome change to the main specification leaves the contemporaneous formal-firm effect essentially unchanged, 0.048 against 0.050. Wages and value added attract significant negative placebo coefficients (-0.164 for average value added), which we read as mean reversion in bank lending: shocks were directed disproportionately toward markets whose financial and real conditions had been deteriorating, so the contemporaneous wage and value-added estimates partly capture recoveries and we do not lean on them. The entry, exit, and formality-transition placebos are uninformative, because the required further lag exists in a single census window and the first stage collapses (a Kleibergen–Paap F of about 1.9), and we report them for transparency rather than as evidence in either direction.

A natural question is whether our findings depend on the Leave-One-Out construction of the Shift-Share instrument. Table 29 re-estimates every outcome with the conventional construction, in which the bank shocks $\hat{\gamma}_{bt}$ are estimated on the full sample including each market's own loans, so that $\hat{Z}_{mt} = \sum_b s_{bm(t-5)}\,\hat{\gamma}_{bt}$ assigns every market the same shock value for a given bank; the sample, controls, weights, and inference are otherwise identical to the main specification. As Borusyak et al. (2021) emphasize, this construction suffers from own-observation overlap: each market's own credit dynamics enter its instrument,

which pulls the IV estimate toward its OLS counterpart. Qualitatively nothing changes, and the modest deviations move exactly as the overlap predicts, toward the attenuated OLS counterparts: the formal-firm elasticity falls from 0.050 to 0.035, with a first stage of similar strength (F of 245 against 266). We therefore retain the Leave-One-Out instrument as our preferred specification: the comparison shows that the refinement, far from generating our results, if anything strengthens the composition margins.

As a complementary identification check, Table 30 moves the analysis to the shock level. The credit supply shocks are re-identified with the cell-fixed-effects approach of Degryse et al. (2019), an extension of the within-firm framework of Khwaja and Mian (2008) in which credit demand is absorbed at the market-sector (LLM $\times$ industry) cell rather than within multi-bank firms. Both multi-bank and single-bank firms now contribute to the shock estimates, a consequential expansion in Mexico, where single-bank firms dominate local credit markets; the unit of observation is the bank-year, outcomes and market credit growth are aggregated across the markets each bank serves using baseline exposure shares, and the shocks, units, aggregation, controls, weights, and samples all differ from the market-level design, so nothing mechanical links the two sets of estimates: if our results reflected a particular scheme for purging demand, the own-market overlap, or fragile specification choices, this regression would be free to disagree.

It does not disagree; it reproduces the composition of the response outcome by outcome: the total stock of firms in a bank's markets does not respond while the formal stock rises (0.036, bootstrap $p < 0.001$, against 0.050 at the market level), turnover again runs through informal exit, the formality transitions match sign by sign, and the fragile value-added responses are absent, consistent with our decision not to lean on them. That a design with different shocks, different units, different aggregation, and different inference reproduces the composition of the response, and not merely its direction, is among the strongest pieces of evidence for the credit supply interpretation of our results.

Our main analysis restricts the estimation sample to market-years for which the Leave-One-Out instrument is constructed from at least half of the local base-period banking exposure. Table 31 re-estimates every outcome at the two extremes: a threshold of zero, which keeps every market-year with a constructible instrument (1,520 observations in 568 markets), and a threshold of 0.8, which keeps only market-years where the instrument pools the great majority of local banking exposure (1,218 observations in 541 markets). The coverage requirement behaves like the measurement restriction it is: the Kleibergen–Paap statistic rises with the threshold, from about 269 to between 288 and 325, while the substantive conclusions do not move, with the formal-firm elasticity in a narrow band around the main estimate (0.053 at zero, 0.050 in the main specification, 0.045 at 0.8) and formal entry at 0.068 and 0.059 at the two extremes; the changes are confined to margins we already treat cautiously. The threshold matters for the precision of the instrument, not

for any substantive conclusion of the paper.

A potential concern with our Leave-One-Out instrument is that its magnitude scales mechanically with coverage: in a market where only half of the base-period banks have an identifiable shock, the instrument is a coverage-weighted sum that is mechanically half as large as it would otherwise be. Table 32 re-estimates the main specification with the renormalized instrument $\tilde{z}_{mt} = z_{mt}/\text{coverage}_{mt}$, which converts the coverage-weighted sum into a coverage-weighted average, standardized as in the main specification and estimated on the identical sample. The estimates are nearly indistinguishable from the main ones, with the formal-firm elasticity at 0.052 against 0.050 and a very strong first stage throughout (Kleibergen–Paap F above 286 in every family); the coverage-driven scaling of the instrument drives no result in the paper.

A remaining concentration concern is that the identifying variation might come disproportionately from a single dominant lender: the shock-level diagnostics show an effective sample of roughly seventeen independent bank-window shocks, with the largest carrying 17 percent of the weight. Table 33 therefore rebuilds the instrument from scratch excluding the bank with the largest share of aggregate baseline credit, removing both its shocks and its exposure shares, and re-estimates the main specification on the identical sample. The cost appears exactly where it should, in the first stage, which falls to a Kleibergen-Paap statistic of about 35, while no conclusion changes: the formal stock rises by a similar magnitude as in the main specification (0.046 against 0.050, with a bootstrap p-value that rises to 0.17 from 0.02), turnover still runs through total and informal exits, and the formality transitions keep their signs with formalization and informalization if anything larger (0.088 and -0.074). What the largest lender contributes is precision, not the result.

The formality composition result at the center of the paper is not the product of any single methodological choice. It survives the balance tests, lagged-outcome controls, and placebo scrutiny, in which the only pre-trends are common to the two sectors or attached to level outcomes we already treat cautiously; it survives replacing the Leave-One-Out construction with the conventional one, renormalizing by coverage, moving the coverage threshold from zero to 0.8, and rebuilding the instrument without the largest lender; and it is reproduced outcome by outcome by a bank-level design with re-identified shocks. Population and weights matter only for precision: the identifying power of the data resides where the credit is, and the employment, wage, and value-added responses, the margins that depend on specification, are the ones we lean on nowhere. Bank credit supply changes the composition of firm activity in Mexican local economies.

H.1 Balance test

Table 24: Balance test: Shift-Share instrument and predetermined market characteristics

	Lagged 5 years		**Lagged 10 years**	
Predetermined variable	$\widehat{\beta}$	SE	$\widehat{\beta}$	SE
Market employment share, sector 1	0.040	(0.026)	0.039	(0.027)
Market employment share, sector 2	−0.008	(0.011)	−0.012	(0.012)
Bank-portfolio sector share, sector 1	−0.007	(0.031)	0.027	(0.053)
Bank-portfolio sector share, sector 2	0.010	(0.034)	−0.057	(0.061)
Credit-weighted interest rate	0.323	(0.256)	0.756	(0.804)
Log number of banks in market	−0.039	(0.072)	−0.072	(0.069)
Log number of credits in market	0.205	(0.314)	−0.034	(0.315)
Observations (range)	1,204–1,360		770–938	
Markets (range)	521–545		459–517	

Notes: Each cell reports the coefficient and market-clustered standard error from a separate regression of the indicated predetermined market characteristic on the standardized leave-one-market-out credit-supply instrument, conditional on year fixed effects and the coverage share interacted with year fixed effects, weighted by baseline market credit. Lags are relative to the baseline period, so the two panels measure each characteristic ten and fifteen years before the outcome wave. Sample restricted to market-years with instrument coverage ≥ 0.5. None of the fourteen coefficients is statistically distinguishable from zero at the 10% level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H.2 Pre-trend test

Table 25: Pre-trend test: main estimates with the lagged outcome change as a control

	Main specification		**With pretrend control**	
Outcome (five-year log change)	$\widehat{\beta}$	SE	$\widehat{\beta}$	SE
Formal firms	0.050***	(0.013)	0.048***	(0.013)
Total firms	0.004	(0.005)	0.003	(0.005)
Informal firms	−0.002	(0.006)	−0.002	(0.006)
Workers (IMSS)	0.020***	(0.005)	0.019***	(0.006)
Total wage bill	0.024**	(0.009)	0.020**	(0.010)
Mean daily wage	−0.012***	(0.004)	−0.008**	(0.003)
Average VA	0.029*	(0.015)	−0.031*	(0.018)
Total VA	0.032*	(0.017)	−0.045**	(0.019)
VA formal	0.028	(0.031)	−0.109***	(0.029)
VA informal	−0.049**	(0.021)	−0.071***	(0.014)
Obs. (range)	1,386–1,394		933–983	
Markets (range)	552–556		527–543	

Notes: The pretrend columns re-estimate the main specification adding the dependent variable's own lagged five-year change as a control; the sample shrinks because the lagged change requires data two windows back. AKM standard errors in parentheses, with stars from the corresponding p-values; the main-specification columns repeat the corresponding entries of the results tables for comparison. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H.3 Reduced-form estimates

Table 26: Reduced-form: Shift-Share instrument and local labor market outcomes

	OLS on credit growth		**Reduced form on $\hat{Z}$**		
Dependent variable	$\widehat{\beta}$	SE	$\widehat{\beta}$	SE	N
Existing firms — Total	0.007	(0.005)	0.004	(0.012)	1,394
Existing firms — Formal	0.025**	(0.012)	0.056***	(0.016)	1,389
Existing firms — Informal	0.006	(0.006)	−0.002	(0.013)	1,394
Entry — Total	0.017*	(0.010)	0.017	(0.022)	996
Entry — Formal	0.033**	(0.016)	0.075**	(0.030)	979
Entry — Informal	0.017	(0.011)	0.013	(0.023)	996
Exit — Total	0.031***	(0.008)	0.055***	(0.011)	996
Exit — Formal	0.025**	(0.011)	0.043*	(0.025)	965
Exit — Informal	0.034***	(0.009)	0.062***	(0.011)	996
Informal→Formal	0.031*	(0.017)	0.059**	(0.027)	957
Formal→Informal	−0.010	(0.013)	−0.044*	(0.026)	958
Formal→Formal	0.027**	(0.014)	0.045***	(0.016)	970
Informal→Informal	0.004	(0.006)	−0.007	(0.013)	996
Total	0.008	(0.018)	0.039	(0.037)	1,394
Formal	0.025	(0.023)	0.046	(0.031)	1,394
Informal	0.016	(0.021)	0.038	(0.051)	1,394
Workers (IMSS)	0.014**	(0.006)	0.022*	(0.012)	1,386
Total wage bill	0.012	(0.010)	0.026	(0.019)	1,386
Mean daily wage	−0.010***	(0.003)	−0.013*	(0.007)	1,386
Average VA	0.034**	(0.014)	0.032	(0.029)	1,394
Total VA	0.041***	(0.016)	0.036	(0.030)	1,394
VA formal	0.032	(0.025)	0.031	(0.046)	1,394
VA informal	0.007	(0.028)	−0.055	(0.063)	1,394

Notes: Each row reports two separate regressions with the same controls, weights, and sample as the main specification: OLS of the outcome on the five-year log change in market credit, and the reduced form of the outcome on the standardized leave-one-market-out instrument. Standard errors clustered at the market (LLM) level. Each reduced-form coefficient equals the product of the corresponding first-stage and IV coefficients. Census worker counts (single-window subsample) are reported in Appendix C. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H.4 Excluding the three largest metropolitan areas

Table 27: Shift-Share IV estimates excluding the three largest metropolitan areas

Dependent variable	$\widehat{\Delta_5 \ln(C)}$	FS F	N
Existing firms — Total	−0.003 (0.006)	82.08	1,385
Existing firms — Formal	0.049*** (0.014)	82.07	1,380
Existing firms — Informal	−0.009 (0.007)	82.08	1,385
Entry — Total	0.018 (0.015)	102.38	990
Entry — Formal	0.067*** (0.019)	102.35	973
Entry — Informal	0.014 (0.015)	102.38	990
Exit — Total	0.041** (0.018)	102.38	990
Exit — Formal	0.067*** (0.018)	102.27	959
Exit — Informal	0.037** (0.018)	102.38	990
Informal→Formal	0.014 (0.020)	101.30	951
Formal→Informal	−0.049** (0.023)	102.26	952
Formal→Formal	0.055*** (0.009)	102.31	964
Informal→Informal	−0.025*** (0.006)	102.38	990
Total	0.114*** (0.020)	82.08	1,385
Formal	0.065*** (0.021)	82.08	1,385
Informal	0.143*** (0.024)	82.08	1,385
Workers (IMSS)	0.028*** (0.008)	82.07	1,377
Total wage bill	0.036*** (0.014)	82.07	1,377
Mean daily wage	0.001 (0.004)	82.07	1,377
Average VA	0.022 (0.021)	82.08	1,385
Total VA	0.018 (0.019)	82.08	1,385
VA formal	0.029 (0.038)	82.08	1,385
VA informal	−0.115*** (0.031)	82.08	1,385

Notes: Same specification and inference as the main tables (AKM standard errors in parentheses, stars from the corresponding p-values; FS F is the Kleibergen–Paap statistic), excluding Mexico City, Monterrey, and Guadalajara. These markets are nine of the 1,194 baseline market-year observations. Wild cluster bootstrap p-values are not recomputed for robustness specifications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H.5 Shift-Share IV Placebo test

Table 28: Placebo tests: Shift-Share IV on pre-period (five-year) outcome changes

Pre-period outcome ($\Delta_5 Y_{t-5}$, log)	$\widehat{\Delta_5 \ln(C)}$	AKM s.e.	First-stage F	N
Log existing firms — Total	0.016*** (0.006)	[0.005]***	290.11	983
Log existing firms — Formal	0.016 (0.013)	[0.008]**	290.06	979
Log existing firms — Informal	0.019** (0.009)	[0.006]***	290.11	983
Log firm entry — Total	0.034 (0.057)	[0.114]	1.89	514
Log firm entry — Formal	−0.078 (0.122)	[0.296]	1.89	508
Log firm entry — Informal	0.048 (0.058)	[0.095]	1.89	514
Log firm exit — Total	−0.043 (0.059)	[0.120]	1.89	514
Log firm exit — Formal	0.146 (0.149)	[0.368]	1.89	495
Log firm exit — Informal	−0.081 (0.077)	[0.108]	1.89	514
Formal–Formal	−0.017 (0.049)	[0.077]	1.59	488
Informal–Formal	−0.112 (0.140)	[0.204]	1.89	496
Formal–Informal	0.075 (0.095)	[0.197]	1.89	485
Informal–Informal	−0.047 (0.053)	[0.127]	1.89	514
Firms with credit — Total	0.079 (0.082)	[0.058]	290.11	983
Firms with credit — Formal	0.091 (0.058)	[0.037]**	290.11	983
Firms with credit — Informal	0.006 (0.157)	[0.120]	290.11	983
Formal workers (IMSS)	−0.011 (0.009)	[0.006]*	290.13	970
Formal wage bill (IMSS)	−0.024** (0.012)	[0.005]***	290.13	970
Mean daily wage (IMSS)	−0.015 (0.010)	[0.006]***	290.13	970
Avg. value added	−0.164*** (0.038)	[0.014]***	290.11	983
Total value added	−0.148*** (0.041)	[0.013]***	290.11	983
Formal value added	−0.174*** (0.063)	[0.021]***	290.11	983
Informal value added	−0.036 (0.036)	[0.019]*	290.11	983

Notes: Each row reports a placebo IV estimate in which the dependent variable is the five-year log change in the indicated outcome *before* the credit-supply shock window, $\Delta_5 Y_{t-5} = \ln Y_{t-5} - \ln Y_{t-10}$. The endogenous regressor, instrument, controls, baseline-credit weights, and sample condition follow the main specification (Table 7); the sample additionally requires a non-missing outcome ten years prior. Standard errors clustered at the market level in parentheses; exposure-robust standard errors following Adão et al. (2019) in brackets. Stars on coefficients are based on the clustered standard errors and stars on brackets on the exposure-robust ones. First-stage F is the Kleibergen–Paap rk Wald statistic on each restricted sample. The entry, exit, transition, and collateral outcomes require lagged flows that exist in a single census window, so their samples halve and the first stage collapses; we report these rows for transparency only. The census-based worker outcomes cannot be estimated in the pre-period at all and are omitted. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H.6 Conventional (Non-Leave-One-Out) IV estimates

Table 29: Shift-Share IV estimates using the conventional (Non-Leave-One-Out) instrument

Dependent variable	$\widehat{\Delta_5 \ln(C)}$	AKM s.e.	Boot. p	First-stage F	N
Log existing firms — Total	0.002 (0.010)	[0.005]	0.930	244.60	1,394
Log existing firms — Formal	0.035** (0.014)	[0.007]***	0.065	244.56	1,389
Log existing firms — Informal	−0.002 (0.012)	[0.006]	0.921	244.60	1,394
Log firm entry — Total	0.007 (0.018)	[0.009]	0.745	258.49	996
Log firm entry — Formal	0.043* (0.024)	[0.009]***	0.132	258.27	979
Log firm entry — Informal	0.007 (0.020)	[0.010]	0.795	258.49	996
Log firm exit — Total	0.049*** (0.010)	[0.009]***	0.003	258.49	996
Log firm exit — Formal	0.021 (0.021)	[0.018]	0.461	258.28	965
Log firm exit — Informal	0.059*** (0.011)	[0.008]***	0.002	258.49	996
Formal–Formal	0.026** (0.013)	[0.005]***	0.145	258.26	970
Informal–Formal	0.049** (0.023)	[0.019]***	0.091	254.67	957
Formal–Informal	−0.043* (0.025)	[0.012]***	0.284	258.26	958
Informal–Informal	−0.003 (0.012)	[0.003]	0.869	258.49	996
Firms with credit — Total	0.024 (0.029)	[0.007]***	0.595	244.60	1,394
Firms with credit — Formal	0.018 (0.025)	[0.011]*	0.572	244.60	1,394
Firms with credit — Informal	0.030 (0.038)	[0.009]***	0.634	244.60	1,394
Formal workers (IMSS)	0.015 (0.010)	[0.004]***	0.184	244.57	1,386
Formal wage bill (IMSS)	0.014 (0.015)	[0.007]**	0.511	244.57	1,386
Mean daily wage (IMSS)	−0.009 (0.006)	[0.003]***	0.258	244.57	1,386
Avg. value added	0.016 (0.024)	[0.016]	0.543	244.60	1,394
Total value added	0.017 (0.024)	[0.016]	0.558	244.60	1,394
Formal value added	−0.001 (0.035)	[0.028]	0.978	244.60	1,394
Informal value added	−0.027 (0.049)	[0.011]**	0.717	244.60	1,394

Notes: Two-stage least squares estimates using the conventional (non-leave-one-out) shift-share instrument, in which the bank shocks are estimated on the full sample including each market's own loans, standardized to unit variance. Specification, controls, baseline-credit weights, and sample condition follow Table 7; the estimation samples are identical to the corresponding leave-one-out regressions. Standard errors clustered at the market level in parentheses, with stars on coefficients based on them; exposure-robust standard errors following Adão et al. (2019) in brackets, with stars on brackets based on exposure-robust p-values; wild cluster bootstrap p-values in the third column. The credit-deepening decomposition sums to one $(0.100 + 0.215 + 0.686)$. First-stage F is the Kleibergen–Paap rk Wald statistic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H.7 Shift-Share IV results at the bank level

Table 30: Bank level IV estimates: Khwaja-Mian-style identification at the bank-year

Dependent variable	$\widehat{\Delta_5 \ln(C)}$	Boot. p	First-stage F	N
Log existing firms — Total	−0.004 (0.005)	0.594	34.91	131
Log existing firms — Formal	0.036*** (0.008)	0.000	34.91	131
Log existing firms — Informal	−0.009* (0.005)	0.213	34.91	131
Log firm entry — Total	−0.001 (0.008)	0.962	30.40	105
Log firm entry — Formal	0.045*** (0.011)	0.000	30.40	105
Log firm entry — Informal	−0.004 (0.009)	0.739	30.40	105
Log firm exit — Total	0.058*** (0.011)	0.003	30.40	105
Log firm exit — Formal	0.020 (0.016)	0.306	30.40	105
Log firm exit — Informal	0.070*** (0.010)	0.000	30.40	105
Formal–Formal	0.021*** (0.008)	0.049	30.40	105
Informal–Formal	0.068*** (0.013)	0.000	30.40	105
Formal–Informal	−0.077*** (0.015)	0.005	30.40	105
Informal–Informal	−0.003 (0.004)	0.685	30.40	105
Firms with credit — Total	−0.003 (0.011)	0.836	34.91	131
Firms with credit — Formal	0.003 (0.016)	0.871	34.91	131
Firms with credit — Informal	0.017 (0.010)	0.158	34.91	131
Formal workers (IMSS)	0.009 (0.007)	0.359	34.91	131
Formal wage bill (IMSS)	0.008 (0.012)	0.636	34.91	131
Mean daily wage (IMSS)	−0.015*** (0.004)	0.004	34.91	131
Avg. value added	0.022 (0.015)	0.256	34.91	131
Total value added	0.018 (0.017)	0.449	34.91	131
Formal value added	−0.006 (0.037)	0.878	34.91	131
Informal value added	−0.013 (0.017)	0.536	34.91	131

Notes: Two-stage least squares estimates at the bank–census-year level. The unit of observation is a bank-year; market outcomes are aggregated to the bank level using the bank's baseline market exposure shares. The endogenous regressor is the bank-level aggregate of five-year market credit growth, instrumented by the bank's credit-supply shock identified from market-sector cell fixed effects following Degryse et al. (2019), an extension of Khwaja and Mian (2008). Controls are census-year fixed effects and the bank-level analogs of the two sector-share controls; estimation is weighted by the bank's baseline credit. The sample keeps bank-years in which the aggregated outcome covers at least half of the bank's baseline credit, so the sample varies across outcomes. Standard errors clustered at the bank level in parentheses, with stars based on them; wild cluster bootstrap p-values in the second column. The credit-deepening decomposition sums to approximately one (0.083+0.209+0.716). First-stage F is the Kleibergen–Paap rk Wald statistic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H.8 Shift-Share IV with different coverage thresholds

Table 31: Sensitivity to instrument coverage threshold

	Coverage ≥ 0				**Coverage** ≥ 0.8			
Dependent variable	$\hat{\beta}$ (s.e.)	F	N	M	$\hat{\beta}$ (s.e.)	F	N	M
Log existing firms — Total	0.006 (0.011)	269.14	1,520	568	0.003 (0.010)	288.27	1,218	541
Log existing firms — Formal	0.053*** (0.016)	269.08	1,514	564	0.045*** (0.014)	288.21	1,213	537
Log existing firms — Informal	−0.000 (0.012)	269.14	1,520	568	−0.001 (0.011)	288.27	1,218	541
Log firm entry — Total	0.016 (0.019)	291.25	1,092	568	0.012 (0.018)	325.07	862	527
Log firm entry — Formal	0.068** (0.026)	290.94	1,071	555	0.059*** (0.023)	324.70	846	516
Log firm entry — Informal	0.013 (0.019)	291.24	1,091	568	0.010 (0.019)	325.07	862	527
Log firm exit — Total	0.051*** (0.010)	291.25	1,092	568	0.047*** (0.011)	325.07	862	527
Log firm exit — Formal	0.037* (0.022)	290.65	1,049	549	0.031* (0.018)	324.89	834	510
Log firm exit — Informal	0.057*** (0.011)	291.25	1,092	568	0.056*** (0.012)	325.07	862	527
Formal–Formal	0.038*** (0.014)	293.29	1,051	553	0.034*** (0.008)	324.77	837	513
Informal–Formal	0.064** (0.025)	290.08	1,042	540	0.051** (0.023)	323.10	827	500
Formal–Informal	−0.047* (0.025)	290.92	1,040	541	−0.035 (0.024)	324.75	829	505
Informal–Informal	−0.004 (0.011)	291.24	1,091	568	−0.004 (0.010)	325.07	862	527
Firms with credit — Total	0.029 (0.029)	269.14	1,520	568	0.026 (0.019)	288.27	1,218	541
Firms with credit — Formal	0.042 (0.027)	269.14	1,520	568	0.031 (0.021)	288.27	1,218	541
Firms with credit — Informal	0.022 (0.040)	269.14	1,520	568	0.023 (0.025)	288.27	1,218	541
Formal workers (IMSS)	0.020** (0.010)	269.09	1,511	563	0.015** (0.007)	288.23	1,211	538
Formal wage bill (IMSS)	0.025 (0.016)	269.09	1,511	563	0.016 (0.010)	288.23	1,211	538
Mean daily wage (IMSS)	−0.013** (0.006)	269.09	1,511	563	−0.011* (0.006)	288.23	1,211	538
Avg. value added	0.025 (0.024)	269.14	1,520	568	0.025 (0.022)	288.27	1,218	541
Total value added	0.030 (0.025)	269.14	1,520	568	0.027 (0.022)	288.27	1,218	541
Formal value added	0.020 (0.043)	269.14	1,520	568	0.014 (0.028)	288.27	1,218	541
Informal value added	−0.040 (0.050)	269.14	1,520	568	−0.031 (0.036)	288.27	1,218	541

Notes: Two-stage least squares estimates re-estimating the main specification at alternative thresholds on instrument coverage. The left block includes all market-years with a non-missing instrument (`cov_z_5` ≥ 0); the right block restricts to market-years for which the leave-one-market-out shift-share instrument pools at least 80% of the local base-period banking exposure (`cov_z_5` ≥ 0.8). The headline specification in Tables 7–9 uses the intermediate threshold `cov_z_5` ≥ 0.5; all other specification details are unchanged. F denotes the first-stage Kleibergen–Paap rk Wald statistic; N denotes observations; M denotes markets. Standard errors clustered at the market level in parentheses, with stars based on them; exposure-robust standard errors following Adão et al. (2019) are similar or smaller in these samples and deliver the same or sharper significance, so the clustered inference reported here is the more conservative choice. The credit-deepening decomposition sums to one in both blocks ($0.146 + 0.181 + 0.673$ and $0.135 + 0.214 + 0.651$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H.9 Shift-Share IV with instrument renormalization

Table 32: Robustness to instrument renormalization

Dependent variable	$\widehat{\Delta_5 \ln(C)}$	AKM s.e.	First-stage F	N
Log existing firms — Total	0.004 (0.010)	[0.006]	286.90	1,394
Log existing firms — Formal	0.052*** (0.016)	[0.016]***	286.84	1,389
Log existing firms — Informal	−0.002 (0.012)	[0.007]	286.90	1,394
Log firm entry — Total	0.016 (0.020)	[0.011]	311.05	996
Log firm entry — Formal	0.070*** (0.027)	[0.021]***	310.76	979
Log firm entry — Informal	0.013 (0.020)	[0.011]	311.05	996
Log firm exit — Total	0.048*** (0.010)	[0.014]***	311.05	996
Log firm exit — Formal	0.041* (0.023)	[0.018]**	310.85	965
Log firm exit — Informal	0.054*** (0.011)	[0.014]***	311.05	996
Formal–Formal	0.042*** (0.016)	[0.011]***	310.80	970
Informal–Formal	0.049* (0.025)	[0.021]**	307.76	957
Formal–Informal	−0.039* (0.023)	[0.018]**	310.73	958
Informal–Informal	−0.008 (0.011)	[0.005]	311.05	996
Firms with credit — Total	0.041 (0.035)	[0.014]***	286.90	1,394
Firms with credit — Formal	0.046 (0.029)	[0.022]**	286.90	1,394
Firms with credit — Informal	0.042 (0.049)	[0.017]**	286.90	1,394
Formal workers (IMSS)	0.021* (0.012)	[0.007]***	286.85	1,386
Formal wage bill (IMSS)	0.026 (0.018)	[0.012]**	286.85	1,386
Mean daily wage (IMSS)	−0.012** (0.006)	[0.005]**	286.85	1,386
Avg. value added	0.032 (0.027)	[0.018]*	286.90	1,394
Total value added	0.035 (0.027)	[0.020]*	286.90	1,394
Formal value added	0.033 (0.044)	[0.039]	286.90	1,394
Informal value added	−0.058 (0.060)	[0.027]**	286.90	1,394

Notes: Two-stage least squares estimates re-estimating the main specification with the renormalized shift-share instrument $\tilde{z}_{mt} = z_{mt}/\texttt{cov_z_5}_{mt}$, which converts the instrument from a coverage-weighted sum of bank shocks into a coverage-weighted average, standardized as in the main specification. The estimation sample, controls, weights, and inference are identical to Table 7. Standard errors clustered at the market level in parentheses, with stars based on them; exposure-robust standard errors following Adão et al. (2019) in brackets, with stars on brackets based on exposure-robust p-values. The credit-deepening decomposition sums to one $(0.174 + 0.200 + 0.626)$. First-stage F is the Kleibergen–Paap rk Wald statistic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H.10 Shift-Share IV estimates excluding the biggest bank

Table 33: Shift-Share IV estimates excluding the biggest bank

Dependent variable	$\widehat{\Delta_5 \ln(C)}$	AKM s.e.	Boot. p	First-stage F	N
Log existing firms — Total	0.001 (0.011)	[0.005]	0.948	35.13	1,394
Log existing firms — Formal	0.046* (0.024)	[0.010]***	0.174	35.13	1,389
Log existing firms — Informal	−0.005 (0.012)	[0.007]	0.803	35.13	1,394
Log firm entry — Total	0.006 (0.021)	[0.011]	0.816	34.97	996
Log firm entry — Formal	0.056 (0.036)	[0.012]***	0.246	34.97	979
Log firm entry — Informal	0.003 (0.021)	[0.012]	0.914	34.97	996
Log firm exit — Total	0.044*** (0.014)	[0.015]***	0.023	34.97	996
Log firm exit — Formal	−0.001 (0.025)	[0.016]	0.976	34.96	965
Log firm exit — Informal	0.059*** (0.015)	[0.015]***	0.004	34.97	996
Formal–Formal	0.028* (0.017)	[0.009]***	0.177	34.96	970
Informal–Formal	0.088** (0.040)	[0.018]***	0.121	35.07	957
Formal–Informal	−0.074** (0.036)	[0.022]***	0.212	34.95	958
Informal–Informal	0.000 (0.012)	[0.004]	0.980	34.97	996
Firms with credit — Total	0.001 (0.032)	[0.012]	0.991	35.13	1,394
Firms with credit — Formal	0.005 (0.036)	[0.021]	0.919	35.13	1,394
Firms with credit — Informal	0.017 (0.035)	[0.016]	0.711	35.13	1,394
Formal workers (IMSS)	0.013 (0.010)	[0.007]*	0.263	35.13	1,386
Formal wage bill (IMSS)	0.009 (0.015)	[0.011]	0.632	35.13	1,386
Mean daily wage (IMSS)	−0.009* (0.006)	[0.005]*	0.130	35.13	1,386
Avg. value added	−0.002 (0.029)	[0.021]	0.955	35.13	1,394
Total value added	−0.002 (0.028)	[0.024]	0.940	35.13	1,394
Formal value added	−0.035 (0.039)	[0.043]	0.440	35.13	1,394
Informal value added	−0.026 (0.048)	[0.022]	0.627	35.13	1,394

Notes: Two-stage least squares estimates re-estimating the main specification with the shift-share instrument reconstructed to exclude the bank holding the largest share of aggregate baseline credit: both its shocks and its exposure shares are removed, and the resulting instrument is standardized as in the main specification. The estimation sample, controls, weights, and sample condition are identical to Table 7. Standard errors clustered at the market level in parentheses, with stars based on them; exposure-robust standard errors following Adão et al. (2019) in brackets, with stars on brackets based on exposure-robust p-values; wild cluster bootstrap p-values in the third column. The credit-deepening decomposition sums to one ($0.090 + 0.235 + 0.675$). First-stage F is the Kleibergen–Paap rk Wald statistic; its decline relative to the main specification reflects the excluded bank's substantial contribution to the identifying variation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

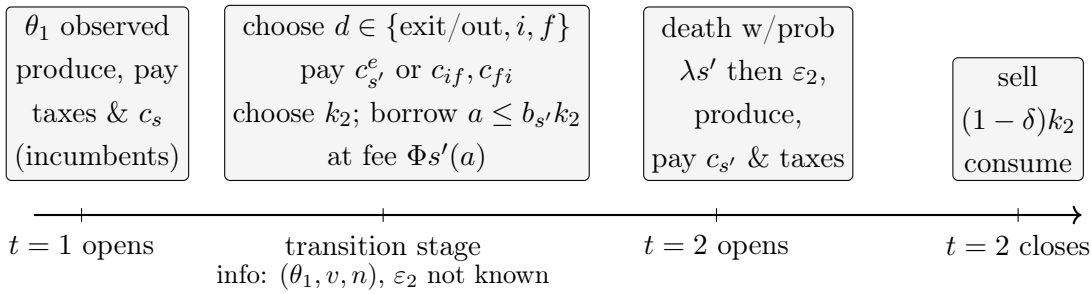
I The two-period model: derivations and additional results

This appendix collects the formal material behind the two-period model of the main text: the timing, the static and dynamic value functions and their envelope properties, the cost ordering behind the threshold structure and the cash-rich region, the master threshold and its four instances, the transition semi-elasticities, and the closed form of the model-implied estimand. Statements keep the notation of the main text, and proofs are in Appendix J.

I.1 Timing

Figure 11 summarizes the timing, which mirrors the recursive model: sector choices for period $t+1$ are made at the end of period t, before the period-$(t+1)$ innovation ε_{t+1} is observed. In period 1, incumbents observe (θ_1, v), hire labor at wage w, produce, settle taxes, and pay the fixed cost c_s. Then each incumbent chooses $d \in \{\text{exit}, i, f\}$ and, if continuing in sector s', next-period capital k_2, borrowing $a = \max\{0,\, k_2 - n\}$ at fee $\Phi s'(a)$ subject to $a \le b_{s'} k_2$, where n denotes internal funds at this stage (defined in (5) below). Potential entrants, knowing (v, k_0) but not ε_2, choose $d \in \{\text{stay out}, i, f\}$, pay $c^e_{s'}$ if entering, and choose k_2 under the same financial technology. Dividends (possibly negative, interpreted as the owner's outside funds absorbed at the financing fee's shadow price) are consumed. Finally, in period 2, death shocks realize (probability $\lambda_{s'}$). A destroyed firm's owner keeps $(1-\delta)k_2$. Surviving firms observe θ_2, hire labor, produce, pay taxes and the fixed cost $c_{s'}$, then sell undepreciated capital $(1-\delta)k_2$. The world ends.

Figure 11: Timing of the two-period model.



I.2 Static problem: labor demand and the profit function

Log productivity combines transitory shocks and a permanent component v, drawn at birth from $\ln v \sim N(0, \sigma^2)$ and known to the firm, such that

$$\ln \theta_1 \;=\; \ln v + \varepsilon_1, \quad \ln \theta_2 \;=\; \rho \ln \theta_1 + (1-\rho)\ln v + \varepsilon_2, \quad \varepsilon_t \overset{iid}{\sim} N(0, \sigma^2_\varepsilon), \quad \rho \in (0,1). \tag{15}$$

Let τ_y^s and τ_ℓ^s denote the taxes effectively paid in sector s: $\tau_y^f = \tau_y$, $\tau_\ell^f = \tau_\ell$, $\tau_y^i = \tau_\ell^i = 0$. Conditional on operating with state (θ, k), the firm solves $\max_{\ell \geq 0} (1-\tau_y^s)\theta k^\alpha \ell^\nu - (1+\tau_\ell^s) w\, \ell$; substituting the optimal ℓ back into the objective yields the operating-profit function

$$\pi_s(\theta, k) \;=\; (1-\nu)\,(1-\tau_y^s)\,\theta k^\alpha \left[\frac{\nu\,(1-\tau_y^s)\,\theta k^\alpha}{(1+\tau_\ell^s)\,w}\right]^{\frac{\nu}{1-\nu}},$$

which the next lemma rewrites in the separable form used throughout.

Lemma I.1 (Labor demand and profits)**.** *Optimal labor and the operating-profit function are*

$$\ell_s^*(\theta, k) = \left[\frac{\nu(1-\tau_y^s)\,\theta k^\alpha}{(1+\tau_\ell^s) w}\right]^{\frac{1}{1-\nu}}, \qquad \pi_s(\theta, k) \;=\; \Omega_s\, x\, k^\gamma, \tag{16}$$

where

$$x \equiv \theta^{\frac{1}{1-\nu}}, \qquad \gamma \equiv \frac{\alpha}{1-\nu} \in (0,1), \qquad \Omega_s \equiv (1-\nu)\,(1-\tau_y^s)^{\frac{1}{1-\nu}} \left[\frac{\nu}{(1+\tau_\ell^s) w}\right]^{\frac{\nu}{1-\nu}}. \tag{17}$$

Labor demand is strictly increasing in θ and k ($\partial \ell^*/\partial \theta > 0$, $\partial \ell^*/\partial k > 0$) and strictly decreasing in the wage and in own-sector taxes ($\partial \ell^*/\partial w < 0$, $\partial \ell_f^*/\partial \tau_y < 0$, $\partial \ell_f^*/\partial \tau_\ell < 0$). The profit function is strictly increasing and strictly concave in k ($\gamma < 1$), and the tax wedge

$$T \;\equiv\; \frac{\Omega_f}{\Omega_i} \;=\; (1-\tau_y)^{\frac{1}{1-\nu}} (1+\tau_\ell)^{-\frac{\nu}{1-\nu}} \;<\; 1 \tag{18}$$

summarizes the fiscal cost of formality per unit of the profitability index x.

All proofs are collected in appendix Section J. The index $x = \theta^{1/(1-\nu)}$ is the natural unit of firm heterogeneity: profits are linear in x, so all expectations below reduce to means of a log-normal random variable. Conditional on the transition-stage information set, define the expected profitability index as

$$\hat{x} \;\equiv\; \mathbb{E}[x_2 \mid \mathcal{I}_1] = \exp\left\{\frac{\rho \ln \theta_1 + (1-\rho)\ln v}{1-\nu} + \frac{\sigma_\varepsilon^2}{2(1-\nu)^2}\right\} \quad \text{(incumbents)}, \tag{19}$$

and $\hat{x}_e(v) = \exp\{\ln v/(1-\nu) + \sigma_\varepsilon^2/(2(1-\nu)^2)\}$ for entrants (whose period-2 productivity is $\ln \theta_2 = \ln v + \varepsilon_2$). $\hat{x}$ is strictly increasing in θ_1 and in v; it is the single sufficient statistic for expected productivity in every decision below.

I.3 Investment and financing: formal statements

Lemma I.2 (Capital and borrowing policy)**.** *The solution to* (7) *is*

$$k_s^*(\hat{x}, n) = \begin{cases} \bar{k}_s(\hat{x}), & n \geq \bar{k}_s(\hat{x}) \qquad \textit{(R1: unconstrained)} \\ n, & \tilde{k}_s(\hat{x}) \leq n < \bar{k}_s(\hat{x}) \qquad \textit{(R2: self-financed at the kink)} \\ \tilde{k}_s(\hat{x}), & (1-b_s)\tilde{k}_s(\hat{x}) \leq n < \tilde{k}_s(\hat{x}) \qquad \textit{(R3: interior borrowing)} \\ \dfrac{n}{1-b_s}, & 0 \leq n < (1-b_s)\tilde{k}_s(\hat{x}) \qquad \textit{(R4: collateral-constrained)} \end{cases} \tag{20}$$

with borrowing $a_s^* = \max\{0, k_s^* - n\}$*, strictly positive exactly in R3-R4. This policy is continuous, weakly increasing in* $\hat{x}$ *and in* n*, and satisfies the following derivatives*

$$\begin{aligned} &\frac{\partial k_s^*}{\partial \phi_s} = -\frac{\tilde{k}_s}{(1-\gamma)R_s} < 0 \text{ in R3}, = 0 \text{ ow}; && \frac{\partial k_s^*}{\partial b_s} = \frac{n}{(1-b_s)^2} > 0 \text{ in R4}, = 0 \text{ ow}; \\ &\frac{\partial k_s^*}{\partial n} = 1 \text{ in R2}, \frac{1}{1-b_s} > 1 \text{ in R4}, = 0 \text{ ow}; && \frac{\partial k_s^*}{\partial \hat{x}} > 0 \text{ in R1, R3}, = 0 \text{ ow}. \end{aligned} \tag{21}$$

Three observations connect Lemma I.2 to the data. First, investment-cash-flow sensitivity ($\partial k^*/\partial n > 0$) is positive only for constrained firms (R2, R4), and is largest exactly where the collateral constraint binds (R4). This is the firm-level evidence of the credit channel that the Shift-Share design picks up at the market level. Second, the marginal cost of external funds ϕ_s acts on the intensive margin of constrained borrowers (R3), while collateral capacity b_s acts on the most severely constrained firms (R4), typically young, low-n entrants. Therefore, the two instruments of a credit expansion reach different parts of the firm distribution. Third, borrowing incidence $\mathbb{P}(a^* > 0)$ rises with expected productivity and falls with internal funds. Finally, notice that this four-region finance-restriction policy is a generalization of the hierarchy of finance with three endogenous regimes of Bond and Meghir (1994).

Lemma I.3 (Value of operating)**.** *Define* $\Psi_s(\hat{x}; R) \equiv (1-\gamma)\,\gamma^{\frac{\gamma}{1-\gamma}}\left(M_s\hat{x}\right)^{\frac{1}{1-\gamma}} R^{-\frac{\gamma}{1-\gamma}}$*, which is strictly increasing and strictly convex in* $\hat{x}$ *and strictly decreasing in* R*. Then*

$$G_s(\hat{x}, n) = \begin{cases} \Psi_s(\hat{x}; R^0), & \textit{R1} \\ M_s\hat{x}\, n^\gamma - R^0 n, & \textit{R2} \\ \phi_s\, n + \Psi_s(\hat{x}; R_s), & \textit{R3} \\ \phi_s\, n + M_s\hat{x}\left(\frac{n}{1-b_s}\right)^\gamma - R_s\,\frac{n}{1-b_s}, & \textit{R4}, \end{cases} \tag{22}$$

where G_s *is continuous, strictly increasing and strictly convex in* $\hat{x}$*, and concave and non-*

decreasing in n. Therefore, the marginal value of internal funds, $q_s \equiv 1+\partial G_s/\partial n$, satisfies

$$q_s = \begin{cases} 1, & R1 \\ 1+\gamma M_s \hat{x} n^{\gamma-1} - R^0 \in (1, 1+\phi_s], & R2 \\ 1+\phi_s, & R3 \\ 1+\phi_s + \frac{\gamma M_s \hat{x}\, k_s^{*\,\gamma-1} - R_s}{1-b_s} \;>\; 1+\phi_s, & R4, \end{cases} \tag{23}$$

The step function q_s is the shadow price of a peso inside the firm. It exceeds one for constrained firms and jumps above $1+\phi_s$ when collateral binds; it is the model object behind the role of entrant wealth $(\mu_{k_0}, \sigma_{k_0})$ in the quantitative model.

Remark I.4 (Borrowing participation and the fixed cost of borrowing)**.** *In the general model the financing fee has an additional fixed component, $\Phi s(a) = \phi_{0s}\mathbf{1}_{a>0} + \phi_s\, a$. However, all results in this section extend directly because conditional on borrowing, the fixed cost is sunk. Thus, all policies and envelope derivatives are unchanged, and thresholds are unchanged up to the bookkeeping $m \to m + \phi_{0s}$ and $C_o \to C_o + \phi_{0f} - \phi_{0i}$ on branches whose continuation choice is to borrow. Define the borrowing participation premium as the surplus gain from capital expansion using external funds over staying at the kink in capital without external funding,*

$$\pi_s^B(\hat{x}, n) \;\equiv\; G_s^{\mathrm{R3/R4}}(\hat{x}, n) - \big[M_s \hat{x}\, n^\gamma - R^0 n\big],$$

which equals zero at the R2/R3 boundary and is strictly increasing in $\hat{x}$ beyond it (its $\hat{x}$-derivative is $M_s(k_s^{*\gamma} - n^\gamma) > 0$ whenever $k_s^* > n$). With $\phi_{0s} > 0$ the firm borrows iff $\pi_s^B \geq \phi_{0s}$, so the R2/R3 boundary shifts to the unique root of $\pi_s^B = \phi_{0s}$, which is a borrowing-participation cutoff in $\hat{x}$ given n. The population mass beyond the cutoff, i.e. the borrowing incidence, is strictly decreasing in ϕ_{0s} given (ϕ_s, b_s). The intensive-margin objects (a^*, the thresholds, their derivatives) in the toy model are unchanged in the quantitative model, conditional on participation.

The next lemma is the analytical workhorse of the paper. By the envelope theorem, the value of choosing a sector responds to i) its financial conditions in proportion to equilibrium borrowing and ii) to collateral capacity in proportion to the shadow value of pledgeability.

Lemma I.5 (Value of financial conditions)**.** *Since G_s, and hence U_s, are differentiable with respect to (ϕ_s, b_s), except at region boundaries, we can derive the following derivatives*

$$\frac{\partial U_s}{\partial \phi_s} = -\,a_s^*(\hat{x}, n) \;\leq\; 0, \qquad \frac{\partial U_s}{\partial b_s} = \mu_s\, k_s^* \;=\; \mathbf{1}_{\mathrm{R4}} \Big[\gamma M_s \hat{x}\, k_s^{*\,\gamma-1} - r_s\Big] \frac{n}{(1-b_s)^2} \;\geq\; 0, \tag{24}$$

where $\mu_s \geq 0$ is the multiplier on (4). Notice that both inequalities are strict exactly when the firm borrows (respectively, when the collateral constraint binds). Finally, observe that the cross-derivatives are zero: $\partial U_s/\partial \phi_{s'} = \partial U_s/\partial b_{s'} = 0$ for $s' \neq s$.

I.4 Formality threshold

Define the origin state of a firm as $o \in \{\text{entrant}, i\text{-incumbent}, f\text{-incumbent}\}$ and its internal funds available before paying the origin-specific fee of each destination as n_o. Thus, the destination s leaves internal funds $n_{o,s} = n_o - \kappa_{os}$ inside the firm, where $\kappa_{os} \in \{c_s^e, c_{if}, c_{fi}, 0\}$ per (5). Then, the firm's formality decision compares

$$\Delta_o(\hat{x}, n_o) \;\equiv\; U_f(\hat{x}, n_o - \kappa_{of}) - U_i(\hat{x}, n_o - \kappa_{oi}). \tag{25}$$

Notice that the empirically relevant region for this comparison is R3 for both destinations. This is growing firms whose desired capital exceeds its internal funds. The statement below is for that region R3, and Section J verifies that the threshold structure extends piecewise to R2/R4 (and characterizes the cash-rich case in Remark I.9). On origin $\times$ destination region R3 $\times$ R3, the following expression holds

$$\Delta_o(\hat{x}, n_o) = A\,\hat{x}^{\frac{1}{1-\gamma}} \;-\; B\,n_o \;-\; C_o, \tag{26}$$

where

$$A \equiv (1-\gamma)\,\gamma^{\frac{\gamma}{1-\gamma}} \left[M_f^{\frac{1}{1-\gamma}} R_f^{-\frac{\gamma}{1-\gamma}} - M_i^{\frac{1}{1-\gamma}} R_i^{-\frac{\gamma}{1-\gamma}} \right], \qquad B \equiv \phi_i - \phi_f \;>\; 0, \tag{27}$$

$$C_o \equiv (1+\phi_f)\,\kappa_{of} - (1+\phi_i)\,\kappa_{oi} + \beta\big[(1-\lambda_f)c_f - (1-\lambda_i)c_i\big]. \tag{28}$$

Assumption 1 (Finance-driven formality)**.** *Define* $T_\lambda \equiv T \cdot \dfrac{1-\lambda_f}{1-\lambda_i}$ *as the tax wedge adjusted by the survival advantage of formality. Then*

$$\left(\frac{R_f}{R_i}\right)^{\gamma} \;<\; T_\lambda. \tag{29}$$

Assumption 1 states that, for a firm that has to borrow, the financing advantage of formality ($\left(\frac{R_f}{R_i}\right)^{\gamma} < 1$) outweighs its fiscal cost net of the survival advantage ($T_\lambda \lesseqgtr 1$). This assumption is equivalent to $A > 0$, and it is the exact sense in which formality is, in this model, access to capital. Absent a financing or survival advantage ($R_f = R_i$, $\lambda f = \lambda i$), $T < 1$ implies $A < 0$ and no borrower would ever formalize.

Assumption 2 (Interior thresholds and cost ordering)**.** *(a)* $C_o + B\,n_o > 0$*, means that the fixed and transition costs of formality are not dominated by its switching benefits. (b) The switching and entry fees satisfy* $c_{if}, c_{fi} > 0$*,* $0 \le c_i^e \le c_f^e$*, and*

$$(1+\phi_f)\,c_f^e - (1+\phi_i)\,c_i^e \;<\; (1+\phi_f)\,c_{if}, \tag{A2-i}$$

$$(1+\phi_f)\,c_f^e + (1+\phi_i)\,c_{fi} \;>\; (1+\phi_i)\,c_i^e. \tag{A2-ii}$$

Conditions (A2-i) and (A2-ii) are exactly equivalent to a cost ordering that places the entrant formality threshold strictly between the two incumbent thresholds, with the formal incumbent facing the lowest cost of formality and the informal incumbent the highest: (A2-i) says that entering formal is cheaper than entering informal and immediately formalizing, and (A2-ii) says that entering informally is cheaper than entering formally and immediately switching down, which holds automatically whenever the switch-down fee is at least the informal entry fee, and in particular whenever informal entry is nearly free, the empirically relevant case (Appendix I states the cost-ordering lemma).

Proposition I.6 (Formality threshold)**.** *Under Assumptions 1–2, for each origin o and internal funds n_o, $\Delta_o(\cdot, n_o)$ is strictly increasing and strictly convex in $\hat{x}$ and crosses zero exactly once, at*

$$\hat{x}_o^F(n_o) \;=\; \left[\frac{B\, n_o + C_o}{A}\right]^{1-\gamma}. \tag{30}$$

Thus, the firm chooses the formal sector if and only if $\hat{x} \geq \hat{x}_o^{\mathrm{F}}(n_o)$. The threshold is strictly increasing in internal funds, so

$$\frac{\partial \hat{x}_o^F}{\partial n_o} = (1-\gamma)\,\frac{B}{A}\left[\frac{Bn_o + C_o}{A}\right]^{-\gamma} \;>\; 0, \tag{31}$$

this is, conditional on expected productivity, firms with more internal funds are less likely to formalize, because self-finance substitutes for the financial advantage of formality.

Proposition I.6 formalizes the sorting logic of the model. The firms that select into formality are those whose growth needs are large relative to their internal resources, i.e. high $\hat{x}$, low n. These are precisely the firms for which a bank credit supply shock is decisive, which is why the Shift-Share elasticities of Section 7 are informative about (ϕ_s, b_s).

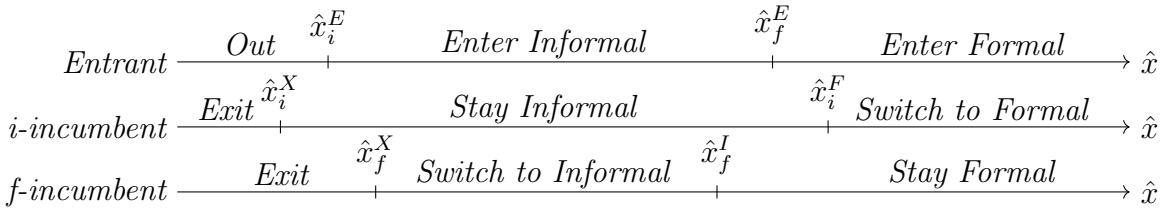
For unconstrained, cash-rich firms the finance advantage is irrelevant, and with taxes as the only non-financial difference a sufficiently wealthy firm prefers informality; a strong enough survival advantage of formality ($T_\lambda \geq 1$) or the size-dependent audit cost of the general model removes this artifact, and the sizable mass of formal-to-informal transitions in the data suggests that for many firms the net non-financial advantage of formality is indeed weak (Appendix I).

I.5 The full transition map and its comparative statics

Each exit and entry margin reduces to one comparison, continuation in the informal sector (the relevant alternative at low $\hat{x}$, by the ordering below) against consuming the outside value, and each is characterized by a condition of the form $G_s(\hat{x}, \tilde{n}) = m$, where $\tilde{n}$ are the funds carried into the continuation and m collects the operating and transition costs. Solving branch by branch of the financing regions yields a single master threshold whose

four instances are exit from either sector and entry into either sector; Appendix I states it and Appendix L develops the complete taxonomy. Two of its properties matter below: a firm without internal funds exits or fails to enter at any productivity (financial exit), and a constrained firm whose quasi-rents on internal funds cover the relevant costs never exits. The thresholds are ordered as follows.

Proposition I.7 (Ordered thresholds)**.** *Under Assumptions 1–2 and $c_i > 0$, for each level of internal funds the policy of every origin state is a partition of the $\hat{x}$ line, such that* $\hat{x}_i^X < \hat{x}_i^E < \hat{x}_f^X < \hat{x}_f^I < \hat{x}_f^E < \hat{x}_i^F$,



I.6 Cost ordering and the cash-rich region

Notice that conditions (A2-i)-(A2-ii) are exactly equivalent (by Lemma I.8 below) to the cost ordering $C_{f\text{-inc}} < C_{\text{entrant}} < C_{i\text{-inc}}$ that places the entrant formality threshold between the two incumbent thresholds. (A2-i) says that enter as formal is cheaper than enter as informal and immediately formalize; (A2-ii) says that entering informally is cheaper than entering formally and immediately switching to formal. (A2-ii) holds automatically whenever $c_{fi} \geq c_i^e$, and in particular whenever informal entry is (nearly) free, the empirically relevant case.

Lemma I.8 (Cost ordering)**.** *Evaluate (28) at each origin's fee pair $(\kappa_{of}, \kappa_{oi})$*[26] *and define $K \equiv \beta[(1-\lambda f)c_f - (1-\lambda i)c_i]$. Then the costs are*

$$C_{i\text{-}inc} = (1+\phi_f)c_{if} + K, \quad C_{f\text{-}inc} = -(1+\phi_i)c_{fi} + K, \quad C_e = (1+\phi_f)c_f^e - (1+\phi_i)c_i^e + K, \tag{32}$$

and $C_e < C_{i\text{-}inc} \iff$ (A2-i), while $C_{f\text{-}inc} < C_e \iff$ (A2-ii). Under $c_{if}, c_{fi} > 0$ alone, $C_{f\text{-}inc} < C_{i\text{-}inc}$ always holds; (A2-i)–(A2-ii) are what places the entrant threshold strictly inside these two costs.

Remark I.9 (Cash-rich firms and the role of the audit margin)**.** *For unconstrained firms (R1 × R1) the finance advantage is irrelevant and $sgn\{\Delta_o\} = sgn\{\Psi_f(\hat{x}; R^0) - \Psi_i(\hat{x}; R^0) - costs\}$, which under $T_\lambda < 1$ is negative for large $\hat{x}$. This means that with taxes as the only non-financial difference, a sufficiently wealthy firm prefers informality. Two observations discipline this artifact. First, with $\lambda i > \lambda f$ large enough, $T_\lambda \geq 1$ and formality remains*

[26] $(c_{if}, 0)$ for informal incumbents, $(0, c_{fi})$ for formal incumbents, (c_f^e, c_i^e) for entrants

absorbing even for the cash-rich. The sizable mass of formal-to-informal transitions in the data, on average 286 *per market and window against* 310 *formalizations (Section 7), suggests that for many firms the net non-financial advantage of formality is weak, that is,* T_λ *close to or below one; the quantitative model lets the data settle this through* $(\tau_y, \tau_\ell, \lambda_s, \eta_i)$.

I.7 The master threshold and its four instances

Combining the exit option with Proposition I.6 yields the complete policy map of each origin state, mirroring the decision rules s_n, s_i, s_f of the quantitative model.

Every exit and entry margin reduces to one comparison, continuation in the informal sector (the relevant alternative at low $\hat{x}$, by the ordering of Proposition I.7) against consuming the outside value. Define

$$A_s \equiv (1-\gamma)\Big(\frac{\gamma}{R_s}\Big)^{\frac{\gamma}{1-\gamma}} M_s^{\frac{1}{1-\gamma}}, \quad A_s^0 \equiv (1-\gamma)\Big(\frac{\gamma}{R^0}\Big)^{\frac{\gamma}{1-\gamma}} M_s^{\frac{1}{1-\gamma}}, \quad \mathcal{K}_s \equiv \beta(1-\lambda_s)c_s, \tag{33}$$

so that $\Psi_s(\hat{x}; R_s) = A_s \hat{x}^{1/(1-\gamma)}$, and define the identities $A = A_f - A_i$ for the formality-margin slope in (27) and $A^0 = A_f^0 - A_i^0$ for the cash-rich slope of Remark I.9. Each margin is characterized by a condition of the form $G_s(\hat{x}, \tilde{n}) = m$, where $\tilde{n}$ are the funds carried into the continuation and m collects the operating and transition costs. Solving branch by branch of (22) defines the following master threshold

$$\mathcal{X}_s(\tilde{n}; m) \;=\; \begin{cases} \big(m/A_s^0\big)^{1-\gamma}, & \text{R1} \\ \dfrac{m + R^0\tilde{n}}{M_s\,\tilde{n}^\gamma}, & \text{R2} \\ \left[\dfrac{m - \phi_s\tilde{n}}{A_s}\right]^{1-\gamma}, & \text{R3, if } m > \phi_s\tilde{n} \\ (1-b_s)^\gamma \dfrac{m + \tilde{n}\,\dfrac{R^0 + \phi_s b_s}{1-b_s}}{M_s\,\tilde{n}^\gamma}, & \text{R4,} \end{cases} \tag{34}$$

where $\mathcal{X}_s$ is continuous across the R1/R2, R2/R3 and R3/R4 boundaries, weakly decreasing in $\tilde{n}$, independent of financial conditions on R1, and satisfies $\mathcal{X}_s \to \infty$ as $\tilde{n} \to 0^+$ on R4. Notice that firms without internal funds exit or fail to enter at any productivity (financial exit). Moreover, on R2, G_s is linear in $\hat{x}$, hence the no power form. If $m \le \phi_s\tilde{n}$ on R3 (or the analogous condition on R4), the constrained firm never exits because the quasi-rents $\phi_s\tilde{n}$ earned by internal funds keep it alive at any productivity.

Hence, the four thresholds are instances of (34):

$$\text{Exit as informal:} \quad \hat{x}_i^X(n) = \mathcal{X}_i\big(n;\ \mathcal{K}_i\big), \qquad n = n^{\text{exit}}, \tag{35}$$

$$\text{Exit as formal:} \quad \hat{x}_f^X(n) = \mathcal{X}_i\big(n - c_{fi};\ \mathcal{K}_i + c_{fi}\big), \qquad n = n^{\text{exit}}, \tag{36}$$

$$\text{Entry as informal:} \quad \hat{x}_i^E(k_0) = \mathcal{X}_i\big(k_0 - c_i^e;\ \mathcal{K}_i + c_i^e\big), \tag{37}$$

$$\text{Entry as formal:} \quad \hat{x}_f^E(k_0) = \hat{x}_e^F(k_0) = \left[\frac{B\,k_0 + C_e}{A}\right]^{1-\gamma} \qquad \text{(Prop. I.6)}. \tag{38}$$

A complete taxonomy of these thresholds is developed in appendix Section L.

I.8 Comparative statics

Proposition I.10 (Comparative statics of all thresholds)**.** *On the borrowing regions, with $D_o \equiv Bn_o + C_o > 0$ and a_s^* evaluated at the relevant margin, the thresholds respond to financial conditions as follows:*

	$\phi_f \uparrow$	$\phi_i \uparrow$	$b_f \uparrow$	$b_i \uparrow$
Formality margins $\hat{x}_e^F, \hat{x}_i^F, \hat{x}_f^F$	$\dfrac{(1-\gamma)\,a_f^*}{D_o} > 0$	$-\dfrac{(1-\gamma)\,a_i^*}{D_o} < 0$	≤ 0	≥ 0
Informal entry/exit $\hat{x}_i^E, \hat{x}_i^X$	0	$\dfrac{a_i^*}{\Psi_i'} > 0$	0	≤ 0
Formal exit $\hat{x}_f^X$	0	$\dfrac{a_i^*}{\Psi_i'} > 0$	0	≤ 0

Note*: entries assume the threshold ordering of Proposition I.7, so the formal incumbent's relevant alternative at its exit margin is the switch-down option; off that ordering the formal-exit row picks up ≥ 0 (ϕ_f) and ≤ 0 (b_f) terms through the value of staying formal. The first row reports $\partial \ln(\text{threshold})/\partial(\text{parameter})$; the second and third rows report level derivatives, with a_i^* evaluated at the margin's funds (n for informal exit/entry, $n - c_{fi}$ for formal exit).*

I.9 Transition semi-elasticities

The thresholds translate into transition probabilities, the objects the Shift-Share design estimates, by integrating over the period-1 population. Consider informal incumbents with permanent component v and internal funds n (the same algebra applies to each origin state). Since $\ln \theta_1 = \ln v + \varepsilon_1$ and $\hat{x}$ is monotone in θ_1, the probability of an informal-to-formal switch is

$$\mathbb{P}(i \to f \mid v, n) = 1 - \Phi(z_{if}), \qquad z_{if} \equiv \frac{\ln \bar{\theta}_{if}(v, n) - \ln v}{\sigma_\varepsilon}, \tag{39}$$

where the productivity threshold inverts (19):

$$\ln \bar{\theta}_{if}(v, n) = \frac{1}{\rho}\Big[(1-\nu)\ln \hat{x}_i^F(n) - (1-\rho)\ln v\Big] - \frac{\sigma_\varepsilon^2}{2\rho(1-\nu)}. \tag{40}$$

Expression (40) is common to all regions. Its derivation uses only the log-linear map (19) and the existence of a cutoff in $\hat{x}$-units, so it converts any threshold, whichever branch of (34) delivers it, closed-form or implicit, into a cutoff for θ_1. The financing region enters only through the value of the threshold and through its derivatives in the chain rule below. Analogous expressions hold for $\mathbb{P}(\text{exit} \mid i) = \Phi(z_{iX})$, $\mathbb{P}(f \to i) = \Phi(z_{fi}) - \Phi(z_{fX})$, entry probabilities (with $\hat{x}_e(v)$ and thresholds in v), and all the other transitions.

I.10 From thresholds to transition rates

Integrating over the period-1 population translates the thresholds into transition probabilities, the objects the Shift-Share design estimates: each transition probability is a normal probability of the standardized gap between log productivity and the operative threshold, so the response of any transition rate to financial conditions is the density of firms at the margin times the threshold shift of the comparative statics above, scaled by borrowing at the relevant margin. Appendix I derives the semi-elasticity of every transition rate in closed form; self-financed firms contribute exactly zero, and aggregate responses are incidence-weighted integrals that concentrate among small, young, low-net-worth firms near their thresholds. The payoff for the empirical strategy is the incidence algebra of the switching margins.

Corollary I.11 (Incidence and the switching margins)**.** *Consider a credit supply expansion that improves both sectors' borrowing terms: spreads fall by the absolute amounts* $\varepsilon_s \equiv \chi_s \phi_s \geq 0$*, i.e.* $\mathrm{d}\phi_s = -\varepsilon_s$*, and/or collateral capacities rise,* $\mathrm{d}b_s \geq 0$*. The conditions below are stated for the spread components; a collateral improvement adds* $\mu_s k_s^* \mathrm{d}b_s$ *to its own sector's side of each inequality. Then:*

(i) Entry and exit. *Under Proposition I.7, the binding entry margin and both exit margins,* $\hat{x}_i^{\mathrm{E}}$*,* $\hat{x}_i^{\mathrm{X}}$*,* $\hat{x}_f^{\mathrm{X}}$*, are informal-continuation comparisons, so they respond only to informal-side terms: all three fall when* $\varepsilon_i > 0$ *or* $\mathrm{d}b_i > 0$*, and are unchanged under a purely formal-side expansion. Hence total entry rises and exit falls at both incumbent margins whenever the expansion reaches the informal side; the formal/informal* composition *of entry is governed by* $\hat{x}_e^{\mathrm{F}}$*, which obeys the same condition as the switching margins of (ii).*

(ii) Switching up. $\mathbb{P}(i \to f)$ *rises if and only if*

$$\varepsilon_f \, a_f^* \; > \; \varepsilon_i \, a_i^* \qquad \text{at the } i \to f \text{ margin.} \tag{41}$$

Because the switch fee travels with the formal option, $a_f^* - a_i^* = \tilde{k}_f - \tilde{k}_i + c_{if} > 0$ *under Assumption 1; hence equal absolute improvements* ($\varepsilon_f = \varepsilon_i$) *always raise informal-to-formal switching. Under equal* proportional *improvements* ($\chi_f = \chi_i$)*,*

(41) *reads* $\phi_f a_f^* > \phi_i a_i^*$*: the borrowing-scale gap must outweigh the spread gap. The same condition, with the* a^**'s evaluated at the entrant's funds* ($a_s^* = \tilde{k}_s - (k_0 - c_s^e)$)*, governs the entrant margin* $\hat{x}_e^{\mathrm{F}}$*: there* $a_f^* - a_i^* = \tilde{k}_f - \tilde{k}_i + c_f^e - c_i^e > 0$*, so equal absolute improvements also raise the formal share of entrants.*

(iii) Switching down. $\mathbb{P}(f \to i)$ *rises if and only if*

$$\varepsilon_i \, a_i^* \;>\; \varepsilon_f \, a_f^* \qquad \text{at the } f \to i \text{ margin,} \tag{42}$$

where now the fee travels with the informal option, $a_i^* - a_f^* = \tilde{k}_i - \tilde{k}_f + c_{fi}$*. Both switching margins can therefore rise together only if the expansion has a sufficiently informal-side component (*ε_i*, or* $b_i \uparrow$*).*

This incidence algebra is the toy model's central payoff for the empirical strategy, and the estimates of Section 6 discipline it. Formalization rises and de-formalization falls; by (ii) and (iii) both movements require the same inequality, that the formal-side improvement dominates at each switching margin, $\varepsilon_f a_f^* > \varepsilon_i a_i^*$, while the size of the de-formalization decline is informative about the informal side, because a larger informal-side improvement attenuates it. The entry and exit block points the same way: informal entry is flat and exit rises, so the informal-side threshold effects of (i) are absent from the data, and the exit increase instead reflects the turnover of marginal cohorts, the subject of the next subsection. The estimated switching pattern is thereby converted into a statement about which sector's borrowing terms the shock reached, and this is the restriction taken to the structural estimation: the first-stage credit response and the collateral response pin the formal-side incidence, the two switching responses discipline the informal-side loading, which the data keep small, and the interest rate is kept as a measurement equation rather than an estimation target.

I.11 Composition: marginal entrants, turnover, and average productivity

The last set of results concerns the composition of the response. This is where the model meets the value-added estimates of Section 7: informal value added falls, formal value added does not move, and the average across firms is at most marginally positive. Allow period-2 firms a shutdown option: after observing θ_2, an active firm operates iff variable profit covers the fixed cost, $\Omega_s x_2 k_2^{\gamma} \geq c_s$; otherwise it stays idle for the period (it still resells its capital). Shutdown is the two-period analogue of endogenous exit for a young cohort. For an entering cohort into sector i (the same algebra applies to a formal entering cohort, the margin the estimated shock moves most), entry occurs iff $v \geq \bar{v}(k_0)$, where $\bar{v}$ solves $\hat{x}_e(\bar{v}) = \hat{x}_i^{\mathrm{E}}(k_0)$, and:

$$\mathbb{P}(\text{shutdown of an entrant with}(v,k_2)) : \pi^X(v) = \Phi\left(\frac{(1-\nu)\ln\big(c_i/(\Omega_i k_2^{\gamma})\big) - \ln v}{\sigma_\varepsilon}\right), \quad (43)$$

which is strictly decreasing in v (directly, and indirectly through k_2).

Proposition I.12 (Composition of a credit expansion)**.** *Let financial conditions improve for the informal sector ($\phi_i \downarrow$ or $b_i \uparrow$), so that $\bar{v}$ falls for every constrained k_0. Then, for the entering cohort:*

(i) Entry rises*: the mass of entrants $\int \mathbb{P}(v \geq \bar{v}(k_0))\,\mathrm{d}F(k_0)$ increases.*

(ii) Average entrant productivity falls*: for $\ln v \sim N(0,\sigma_v^2)$, $\mathbb{E}[v \mid v \geq \bar{v}]$ is strictly increasing in $\bar{v}$; hence a lower threshold mechanically lowers the mean (and every quantile) of entrant productivity, and with it average value added per entrant.*

(iii) Cohort turnover rises*: the cohort shutdown rate $\mathbb{E}[\pi^X(v) \mid v \geq \bar{v}]$ is strictly decreasing in $\bar{v}$; the marginal entrants admitted by the expansion fail at higher rates than infra-marginal ones, so measured exit rises with a lag even as incumbent exit thresholds fall.*

Part (iii) of the proposition isolates the composition channel: for the collateral- constrained entrants where the margin lives, capital is pinned by wealth and collateral, so a pure spread reduction leaves their capital unchanged and the composition effect is the entire response, while a collateral relaxation deepens their capital and partially offsets it. The proposition thus reproduces, inside the two-period model, the composition signature of Section 6: the expansion admits marginal, lower-v entrants and raises their subsequent turnover, the model counterpart of the estimated exit response, while on the informal side the formality threshold falls, the most productive informal firms leave the pool, and informal average value added declines. The two forces lower average productivity within the informal sector, the response that is robust in the data, and leave the economy-wide average roughly flat, exactly as estimated: the model reads the value-added results as reallocation plus selection, not as productivity gains within infra-marginal firms.

Proposition I.13 (Transition semi-elasticities)**.** *Define $p = \mathbb{P}(i \to f \mid v,n)$ from (39). Then, when both destinations borrow at an interior optimum (R3) at the margin,*

$$\frac{\partial p}{\partial \phi_f} = -\,\phi(z_{if})\,\frac{(1-\nu)(1-\gamma)}{\rho\,\sigma_\varepsilon}\,\frac{a_f^*}{D_i} \;<\; 0, \qquad \frac{\partial p}{\partial \phi_i} = \phi(z_{if})\,\frac{(1-\nu)(1-\gamma)}{\rho\,\sigma_\varepsilon}\,\frac{a_i^*}{D_i} \;>\; 0, \quad (44)$$

while, when the relevant destination is collateral-constrained (R4) at the margin, the collateral responses are

$$\frac{\partial p}{\partial b_f} = \phi(z_{if})\,\frac{1-\nu}{\rho\,\sigma_\varepsilon}\,\frac{\mu_f k_f^*}{N_i} \;\geq\; 0, \qquad \frac{\partial p}{\partial b_i} = -\,\phi(z_{if})\,\frac{1-\nu}{\rho\,\sigma_\varepsilon}\,\frac{\mu_i k_i^*}{N_i} \;\leq\; 0, \tag{45}$$

where N_i is the numerator of the operative branch of $\hat{x}_i^F$ at the crossing[27] *and $\mu_s k_s^*$ is the shadow value of collateral of Lemma I.5. The same structure, with the appropriate threshold and sign from Proposition I.10, applies to every other transition probability. The factor $(1-\gamma)$ is specific to the power branch (R3) and drops on the linear branches; all responses are scaled by $\phi(z)/\sigma_\varepsilon$, which is the density of firms at the margin, so aggregate responses are largest where the firm distribution is thick near thresholds.*

Two aggregation implications worth highlighting. First, because (40) is region-free, the chain rule composes with whichever region's log-derivative is operative at the margin for the given (v, n). Thus, the ϕ-formulas displayed are the R3 instances, the collateral formulas the R4 instances, and R1/R2 self-financed firms contribute exactly zero. Second, aggregate transition responses are therefore incidence-weighted integrals. This is $\phi(z)$ times the region-specific term, integrated over the joint distribution of (v, n), that concentrates predicted responses among small, young, low-net-worth firms. Which is the sample counterpart of the heterogeneity documented in Section 7.

I.12 The model-implied estimand in closed form

Finally, the toy model makes precise what the Leave-One-Out Shift-Share coefficient identifies through the lens of the model. Appendix K shows as an example the detailed case of the effect on the mass of informal firms. The quantitative model computes this object numerically; the generic case presented here is schematic and serves to draw general conclusions. Let the bank-supply shock in market m shift financial conditions by $\mathrm{d}\Xi_m = (-\chi_i\phi_i, -\chi_f\phi_f, \kappa_i b_i, \kappa_f b_f)'\,\zeta_m$, where ζ_m is the (exogenous) shock intensity. Aggregate borrowing is $C = \int a^*\,\mathrm{d}\mu$ over active firms, and each firm-count outcome Y (stocks, entry, exit, transitions by sector) aggregates the threshold rules. To first order, the model counterpart of the IV estimand is the ratio of two equilibrium responses to the *same* financial-conditions shift:

$$\beta_Y^{\text{model}} \;=\; \frac{\mathrm{d}\ln Y/\mathrm{d}\zeta}{\mathrm{d}\ln C/\mathrm{d}\zeta} \;=\; \frac{\displaystyle\sum_{j\in\text{margins of } Y} \frac{\phi(z_j)}{Y}\,\frac{(1-\nu)(1-\gamma)}{\rho\sigma_\varepsilon}\,\frac{\phi\,a^*\chi}{D_j}}{\dfrac{1}{C}\Big[\underbrace{\displaystyle\int_{borrowers}\frac{\chi_{s'}\phi_{s'}\tilde{k}_{s'}(\hat{x})}{(1-\gamma)R_{s'}}\mathrm{d}\mu}_{\text{intensive: loan deepening}} - \underbrace{\displaystyle\sum_j\int w_M s_M \Delta a_j^*\mathrm{d}\mu}_{\text{extensive: Created/swapped loans}}\Big]}, \tag{46}$$

[27] Which is equal to $D_i\hat{x}^*$ on R4×R4. See appendix Section L for more details.

where w_M and s_M are as defined in Appendix K. Three lessons that carry to the quantitative model. First, the numerator is a density-weighted sum of threshold shifts: the IV coefficient is large where many firms sit near their thresholds, young, small, constrained firms, which rationalizes why the gross-flow elasticities (entry, exit, switching) exceed the net stock response, since flows add up threshold crossings while the stock nets them out, and why the responses concentrate on the formal side, the sector whose borrowing terms the shock improves. Second, the denominator mixes intensive responses (continuing borrowers deepen loans by $\chi_s\phi_s\tilde{k}_s/[(1-\gamma)R_s]$ each) with extensive ones (loans created and swapped at the margins), and both are built from parameters estimated off cross-sectional moments, spread levels, participation costs, collateral caps. Matching the observed first stage disciplines that financing block directly: the quantity response pins $\chi_f\phi_f$ and the collateral response pins κ_f, while the loan-rate response is kept as a measurement check rather than a target, because the observed book rate mixes repricing with near-complete turnover of the loan stock. Third, because ζ enters numerator and denominator symmetrically, β_Y^{model} is invariant to the scale of the shock for small ζ: the model counterpart can be computed from any small perturbation, and only the incidence $(\chi_i, \chi_f, \kappa_f)$, not the size, of the credit supply shock needs to be estimated $\kappa_i = 0$ in the baseline: bank credit supply shocks do not shift informal collateral technology).

J Proofs

Proof of Lemma I.1. The objective $(1-\tau_y^s)\theta k^\alpha \ell^\nu - (1+\tau_\ell^s)w\ell$ is strictly concave in ℓ ($\nu < 1$); the first-order condition $\nu(1-\tau_y^s)\theta k^\alpha \ell^{\nu-1} = (1+\tau_\ell^s)w$ gives ℓ_s^* in (16). Substituting, using $(1+\tau_\ell^s)w\,\ell^* = \nu(1-\tau_y^s)\theta k^\alpha \ell^{*\nu}$, yields $\pi_s = (1-\nu)(1-\tau_y^s)\theta k^\alpha \ell^{*\nu}$ and, after substituting $\ell^{*\nu}$, expression (17). Monotonicity claims follow by inspection of the closed forms: $\partial \ln \ell^*/\partial \ln \theta = 1/(1-\nu) > 0$, $\partial \ln \ell^*/\partial \ln k = \alpha/(1-\nu) > 0$, $\partial \ln \ell^*/\partial \ln w = -1/(1-\nu) < 0$, and both tax terms enter with negative sign. Strict concavity of π_s in k follows from $\gamma = \alpha/(1-\nu) < 1$, itself implied by $\alpha + \nu < 1$. $T < 1$ is immediate from $\tau_y, \tau_\ell > 0$. □

Proof of Lemma I.2. The objective in (7) is continuous and, on each branch of $a = \max\{0, k-n\}$, strictly concave in k (the return term has curvature $\gamma < 1$; the cost terms are linear). Marginal return to capital is $g'(k) = \gamma M_s \hat{x} k^{\gamma-1} + \beta(1-\delta)$, strictly decreasing, with range $(\beta(1-\delta), \infty)$. Marginal cost is 1 for $k < n$ and $1+\phi_s$ for $k > n$. If $g'(n) \le 1$, i.e. $n \ge \bar{k}_s$, the self-financed optimum $\bar{k}_s$ (where $g' = 1$) is attainable without borrowing: region R1. If $1 < g'(n) \le 1+\phi_s$, i.e. $\tilde{k}_s \le n < \bar{k}_s$, the kink at $k = n$ is optimal: region R2. If $g'(n) > 1+\phi_s$, i.e. $n < \tilde{k}_s$, the firm borrows to the interior optimum $\tilde{k}_s$ (where $g' = 1+\phi_s$) provided the collateral constraint $k - n \le b_s k \iff k \le n/(1-b_s)$ permits, i.e. $\tilde{k}_s \le n/(1-b_s)$: region R3. Otherwise the constraint binds and $k = n/(1-b_s)$: region R4. The boundaries in (20) restate these conditions. The derivative signs follow from

$\tilde{k}_s = (\gamma M_s \hat{x}/R_s)^{1/(1-\gamma)}$ and $\bar{k}_s = (\gamma M_s \hat{x}/R^0)^{1/(1-\gamma)}$, so that $\partial \tilde{k}_s/\partial \phi_s = -\tilde{k}_s/[(1-\gamma)R_s]$, $\partial \tilde{k}_s/\partial \hat{x} = \tilde{k}_s/[(1-\gamma)\hat{x}]$ and $\partial \bar{k}_s/\partial \hat{x} = \bar{k}_s/[(1-\gamma)\hat{x}]$; from $k = n/(1-b_s)$ in R4, whose b_s-derivative is $n/(1-b_s)^2$; and by inspection elsewhere. Continuity at boundaries is direct substitution. □

Proof of Lemma I.3. Substitute the policy (20) into (7). R1: $G = M_s \hat{x} \bar{k}^\gamma - R^0 \bar{k}$; using $\gamma M_s \hat{x} \bar{k}^{\gamma-1} = R^0$, $G = R^0 \bar{k}(1-\gamma)/\gamma = \Psi_s(\hat{x}; R^0)$ after substituting $\bar{k}$. R2: $k = n$, $a = 0$: $G = M_s \hat{x} n^\gamma - R^0 n$. R3: $G = -\tilde{k} - \phi_s(\tilde{k} - n) + M_s \hat{x} \tilde{k}^\gamma + \beta(1-\delta)\tilde{k} = \phi_s n + M_s \hat{x} \tilde{k}^\gamma - R_s \tilde{k}$, and the same substitution with R_s gives $\phi_s n + \Psi_s(\hat{x}; R_s)$. R4: substitute $k = n/(1-b_s)$ directly. Monotonicity and convexity of Ψ_s in $\hat{x}$ follow from the exponent $1/(1-\gamma) > 1$; G inherits them. The formulas for q_s are the n-derivatives of (22); the R2 bounds follow because $g'(n) \in (1, 1+\phi_s]$ there, and the R4 inequality because the bracketed marginal-return gap is strictly positive when the collateral constraint binds. Concavity of G in n holds piecewise and at kinks because q_s is (weakly) decreasing in n region by region as n rises through R4→R3→R2→R1. □

Proof of Lemma I.5. On R1–R2, $a^* = 0$ and (ϕ_s, b_s) do not enter the realized objective: derivatives are zero, consistent with (24). On R3–R4 apply the envelope theorem to (7): the derivative of the value with respect to ϕ_s at the optimum is the partial derivative of the objective, $-a_s^*$; for b_s, the constraint $a \le b_s k$ has multiplier $\mu_s \ge 0$ and the envelope derivative is $\mu_s k_s^*$, zero when slack (R3) and equal to the stated expression in R4, where the Kuhn–Tucker condition gives $\mu_s = \left[\gamma M_s \hat{x} k^{*\gamma-1} - R_s\right]/(1-b_s)\cdot$(normalization); differentiating (22) in R4 with respect to b_s directly yields $\left[\gamma M_s \hat{x} k_s^{*\gamma-1} - R_s\right] n/(1-b_s)^2 > 0$. Cross-sector derivatives are zero because s'-parameters do not enter sector-s payoffs. □

Proof of Proposition I.6. On R3×R3, (26) follows by subtracting the R3 branch of (22) for each destination and collecting the $\hat{x}^{1/(1-\gamma)}$, n_o, and constant terms, using (5). $A > 0$ iff $M_f^{1/(1-\gamma)} R_f^{-\gamma/(1-\gamma)} > M_i^{1/(1-\gamma)} R_i^{-\gamma/(1-\gamma)}$, i.e. iff $M_f/M_i > (R_f/R_i)^\gamma$; since $M_f/M_i = T_\lambda$ by definition, this is Assumption 1. Strict monotonicity and convexity in $\hat{x}$ follow from $A > 0$ and the exponent $1/(1-\gamma) > 1$; since $\Delta_o \to -(Bn_o + C_o) < 0$ as $\hat{x} \to 0$ (by Assumption 2) and $\Delta_o \to \infty$ as $\hat{x} \to \infty$, a unique zero exists and solves (30). Differentiating (30) in n_o gives (31). For completeness, off R3×R3 the difference Δ_o remains continuous and strictly increasing in $\hat{x}$ wherever the formal destination is in a (weakly) more constrained region than the informal one, which holds at the crossing point under Assumption 1 because $\tilde{k}_f > \tilde{k}_i$ (shown in the proof of Corollary I.11); hence the crossing is unique globally on the borrowing range. □

Proof of Lemma I.8. Substituting each origin's fee pair into (28) gives (32) directly; the common constant K cancels from every pairwise difference. Upper comparison:

$$C_{i\text{-inc}} - C_e = (1+\phi_f)c_{if} - \left[(1+\phi_f)c_f^e - (1+\phi_i)c_i^e\right] \;>\; 0 \iff \text{(A2-i)}.$$

Lower comparison:

$$C_e - C_{f\text{-inc}} = (1+\phi_f)c_f^e - (1+\phi_i)c_i^e + (1+\phi_i)c_{fi} \;>\; 0 \iff \text{(A2-ii)}.$$

Note that $c_i^e \le c_f^e$ alone does *not* imply (A2-ii), because $\phi_i > \phi_f$ can make $(1+\phi_i)c_i^e$ exceed $(1+\phi_f)c_f^e$; if $c_{fi} \ge c_i^e$, however, $(1+\phi_i)(c_{fi} - c_i^e) \ge 0$ and $(1+\phi_f)c_f^e > 0$ deliver (A2-ii). Finally, $C_{i\text{-inc}} - C_{f\text{-inc}} = (1+\phi_f)c_{if} + (1+\phi_i)c_{fi} > 0$ under positive switching fees alone. □

Proof of Proposition I.7. The three-interval structure for each origin follows from: (i) values of continuation are continuous and strictly increasing in $\hat{x}$ while exit/out values are constant in $\hat{x}$, giving unique exit/entry cutoffs; (ii) Proposition I.6 gives the unique formality cutoff above them whenever $\hat{x}^{\mathrm{X}} < \hat{x}^{\mathrm{F}}$, which holds under Assumption 2 (the formality margin requires strictly positive net cost $D_o > 0$ while the exit margin turns on c_i); the formal incumbent's middle interval is switching down rather than staying informal by definition of its choice set. The ordering $\hat{x}_f^{\mathrm{F}} < \hat{x}_e^{\mathrm{F}} < \hat{x}_i^{\mathrm{F}}$ follows from (30): thresholds are increasing in C_o, and $C_{f\text{-inc}} < C_{\text{entrant}} < C_{i\text{-inc}}$ by Assumption 2. □

Proof of Proposition I.10. Each threshold is defined by an indifference condition $\Delta(\hat{x}^*, \cdot) = 0$ between two options. By the implicit function theorem, $\partial\hat{x}^*/\partial\xi = -(\partial\Delta/\partial\xi)/(\partial\Delta/\partial\hat{x})$, and $\partial\Delta/\partial\hat{x} > 0$ at an upward crossing. The numerators are given by Lemma I.5: for the formality margins, $\partial\Delta/\partial\phi_f = -a_f^*$, $\partial\Delta/\partial\phi_i = +a_i^*$, $\partial\Delta/\partial b_f = \mu_f k_f^* \ge 0$, $\partial\Delta/\partial b_i = -\mu_i k_i^* \le 0$. Using $\partial\Delta/\partial\hat{x} = A\hat{x}^{\gamma/(1-\gamma)}/(1-\gamma) = D_o/[(1-\gamma)\hat{x}^*]$ at the threshold (substituting $A\hat{x}^{*1/(1-\gamma)} = D_o$) delivers the log-derivatives in the first row. For the informal entry/exit margins the indifference is $G_i = \text{const}$, so $\partial\Delta/\partial\phi_i = -a_i^*$ and $\partial\Delta/\partial\hat{x} = \Psi_i' > 0$, giving the stated positive derivative (thresholds rise as informal finance becomes dearer); formal-side parameters do not enter, hence the zeros. For formal exit, the relevant comparison at low $\hat{x}$ under the ordering of Proposition I.7 is exit versus switching down, whose value is $U_i(\hat{x}, n - c_{fi})$; its derivatives in informal financial parameters are those of U_i (Lemma I.5), delivering the third row's informal-side entries, and formal-side parameters do not enter the comparison at all, hence its zeros. Off that ordering — if staying formal were the binding alternative at the exit margin — the cutoff would be $\mathcal{X}_f(n; \mathcal{K}_f)$ and easier formal terms would weakly lower it (≥ 0 in ϕ_f, ≤ 0 in b_f), which is the caveat recorded in the table note. □

Proof of Proposition I.13. $p = 1 - \Phi(z)$ with $z = (\ln\bar{\theta} - \ln v)/\sigma_\varepsilon$ and, from (40), $\partial \ln\bar{\theta}/\partial\xi = \frac{1-\nu}{\rho}\,\partial \ln\hat{x}_i^{\mathrm{F}}/\partial\xi$ for any financial parameter ξ. Chain-differentiate and substitute the log-derivatives from Proposition I.10; $\phi(\cdot) > 0$ delivers the signs. For (45): on the linear branches the threshold is the ratio of a linear numerator to a slope (Section L); the quotient rule with the root identity $N_i = D_4\hat{x}^*$ gives $\partial \ln\hat{x}_i^{\mathrm{F}}/\partial b_f = -\mu_f k_f^*/N_i$ and $\partial \ln\hat{x}_i^{\mathrm{F}}/\partial b_i = +\mu_i k_i^*/N_i$, and the same chain rule applies; the $(1-\gamma)$ factor is absent because the operative branch is linear in $\hat{x}$. □

Proof of Corollary I.11. *(i)* Under Proposition I.7 the binding entry margin and both exit margins compare an *informal* continuation with the outside value, so by Lemma I.5 they respond only to informal-side terms (second and third rows of Proposition I.10): all three thresholds fall when $\varepsilon_i > 0$ or $\mathrm{d}b_i > 0$, and total entry, the mass of entrants above $\hat{x}_i^{\mathrm{E}}$, rises.

(ii)–(iii) By Lemma I.5, total differentiation of the formality margin of origin o gives

$$\mathrm{d}\Delta_o = -a_f^* \,\mathrm{d}\phi_f + a_i^* \,\mathrm{d}\phi_i = \varepsilon_f a_f^* - \varepsilon_i a_i^*,$$

so $\hat{x}_o^{\mathrm{F}}$ falls, formality gains ground, if and only if $\varepsilon_f a_f^* > \varepsilon_i a_i^*$. Since $p_{i\to f}$ falls in $\hat{x}_i^{\mathrm{F}}$ while $p_{f\to i}$ rises in $\hat{x}_f^{\mathrm{F}}$, this is (41) at the $i \to f$ margin and, reversed, (42) at the $f \to i$ margin. For the borrowing gaps: at the $i \to f$ margin the formal option carries funds $n - c_{if}$ and the informal option n, so

$$a_f^* - a_i^* = \left[\tilde{k}_f - (n - c_{if})\right] - \left[\tilde{k}_i - n\right] = \tilde{k}_f - \tilde{k}_i + c_{if}.$$

For the entrant margin the fee pair (c_f^e, c_i^e) gives $a_f^* - a_i^* = \tilde{k}_f - \tilde{k}_i + c_f^e - c_i^e$, so the same argument applies to $\hat{x}_e^{\mathrm{F}}$.

Under Assumption 1, $T_\lambda > (R_f/R_i)^\gamma$ implies $T_\lambda R_i/R_f > (R_i/R_f)^{1-\gamma} > 1$, hence $\tilde{k}_f/\tilde{k}_i = \left(T_\lambda R_i/R_f\right)^{1/(1-\gamma)} > 1$: the gap is strictly positive, and $\varepsilon_f = \varepsilon_i$ delivers (41). At the $f \to i$ margin the fee switches sides, $a_i^* - a_f^* = \tilde{k}_i - \tilde{k}_f + c_{fi}$, giving the statement in (iii). □

Proof of Proposition I.12. (i) The entry threshold $\bar{v}$ falls by Proposition I.10 (second row), so the entry probability $1 - F_v(\bar{v})$ rises for every k_0 with a constrained margin. (ii) For a log-normal v the truncated mean $m(\bar{v}) \equiv \mathbb{E}[v \mid v \geq \bar{v}]$ satisfies $m'(\bar{v}) = h(\bar{v})\,[m(\bar{v}) - \bar{v}] > 0$, where $h = f_v/(1 - F_v)$ is the hazard and $m(\bar{v}) > \bar{v}$; hence a lower threshold strictly lowers the truncated mean, and first-order stochastic dominance of the truncated distributions delivers the same for every quantile and for average value added (an increasing transform of v). (iii) Write the cohort shutdown rate as $S(\bar{v}) = \mathbb{E}[\pi^X(v) \mid v \geq \bar{v}]$ with π^X strictly decreasing (by (43) and $\partial k_2/\partial v \geq 0$). Then $S'(\bar{v}) = h(\bar{v})\left[S(\bar{v}) - \pi^X(\bar{v})\right] < 0$ because $\pi^X(\bar{v}) > \pi^X(v)$ for all $v > \bar{v}$ implies $\pi^X(\bar{v}) > S(\bar{v})$. A lower $\bar{v}$ therefore raises the cohort shutdown rate. □

K Explicit instance of the estimand: informality mass

Fix the outcome $Y = N_i'$, the mass choosing the informal sector at the transition stage (survival $1 - \lambda_i$ multiplies the measured stock but cancels from $\mathrm{d}\ln Y$). Six thresholds delimit it: informal incumbents contribute the band $[\hat{x}_i^{\mathrm{X}}, \hat{x}_i^{\mathrm{F}})$, formal incumbents the switch-down band $[\hat{x}_f^{\mathrm{X}}, \hat{x}_f^{\mathrm{F}})$, and entrants the interval $[\bar{v}^E, \bar{v}^F)$ in type units. Assume the empirically relevant financing regions (R3 at every one-sided margin, R3×R3 at the formality margins);

on R4-financed margins the formulas below pick up the collateral terms (μk^*, $\kappa_s b_s$) per the taxonomy appendix. For each margin M, contracting Proposition I.10 with the shock $(\mathrm{d}\phi_s, \mathrm{d}b_s) = (-\chi_s\phi_s, \kappa_s b_s)\,\mathrm{d}\zeta$ gives the log-shift per unit of intensity,

$$s_M = -(1-\gamma)\,\chi_i\phi_i\,\frac{a^*_{i,M}}{N_M}, \qquad M \in \{X_i,\, X_f,\, E\} \text{ (one-sided, i-continuation)},$$
$$s_{F,o} = -\frac{1-\gamma}{D_o}\Big[\chi_f\phi_f\,a^*_{of} - \chi_i\phi_i\,a^*_{oi}\Big], \quad o \in \{i, f, e\} \text{ (formality margins)},$$

with $(\tilde{n}_M, m_M)$ each margin's funds–cost pair, $N_M = m_M - \phi_i\tilde{n}_M$ the operative R3 numerator, and all a^*'s evaluated at the crossing. The density weights at the margins are $w_M = \frac{1-\nu}{\rho\sigma_\varepsilon}\,\phi(z_M)$ for incumbent margins and $w_M = (1-\nu)\,f_v(\bar{v}_M)\,\bar{v}_M$ for entrant margins (entry decisions are deterministic in v, so the relevant density is that of the type distribution). Writing $\iota_M = +1$ if an upward crossing of M moves a firm *into* Y and -1 if out of it ($\iota = +1$ at X_i, X_f, E; $\iota = -1$ at F_i, F_f, F_e), the reduced form is

$$\frac{\mathrm{d}\ln N_i'}{\mathrm{d}\zeta} \;=\; -\frac{1}{N_i'}\sum_M \iota_M \int w_M\, s_M\, \mathrm{d}\mu_{o(M)}, \tag{47}$$

each integral taken over the origin population carrying margin M. The first stage decomposes into the loan-size response of continuing borrowers and the loans created or swapped at the same six margins:

$$\frac{\mathrm{d}\ln C}{\mathrm{d}\zeta} \;=\; \frac{1}{C}\left[\sum_{s'\in\{i,f\}}\int_{\text{borrowers to } s'} \frac{\chi_{s'}\phi_{s'}\,\tilde{k}_{s'}(\hat{x})}{(1-\gamma)\,R_{s'}}\,\mathrm{d}\mu \;-\; \sum_M \int w_M\, s_M\, \Delta a^*_M\, \mathrm{d}\mu_{o(M)}\right], \tag{48}$$

where Δa^*_M is the borrowing gained on an upward crossing: the new loan $a^*_{i,M}$ at X_i, X_f, E (exit and non-entry carry no loan), and the destination swap $\Delta a^*_{F,o} = \tilde{k}_f - \tilde{k}_i + \kappa_{of} - \kappa_{oi}$ at the formality margins — precisely the borrowing gaps of Corollary I.11. The model coefficient is the ratio $\beta^{\text{model}}_{N_i} = \big(\mathrm{d}\ln N_i'/\mathrm{d}\zeta\big)\big/\big(\mathrm{d}\ln C/\mathrm{d}\zeta\big)$; every term in (47)–(48) is linear in the incidence $(\chi_i, \chi_f, \kappa_f)$, so the scale of ζ cancels.

L Complete threshold taxonomy

This appendix reports the fully substituted closed form of every exit and entry threshold on every financing region. Throughout, $\mathcal{K}_s \equiv \beta(1-\lambda_s)c_s$ (so that $K = \mathcal{K}_f - \mathcal{K}_i$), $A_s = (1-\gamma)(\gamma/R_s)^{\frac{\gamma}{1-\gamma}} M_s^{\frac{1}{1-\gamma}}$ and $A^0_s = (1-\gamma)(\gamma/R^0)^{\frac{\gamma}{1-\gamma}} M_s^{\frac{1}{1-\gamma}}$, with $A = A_f - A_i$ and $A^0 = A^0_f - A^0_i$. Each one-sided margin (exit or entry against the outside option, informal continuation being the relevant alternative by Proposition I.7) is the solution of $G_i(\hat{x}, \tilde{n}) = m$ with the margin-specific pair $(\tilde{n}, m)$:

margin	$\tilde{n}$	m	comparison
exit as informal	n	$\mathcal{K}_i$	$U_i(\hat{x}, n) \gtrless n$
exit as formal	$n - c_{fi}$	$\mathcal{K}_i + c_{fi}$	$U_i(\hat{x}, n - c_{fi}) \gtrless n$
entry as informal	$k_0 - c_i^e$	$\mathcal{K}_i + c_i^e$	$U_i(\hat{x}, k_0 - c_i^e) \gtrless k_0$

with $n = n^{\text{exit}}$ of (5). Substituting the four branches of (22):

Exit as informal.

$$\hat{x}_i^{\mathrm{X}}(n) = \begin{cases} \left(\mathcal{K}_i / A_i^0\right)^{1-\gamma}, & \text{R1} \\ \dfrac{\mathcal{K}_i + R^0 n}{M_i\, n^{\gamma}}, & \text{R2} \\ \left[\dfrac{\mathcal{K}_i - \phi_i n}{A_i}\right]^{1-\gamma}, & \text{R3, if } \mathcal{K}_i > \phi_i n \\ (1-b_i)^{\gamma-1}\, \dfrac{\mathcal{K}_i(1-b_i) + n\big(R^0 + \phi_i b_i\big)}{M_i\, n^{\gamma}}, & \text{R4.} \end{cases} \tag{49}$$

Exit as formal (against switching down).

$$\hat{x}_f^{\mathrm{X}}(n) = \begin{cases} \left[(\mathcal{K}_i + c_{fi}) / A_i^0\right]^{1-\gamma}, & \text{R1} \\ \dfrac{\mathcal{K}_i + c_{fi} + R^0 (n - c_{fi})}{M_i\, (n - c_{fi})^{\gamma}}, & \text{R2} \\ \left[\dfrac{\mathcal{K}_i + (1+\phi_i) c_{fi} - \phi_i n}{A_i}\right]^{1-\gamma}, & \text{R3, if numerator} > 0 \\ (1-b_i)^{\gamma-1}\, \dfrac{(\mathcal{K}_i + c_{fi})(1-b_i) + (n - c_{fi})\big(R^0 + \phi_i b_i\big)}{M_i\, (n - c_{fi})^{\gamma}}, & \text{R4.} \end{cases} \tag{50}$$

Entry as informal. Replace (n, c_{fi}) by (k_0, c_i^e) in (50); in particular on R3,

$$\hat{x}_i^{\mathrm{E}}(k_0) = \left[\frac{\mathcal{K}_i + (1+\phi_i) c_i^e - \phi_i k_0}{A_i}\right]^{1-\gamma}, \tag{51}$$

and in type units $\bar{v}(k_0) = \hat{x}_i^{\mathrm{E}}(k_0)^{\,1-\nu} \exp\{-\sigma_\varepsilon^2 / (2(1-\nu))\}$.

Entry as formal (the entrant formality margin). This margin compares two continuation values, each on its own region, with funds $\tilde{n}_f = k_0 - c_f^e$ and $\tilde{n}_i = k_0 - c_i^e$. Because $G_s(\cdot, \tilde{n})$ is a power function of $\hat{x}$ on R1/R3 but *linear* in $\hat{x}$ on R2/R4, closed forms

obtain on the diagonal region pairs:

$$\text{R3} \times \text{R3}: \quad \hat{x}_e^{\mathrm{F}}(k_0) = \left[\frac{B\,k_0 + C_e}{A}\right]^{1-\gamma}, \qquad C_e = (1+\phi_f)c_f^e - (1+\phi_i)c_i^e + K; \tag{52}$$

$$\text{R1} \times \text{R1}: \quad \hat{x}_e^{\mathrm{F}} = \left(C_e^0/A^0\right)^{1-\gamma}, \qquad C_e^0 = c_f^e - c_i^e + K, \tag{53}$$

(direction of the inequality per the trichotomy of Remark I.9);

$$\text{R2} \times \text{R2}: \quad \hat{x}_e^{\mathrm{F}} = \frac{K + \beta(1-\delta)\left(c_f^e - c_i^e\right)}{M_f \tilde{n}_f^{\gamma} - M_i \tilde{n}_i^{\gamma}}, \qquad \text{formal above iff the denominator} > 0 \tag{54}$$

(at kink, only cost is forgone resale value, hence $\beta(1-\delta)$, not R^0);

$$\text{R4} \times \text{R4}: \quad \hat{x}_e^{\mathrm{F}} = \frac{K + R_f h_f - R_i h_i + (1+\phi_i)\tilde{n}_i - (1+\phi_f)\tilde{n}_f}{M_f h_f^{\gamma} - M_i h_i^{\gamma}}, \tag{55}$$

with $h_s \equiv \tilde{n}_s/(1-b_s)$ the capital installed at the binding cap.

Off-diagonal pairs (one destination borrowing while the other self-finances, e.g. R3×R2) mix $\hat{x}^{1/(1-\gamma)}$ and $\hat{x}$ terms: the indifference condition remains strictly monotone in $\hat{x}$ at the crossing and has a unique root, but no closed form; it is solved implicitly. The same diagonal/off-diagonal taxonomy applies verbatim to the incumbent formality margins $\hat{x}_i^{\mathrm{F}}, \hat{x}_f^{\mathrm{F}}$ with the fee pairs of Lemma I.8.

Branch selection. Which branch applies is a self-consistency requirement: compute the candidate threshold from each branch and retain the one whose funds fall in that branch's bracket *evaluated at the candidate*,

$$\text{R1: } \tilde{n} \geq \bar{k}_s(\hat{x}^*); \quad \text{R2: } \tilde{k}_s(\hat{x}^*) \leq \tilde{n} < \bar{k}_s(\hat{x}^*); \quad \text{R3: } (1-b_s)\tilde{k}_s(\hat{x}^*) \leq \tilde{n} < \tilde{k}_s(\hat{x}^*); \quad \text{R4: below.}$$

Exactly one branch is self-consistent: G_s is continuous and strictly increasing in $\hat{x}$ ($\partial G_s/\partial \hat{x} = M_s k^{*\gamma} > 0$ by the envelope theorem), and at fixed $\tilde{n}$ the firm moves monotonically through R1→R2→R3→R4 as $\hat{x}$ rises, so $G_s(\cdot, \tilde{n})$ crosses any constant m exactly once.

M Discussion of modeling assumptions

Holding the wage fixed is the exact analogue of the empirical identification, which is within market and year and purged of aggregate shocks, and it is additionally supported by local wage stickiness at the five year horizon. Debt is within period. Firms borrow to install capital and repay out of the following period's resources, so net worth stays out of the state vector, at the cost of abstracting from leverage dynamics. The measurement counterpart is that model borrowing maps to gross new credit within a year and the cap b_s to the ratio of credit lines to capital among borrowers. Loans are safe because the cap guarantees repayment, so ϕ_s is an intermediation cost rather than a default premium, and

reinterpreting part of it as a default premium would change the welfare accounting but not the firm's problem. The financial primitives themselves are measured rather than derived from deeper frictions.[28] Allowing the productivity process to differ by sector is essential, since the census autocovariances imply higher innovation variance for informal firms, and imposing a common process would force the switching costs to absorb the difference.

[28]Limited enforcement generates endogenous borrowing limits (Albuquerque and Hopenhayn, 2004; Clementi and Hopenhayn, 2006), costly intermediation and default generate endogenous spreads (Cooley and Quadrini, 2001; Arellano et al., 2012), and D'Erasmo and Boedo (2012) embed endogenous credit in an economy with informality. Each such theory delivers exactly the two objects assumed here, a quantity limit and a price wedge, and predicts the estimated sector ranking. Treating them as measured primitives, as in Midrigan and Xu (2014) and Moll (2014), keeps the mapping from the quasi-experimental evidence transparent. The cost is a caveat on the counterfactuals. Under a large formalization the informal primitives would themselves improve, so experiments holding them fixed likely understate the financial channel, in the direction Brooks and Dovis (2020) document.